

AN INVERSE CALCULUS FOR THE ODD LAYER OF THE COLLATZ MAP

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ABSTRACT. We develop a finite-state inverse calculus for the odd layer of the Collatz map. The construction assigns to each non-ghost inverse branch a row, a token, and a lift parameter $p \geq 0$, producing a complete row/token/ p -classification of inverse transitions. This classification turns the inverse Collatz graph into a symbolic dynamical system equipped with explicit affine formulas, residue targets, base witnesses, and transition rules. It also gives a natural state-transition network and a stochastic interpretation of inverse branching, including stationary and entropy-based descriptions of the resulting inverse dynamics.

The central deterministic application of the calculus is forward descent. The guiding principle is that inverse growth is equivalent to descent in the forward odd orbit. Using the drift equation, we show that every row with $p \geq 1$ is strictly inverse-expansive and that every non-exceptional $p = 0$ row is strictly inverse-expansive away from the terminal fixed point $1 \mapsto 1$. Thus any persistent obstruction to forward descent must lie in the exceptional $p = 0$ anti-growth mechanism.

We isolate this exceptional mechanism in the $(o, 2)$ -row. A pure $(o, 2)$ -run is finite for every finite positive state. The only remaining local obstruction is therefore the compensated exceptional $(o, 2)/(o, 1)$ -segment, where $(o, 1)$ is the least compensating row in the $p = 0$ table. For finite prefixes of such segments, the residue calculus gives nested constraints

$$m_0 = A_N + 3^{L_N} y_N, \quad A_{N+1} = A_N + 3^{L_N} c_N.$$

If such a segment were extendable indefinitely from a finite positive m_0 , then either the residues define only a 3-adic object, or they stabilize with

$$A_N = m_0, \quad y_N = 0, \quad c_N = 0.$$

In the stabilizing case the exceptional residue-building process is exhausted.

Since the compensated $(o, 2)/(o, 1)$ -segment is the worst local anti-growth model, all forward non-descent obstructions detected by the row calculus are exhausted. It follows that every odd integer $x > 1$ reaches a smaller odd iterate. By infinite descent, every positive integer reaches 1 under the Collatz map.

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Part 1. The Inverse Machine: Definitions and Mechanics

1. INTRODUCTION AND RELATED WORK

The Collatz conjecture asserts that the dynamics of the $3x + 1$ map on positive integers eventually reaches the cycle $1 \rightarrow 4 \rightarrow 2 \rightarrow 1$. In this paper, we focus on the *odd layer* of the dynamics, governed by the accelerated map:

$$U(y) = \frac{3y + 1}{2^{\nu_2(3y+1)}},$$

where $\nu_2(n)$ is the 2-adic valuation of n . Our approach develops a finite-state, word-based framework for the inverse of this map. By establishing a unified table of inverse operations (“rows”) and a calculus of their composition (“words”), we transform the problem of reachability into a deterministic procedure of solving linear congruences.

Our framework rests on three novel components: a *unified inverse table* that generates certified preimages $U(x') = x$; a *column-lift parameter* p that scales the 2-adic slope while preserving modular routing; and *explicit steering gadgets* that manipulate the affine parameters of a word to ensure 2-adic congruences are always solvable.

1.1. Relation to Prior Techniques. Our approach—finite word semantics on the odd layer, certified one-step inverses, and congruence-based “steering” to lift residues from $M_K = 3 \cdot 2^K$ to M_{K+1} —sits alongside several established techniques in the literature.

Mod- 2^k analysis and lifting. Garner studied the $3n+1$ dynamics modulo powers of two, organizing inverse branches by congruence classes and effectively “lifting” structure from 2^k to 2^{k+1} [2]. Our use of the unified rows with a column-lift parameter p (which multiplies the 2-adic slope by 2^{6p}) and the residue steering gadgets plays a similar role: we solve linear congruences for m to pass from M_K to M_{K+1} while preserving certified inverses at each step.

Inverse trees and predecessor sets. Wirsching’s monograph develops the inverse (predecessor) tree of the $3n+1$ function as a dynamical system, with emphasis on structure, measures, and asymptotics on inverse branches [9]. Conceptually, our move alphabet and per-row affine forms are a finite-state presentation of those inverse branches: each token certifies $U(x') = x$ and the composition yields an affine map in the “index” m , which we then route by residues M_K .

The 2-adic viewpoint and conjugacies. Bernstein and Lagarias constructed a 2-adic conjugacy map relating the odd-accelerated Collatz dynamics to a Bernoulli-like shift [1]. Our p -lift (multiplying by 2^{6p}) and the parity/valuation steering reflect this same 2-adic continuity: column-lifts shift 2-adic scale, while steering gadgets tune intercept parity to land on prescribed residue classes.

Symmetries and autoconjugacy. Monks and Yazinski analyzed autoconjugacies of the $3x+1$ function and their implications for orbit structure [5]. While our framework is more combinatorial/affine, the way we keep the family pattern fixed and exploit same-family padding resonates with their use of structural symmetries.

Surveys and context. For broad background and additional modular/density insights, see [4, 7, 8]; for 2-adic heuristics and continuity themes, see [3, 6]. These perspectives motivate our use of 2-adic “padding” and linear congruences as lifting mechanisms.

1.2. Contributions. The main contribution of this work is a unified, certified inverse-word calculus on the odd layer together with explicit steering gadgets that turn residue targeting into solvable congruences. Specifically:

- **One-table, word-driven inverse calculus.** We give a unified $p=0$ row table with closed forms $x' = 6F(0, m) + \delta$ indexed only by (s, j, m) . Once a token and (s, j) are fixed, the step is fully determined and the forward identity $3x' + 1 = 2^\alpha x$ holds by construction.
- **Column-lift p .** The parameter p multiplies the slope by 2^{6p} without changing the token type or output family, yielding a single mechanism that subsumes whole towers of congruence tables.
- **CRT tag for transparent indexing.** The tag $t = (x - 1)/2$ (equivalently $(3x + 1 - 4)/6$) makes family detection and indices (s, j, m) linear in t , simplifying routing proofs.

- **Steering gadgets.** Short same-family words provably boost the slope's 2-adic valuation and toggle the affine intercept $B \bmod 2$ (and $B \bmod 3$), ensuring solvability of the lifting congruence at each modulus.
- **From small witnesses to exact integers.** Starting at $M_3=24$, we give a deterministic induction $M_K \rightarrow M_{K+1}$ that reaches every odd residue with certified steps, and then a 2-adic refinement to hit any prescribed odd integer exactly.
- **Executable certificates.** A reference implementation emits step traces and verifies $U(x') = x$ at each step, making all claims reproducible.
- **Topological and Probabilistic Analysis.** We formalize the network as a Finite State Automaton and compute the stationary distribution of the resulting Markov chain. This yields a precise Lyapunov exponent for the inverse map ($\Lambda \approx 3.51$), explaining the difficulty of the problem as a competition between global expansion and topologically constrained contraction channels.

1.3. Main Claim and Method. Our main claim (Theorem 8) is that every odd $x \equiv 1, 5 \pmod{6}$ reaches 1 in finitely many accelerated odd Collatz steps. The method is modular: (i) certify row-level inverses $U(x') = x$; (ii) show any admissible word yields an affine form in m with controlled terminal family; (iii) furnish base witnesses modulo 24; (iv) use same-family *steering gadgets* to raise $v_2(A)$ and control $B \bmod 2$ and $B \bmod 3$; (v) lift residues $M_K \rightarrow M_{K+1}$; and (vi) pass from residues to exact integers by 2-adic refinement.

2. PRELIMINARIES: NOTATION AND INDICES

We enumerate the ambient assumptions and notation used throughout. All variables are integers unless noted. We work exclusively on the *odd layer*: inputs x are always positive odd integers.

2.1. The Accelerated Odd Map. We use the accelerated odd Collatz map $U(y)$, standard in the literature [4]:

$$U(y) = \frac{3y + 1}{2^{\nu_2(3y+1)}},$$

where $\nu_2(n)$ denotes the 2-adic valuation of n . Since $3y + 1$ is always even for odd y , $U(y)$ returns an odd integer. Furthermore, for any odd y , $3y + 1 \equiv 4 \pmod{6}$. Dividing by powers of 2 (which are $\equiv 2, 4 \pmod{6}$) yields an output $U(y)$ congruent to either 1 or 5 modulo 6. Thus, the residue class 3 (mod 6) never appears in the image of U .

2.2. Families and Indices. We classify odd integers $x \not\equiv 3 \pmod{6}$ into two families:

$$(1) \quad s(x) = \begin{cases} \text{e}, & \text{if } x \equiv 1 \pmod{6}, \\ \text{o}, & \text{if } x \equiv 5 \pmod{6}. \end{cases}$$

To enable a finite-state calculus, we decompose x into three hierarchical indices:

(1) The **Coarse Index** r :

$$r = \left\lfloor \frac{x}{6} \right\rfloor.$$

(2) The **Router** j (determines the next table row):

$$j = r \bmod 3 \in \{0, 1, 2\}.$$

(3) The **Internal Index** m (the operand for the affine form):

$$m = \left\lfloor \frac{x}{18} \right\rfloor.$$

From these, x can be uniquely reconstructed as:

$$x = 18m + 6j + p_6, \quad \text{where } p_6 \in \{1, 5\} \text{ is determined by } s(x).$$

2.3. The CRT Tag and Re-indexing. To simplify the arithmetic of families and indices, we introduce the **CRT Tag** $t(x)$.

Lemma 1 (CRT tag). *For odd x , the quantity*

$$t = \frac{x-1}{2}$$

is an integer. The map $x \mapsto t$ is a bijection between odd integers $x \geq 1$ and non-negative integers $t \geq 0$.

The tag t linearizes the family and index calculations, avoiding nested floors in many proofs.

Corollary 1 (Indices from tag). *The indices $s(x)$, j , and m are determined by t modulo 3 and 9:*

$$x \bmod 6 = 2(t \bmod 3) + 1,$$

$$m = \left\lfloor \frac{t}{9} \right\rfloor,$$

$$j = \left\lfloor \frac{t}{3} \right\rfloor \bmod 3.$$

Specifically, if $t \equiv 0 \pmod{3}$, then $x \in \text{e}$; if $t \equiv 2 \pmod{3}$, then $x \in \text{o}$. The case $t \equiv 1 \pmod{3}$ corresponds to $x \equiv 3 \pmod{6}$, which is excluded from the odd layer.

2.4. Move Alphabet. We define a set of tokens $\mathcal{A} = \{\Psi, \psi, \omega, \Omega\}$ representing the valid transitions between families:

- Ψ (type ee): Maps family e $\rightarrow$ e.
- ψ (type eo): Maps family e $\rightarrow$ o.
- ω (type oe): Maps family o $\rightarrow$ e.
- Ω (type oo): Maps family o $\rightarrow$ o.

A sequence of these tokens is called a *word* W .

2.5. Summary of Notation (Quick Reference). To aid the reader, we summarize the hierarchy of indices used to classify odd integers in the inverse map.

Table 1. Nomenclature and Variable Hierarchy.

Symbol	Definition	Role / Intuition
x	$x \in 2\mathbb{Z} + 1$	The Odd Integer (the value itself).
t	$(x - 1)/2$	The CRT Tag . A continuous index $0, 1, 2 \dots$ that linearizes the geometry.
ρ	$t \bmod 9$	The Topological Node . Determines the parameter family and network role (e.g., Attractor, Heart).
m	$\lfloor t/9 \rfloor$	The Internal Index . The operand for the affine map $x' = Am + B$.
j	$\lfloor t/3 \rfloor \bmod 3$	The Router . Determines the "next-hop" admissibility modulo 3.
$s(x)$	$x \bmod 6$	The Parity Family . e (if $x \equiv 1$) or o (if $x \equiv 5$).
p	$p \geq 0$	The Column Lift . Scales the step size (multiplies slope by 64^p).
α	Table constant	The Base Valuation . The minimal power of 2 removed in a forward step.

We use the symbol σ for a 2-adic valuation exponent; its precise meaning (e.g. $\sigma = v_2(A_W)$, $\sigma = v_2(3x' + 1)$ or $\sigma = \alpha + 6p$) will be specified locally when used.

3. THE UNIFIED INVERSE TABLE

To unify all Collatz inverse orbits, we parametrize every possible step using a fixed set of row parameters $(\alpha, \beta, c, \delta)$ and a dynamic column-lift $p \in \mathbb{Z}_{\geq 0}$. This allows us to treat the inverse map as a table lookup determined solely by the indices $s(x)$ and j .

3.1. The Parameter Table. Table 2 lists the twelve canonical rows derived from the constraints above. The type indicates the transition from input family to output family (e.g., eo means input e, output o).

Table 2. Row parameters $(\alpha, \beta, c, \delta)$. Keys: $eej \leftrightarrow \Psi_j$, $eo j \leftrightarrow \psi_j$, $oej \leftrightarrow \omega_j$, $ooj \leftrightarrow \Omega_j$.

Row	(s, j)	Type	α	β	c	(δ)
Ψ_0	(e, 0)	ee	2	2	-2	(1)
Ψ_1	(e, 1)	ee	4	56	-2	(1)
Ψ_2	(e, 2)	ee	6	416	-2	(1)
ω_0	(o, 0)	oe	3	20	-2	(1)
ω_1	(o, 1)	oe	1	11	-2	(1)
ω_2	(o, 2)	oe	5	272	-2	(1)
ψ_0	(e, 0)	eo	4	8	-8	(5)
ψ_1	(e, 1)	eo	6	224	-8	(5)
ψ_2	(e, 2)	eo	2	26	-8	(5)
Ω_0	(o, 0)	oo	5	80	-8	(5)
Ω_1	(o, 1)	oo	3	44	-8	(5)
Ω_2	(o, 2)	oo	1	17	-8	(5)

3.2. The Unified p -Lifted Form. To reach arbitrarily high powers of 2, we extend the base table with a column–lift parameter $p \geq 0$. This parameter scales the 2-adic slope by 2^{6p} while preserving the routing logic.

For any row in Table 2, define the lifted transform:

$$(2) \quad F(p, m) := \frac{(9m 2^\alpha + \beta) 64^p + c}{9}, \quad x' := 6 F(p, m) + \delta.$$

Remark (Integrality). Since $64 \equiv 1 \pmod{9}$, we have $\beta 64^p + c \equiv \beta + c \pmod{9}$. Since $\beta + c$ is divisible by 9 for all valid rows, $F(p, m)$ is always an integer.

3.3. The Base $p = 0$ Table (Straight Substitution). At $p = 0$, the formula simplifies to the affine forms shown in Table 3. These are the fundamental building blocks of the inverse calculus.

4. ROW CORRECTNESS AND DRIFT

Having defined the unified parameter table and the lifted transform, we now verify two fundamental properties:

- (1) **Correctness:** Every step $x \mapsto x'$ produced by the table is a valid inverse of the accelerated Collatz map U .
- (2) **Linearity:** The change in the CRT tag (the “drift”) is a linear function of the coarse index r .

Table 3. Unified $p = 0$ forms with $x'(m) = 6F(0, m) + \delta$.

(s, j)	Type	Token	$x'(m)$
(e, 0)	ee	Ψ_0	$24m + 1$
(e, 1)	ee	Ψ_1	$96m + 37$
(e, 2)	ee	Ψ_2	$384m + 277$
(o, 0)	oe	ω_0	$48m + 13$
(o, 1)	oe	ω_1	$12m + 7$
(o, 2)	oe	ω_2	$192m + 181$
(e, 0)	eo	ψ_0	$96m + 5$
(e, 1)	eo	ψ_1	$384m + 149$
(e, 2)	eo	ψ_2	$24m + 17$
(o, 0)	oo	Ω_0	$192m + 53$
(o, 1)	oo	Ω_1	$48m + 29$
(o, 2)	oo	Ω_2	$12m + 11$

4.1. The Forward Identity. We first prove that the row design constraints derived in Section 3 guarantee the forward identity $3x' + 1 = 2^\sigma x$ for all $p \geq 0$ where $\sigma := \alpha + 6p$.

Lemma 2 (Row Correctness). *Fix any admissible row with parameters $(\alpha, \beta, c, \delta)$ and a column-lift $p \geq 0$. Let $x = 18m + 6j + \delta_6$ be an admissible input. Let $x' = 6F(p, m) + \delta$ be the value generated by the unified table. Then:*

$$(3) \quad 3x' + 1 = 2^{\alpha+6p} x.$$

Consequently, $\nu_2(3x' + 1) = \alpha + 6p$ and $U(x') = x$.

Proof. Substitute the definition of x' and $F(p, m)$:

$$\begin{aligned} 3x' + 1 &= 3 \left(6 \left[\frac{(9m 2^\alpha + \beta) 64^p + c}{9} \right] + \delta \right) + 1 \\ &= 2((9m 2^\alpha + \beta) 64^p + c) + 3\delta + 1. \end{aligned}$$

Expanding terms:

$$3x' + 1 = 18m 2^\alpha 64^p + 2\beta 64^p + (2c + 3\delta + 1).$$

Recall the row design constraint $c = -(3\delta + 1)/2$, which implies $2c + 3\delta + 1 = 0$. The constant term vanishes, leaving:

$$3x' + 1 = 18m 2^{\alpha+6p} + 2\beta 2^{6p}.$$

We determine the value of β by matching the residue part of the target odd integer. Write the target as

$$x = 18m + 6j + \delta_{\text{tar}}.$$

The inverse row is required to satisfy

$$3x' + 1 = 2^{\alpha+6p} x.$$

Hence

$$2^{\alpha+6p}x = 2^\alpha 64^p(18m + 6j + \delta_{\text{tar}}).$$

Expanding the right-hand side gives

$$2^{\alpha+6p}x = 18m 2^\alpha 64^p + 2^\alpha(6j + \delta_{\text{tar}})64^p.$$

On the other hand, using

$$x' = 6F(p, m) + \delta_{\text{src}},$$

we have

$$3x' + 1 = 18F(p, m) + 3\delta_{\text{src}} + 1.$$

With

$$F(p, m) = \frac{(9m2^\alpha + \beta)64^p + c}{9},$$

this becomes

$$3x' + 1 = 18m 2^\alpha 64^p + 2\beta 64^p + 2c + 3\delta_{\text{src}} + 1.$$

The m -terms already match and $2c + 3\delta_{\text{src}} + 1 = 0$. Therefore the residue terms must satisfy

$$2\beta 64^p = 2^\alpha(6j + \delta_{\text{tar}})64^p.$$

Cancelling 64^p , we obtain

$$2\beta = 2^\alpha(6j + \delta_{\text{tar}}).$$

Therefore

$$\boxed{\beta = 2^{\alpha-1}(6j + \delta_{\text{tar}})}.$$

Substitute $\beta = 2^{\alpha-1}(6j + \delta_6)$:

$$\begin{aligned} 3x' + 1 &= 18m 2^{\alpha+6p} + 2(2^{\alpha-1}(6j + \delta_6))2^{6p} \\ &= 18m 2^{\alpha+6p} + 2^\alpha(6j + \delta_6)2^{6p} \\ &= 2^{\alpha+6p}(18m + 6j + \delta_6) \\ &= 2^{\alpha+6p}x. \end{aligned}$$

Since x is odd, the valuation is exactly $\alpha + 6p$. $\square$

Example 1 (Verification of ω_1 at $p = 0$). Consider the row ω_1 (family o, $j = 1$). Parameters are $\alpha = 1, \beta = 11, c = -2, \delta = 1$. Let $x = 29$.

- **Input Analysis:** $x \equiv 5 \pmod{6}$ (family o), $r = \lfloor 29/6 \rfloor = 4$, $j = 4 \bmod 3 = 1$. The row is admissible. $m = \lfloor 29/18 \rfloor = 1$.
- **Calculate x' :** $F(0, 1) = (9(1)2^1 + 11 - 2)/9 = (18 + 9)/9 = 3$.
 $x' = 6(3) + 1 = 19$.
- **Check Identity:** $3(19) + 1 = 58$. The formula predicts $2^\alpha x = 2^1(29) = 58$. It matches.
- **Forward Map:** $U(19) = (57 + 1)/2^1 = 29 = x$. Verified.

Lemma 3 (Completeness of the row table for non-ghost inverses). *Let $U : \mathbb{Z}_{>0}^{\text{odd}} \rightarrow \mathbb{Z}_{>0}^{\text{odd}}$ be the accelerated Collatz map*

$$U(x') = \frac{3x' + 1}{2^{\nu_2(3x'+1)}}.$$

Fix an active odd target $x \not\equiv 0 \pmod{3}$, and write

$$x = 18m + 6j + \delta_6, \quad m \in \mathbb{Z}_{\geq 0}, j \in \{0, 1, 2\}, \delta_6 \in \{1, 5\},$$

so $\delta_6 = 1$ corresponds to $s = \text{e}$ (i.e. $x \equiv 1 \pmod{6}$) and $\delta_6 = 5$ corresponds to $s = \text{o}$ (i.e. $x \equiv 5 \pmod{6}$).

For each integer $A \geq 1$ with $2^A x \equiv 1 \pmod{3}$, define the (formal) inverse

$$x_A := \frac{2^A x - 1}{3}.$$

Call x_A a ghost inverse if $x_A \equiv 0 \pmod{3}$ (equivalently, x_A is an inactive predecessor), and a non-ghost inverse otherwise.

Then the following hold.

(i) Unique forbidden class modulo 6. *For fixed (s, j) , among the admissible residue classes $A \pmod{6}$*

$$A \pmod{6} \in \begin{cases} \{0, 2, 4\}, & s = \text{e}, \\ \{1, 3, 5\}, & s = \text{o}, \end{cases}$$

there is a unique residue class $a_{\text{bad}}(s, j)$ such that x_A is ghost iff $A \equiv a_{\text{bad}}(s, j) \pmod{6}$.

(ii) The row table selects exactly the two non-ghost classes. *Let $\mathcal{A}(s, j)$ be the pair of α -values appearing in Table 2 for the two rows with key (s, j) (e.g. for $(\text{e}, 0)$ the rows are Ψ_0 and ψ_0 , etc.) Interpreting $\alpha = 6$ as $\alpha \equiv 0 \pmod{6}$, the set $\mathcal{A}(s, j)$ is precisely the set of admissible classes excluding $a_{\text{bad}}(s, j)$. Equivalently, for each (s, j) the table contains exactly the two classes $A \pmod{6}$ yielding non-ghost inverses.*

(iii) Surjectivity onto non-ghost inverses. *For every non-ghost inverse x_A (i.e. $x_A \not\equiv 0 \pmod{3}$), there exists a unique choice of row in Table 2 with key (s, j) and $\alpha \in \mathcal{A}(s, j)$, together with*

$$p = \frac{A - \alpha}{6} \in \mathbb{Z}_{\geq 0},$$

such that the unified lifted form (2) produces exactly x_A :

$$x' = 6F(p, m) + \delta = x_A.$$

Conversely, every value produced by (2) is a non-ghost inverse.

Proof. We use two elementary congruences.

Step 1: ghost condition is a single class $A \bmod 6$. By definition,

$$\begin{aligned} x_A &\equiv 0 \pmod{3} \\ \iff \frac{2^A x - 1}{3} &\equiv 0 \pmod{3} \\ \iff 2^A x &\equiv 1 \pmod{9}. \end{aligned}$$

Since $2^A \bmod 9$ has period 6, the congruence $2^A x \equiv 1 \pmod{9}$, for fixed $x \bmod 9$, determines a unique residue class $A \bmod 6$. Moreover, because $x \not\equiv 0 \pmod{3}$, admissibility $2^A x \equiv 1 \pmod{3}$ forces A to have the parity dictated by $x \bmod 3$, i.e. A even for $s = e$ and A odd for $s = o$. This proves (i).

Step 2: identify the forbidden class and match the table. From $x = 18m + 6j + \delta_6$ we have $x \equiv 6j + \delta_6 \pmod{9}$. A short check gives:

	$j = 0$	$j = 1$	$j = 2$
$s = e$ ($\delta_6 = 1$): $x \bmod 9$	1	7	4
a_{bad} in $\{0, 2, 4\}$	0	2	4
$s = o$ ($\delta_6 = 5$): $x \bmod 9$	5	2	8
a_{bad} in $\{1, 3, 5\}$	1	5	3

Indeed, for example when $s = e, j = 1$ we have $x \equiv 7 \pmod{9}$, and among $a \in \{0, 2, 4\}$ the unique solution to $2^a \cdot 7 \equiv 1 \pmod{9}$ is $a = 2$ (since $2^2 \equiv 4$ and $4 \cdot 7 \equiv 1 \pmod{9}$). The other entries are similar.

Now compare with Table 2: for each fixed (s, j) the table lists exactly two α values, which (reducing $\alpha = 6$ to 0 modulo 6) are precisely the two admissible classes *different* from $a_{\text{bad}}(s, j)$. This proves (ii).

Step 3: surjectivity onto non-ghost inverses. Let x_A be a non-ghost inverse. Then $A \bmod 6$ is admissible and not equal to $a_{\text{bad}}(s, j)$, hence by (ii) there is a unique $\alpha \in \mathcal{A}(s, j)$ with $\alpha \equiv A \pmod{6}$. Set $p = (A - \alpha)/6 \geq 0$.

Choose the corresponding row parameters $(\alpha, \beta, c, \delta)$ from Table 2 and form $x' = 6F(p, m) + \delta$ as in (2). By the Row Correctness Lemma (Lemma 2),

$$3x' + 1 = 2^{\alpha+6p} x = 2^A x,$$

hence $x' = (2^A x - 1)/3 = x_A$. Since x_A was assumed non-ghost, we also have $x' \not\equiv 0 \pmod{3}$, so the produced inverse is non-ghost.

Conversely, any x' produced by (2) satisfies $3x' + 1 = 2^{\alpha+6p} x$, so it is an inverse with exponent $A = \alpha + 6p$; and by (ii) the row table never selects the forbidden class $a_{\text{bad}}(s, j)$, hence $2^A x \not\equiv 1 \pmod{9}$ and $x' \not\equiv 0 \pmod{3}$. This completes (iii). $\square$

Illustration of the exponent-parity rule. The preceding proof shows that admissibility of

$$x_A = \frac{2^A x - 1}{3}$$

already forces the parity of A : if $x \equiv 1 \pmod{6}$, then A is even, while if $x \equiv 5 \pmod{6}$, then A is odd. The following tables show only the first few admissible columns. In principle the columns continue indefinitely, since the admissible powers of 2 do not terminate. Thus a fixed odd target x does not have a single formal inverse predecessor, but an infinite family of inverse candidates indexed by the admissible exponents A . This is one source of the branching, or multidimensional, character of the inverse Collatz calculus.

$x \equiv 1 \pmod{6}$	$A = 2$	$A = 4$	$A = 6$	$A = 8$
1	1	5	21	85
7	9	37	149	597
13	17	69	277	1109
19	25	101	405	1621
25	33	133	533	2133

Table 4. Illustration of the inverse formula $x_A = (2^A x - 1)/3$ for targets $x \equiv 1 \pmod{6}$. Only even exponents A are admissible, and the table records the corresponding odd inverse values x_A .

$x \equiv 5 \pmod{6}$	$A = 1$	$A = 3$	$A = 5$	$A = 7$
5	3	13	53	213
11	7	29	117	469
17	11	45	181	725
23	15	61	245	981
29	19	77	309	1237

Table 5. Illustration of the inverse formula $x_A = (2^A x - 1)/3$ for targets $x \equiv 5 \pmod{6}$. Only odd exponents A are admissible, and the table records the corresponding odd inverse values x_A .

Although only finitely many entries are displayed, each table represents an infinite array: the left column continues through all odd targets in the indicated residue class, while the exponent columns continue indefinitely through the admissible powers of 2.

4.2. The Drift Equation. While the map U^{-1} appears complex on the integers, it simplifies significantly when viewed through the lens of the CRT tag $t(x) = (x - 1)/2$.

Definition 1 (Drift). For a single inverse step $x \xrightarrow{U^{-1}} x'$, the **Drift** d is the change in the tag potential:

$$d := t(x') - t(x) = \frac{x' - x}{2}.$$

The drift measures the "velocity" of the orbit. Positive drift implies the orbit moves upward ($x' > x$); negative drift implies descent ($x' < x$).

Proposition 1 (The Drift Equation). *Let $x = 6r + \varepsilon$ with $\varepsilon \in \{1, 5\}$. For any row in the unified table at column p , the drift is linear in the coarse index r :*

$$(4) \quad d(r, p) = r \cdot K + \Delta_\varepsilon,$$

where the slope K depends on the total exponent $A = \alpha + 6p$:

$$K = 2^A - 3, \quad \Delta_\varepsilon = \frac{\varepsilon(2^A - 3) - 1}{6}.$$

Proof. From Lemma 2, $x' = \frac{2^A x - 1}{3}$. Substitute $x = 6r + \varepsilon$:

$$x' - x = \frac{2^A(6r + \varepsilon) - 1}{3} - (6r + \varepsilon) = 2r(2^A - 3) + \frac{\varepsilon(2^A - 3) - 1}{3}.$$

Dividing by 2 gives the drift d . □

Example 2 (Drift Calculation for ω_1). Using $x = 29$ ($r = 4, \varepsilon = 5$) and $\alpha = 1, p = 0$ ($A = 1$).

- **Tags:** $t(29) = 14, t(19) = 9$. Actual Drift $d = 9 - 14 = -5$.
- **Formula:** $K = 2^1 - 3 = -1$. Offset $\Delta_5 = (5(-1) - 1)/6 = -1$. Predicted $d = 4(-1) + (-1) = -5$. Matches.

Corollary 2 (Drift Bounds). • For family e ($\varepsilon = 1$), the drift is non-negative ($d \geq 0$), with equality only at the terminal fixed point $1 \mapsto 1$.

- Away from this terminal fixed point, every non-exceptional $p = 0$ row has positive inverse drift.
- For family o ($\varepsilon = 5$), the drift is negative ($d < 0$) if and only if $p = 0$ and the token is either ω_1 or Ω_2 .
- For $p \geq 1$, K grows exponentially ($K \approx 2^{6p}$), making the map strongly expansive.

Example 3 (High-Lift Expansiveness). Consider Ψ_0 at $p = 1$ applied to $x = 1$.

- **Parameters:** $\alpha = 2, p = 1 \implies A = 2 + 6 = 8$.
- **Step:** $x = 1 \implies m = 0$. $F(1, 0) = (0 + 2(64) - 2)/9 = 14$. $x' = 6(14) + 1 = 85$.
- **Drift:** $t(1) = 0, t(85) = 42$. $d = 42$.
- **Formula:** $r = 0$. $K = 2^8 - 3 = 253$. $\Delta_1 = (1(253) - 1)/6 = 42$. Matches.

A single step at $p = 1$ multiplied the value by roughly $2^8/3 \approx 85$.

Part 2. The Control Logic: Steering and Routing

5. LIFTED CONTROL SUMMARIES FOR $p \geq 0$

Row correctness shows that each branch row admits the affine form

$$x' \equiv 6F(p, u) + \delta \pmod{M_K}, \quad M_K = 3 \cdot 2^K,$$

with $F(p, u)$ linear in the free index u . For control arguments it is convenient to reorganize the row parameters into derived quantities that depend on the lift level $p \geq 0$ and directly govern pinning and solvability.

Lifted affine form. For a fixed row with parameters $(\alpha, \beta, c, \delta)$ define

$$\alpha_p := \alpha + 6p, \quad k^{(p)} := \frac{\beta 64^p + c}{9} \in \mathbb{Z}.$$

Using $64^p = 2^{6p}$, the row map can be written as

$$(5) \quad x' \equiv 6(2^{\alpha_p} u + k^{(p)}) + \delta \pmod{M_K}.$$

Thus pinning and solving depend only on the comparison of α_p with K , and the constant term $k^{(p)}$.

Router transitions are p -invariant. Reducing (5) modulo 3 shows that the router transition triple $j'(r = 0, 1, 2)$ depends only on the base row, not on p : since $64^p \equiv 1 \pmod{3}$ and $6p$ is even, both $k^{(p)} \pmod{3}$ and $2^{\alpha_p} \pmod{3}$ are independent of p .

Table 12 records, for each input state (s, j) , the two available choices $s' = e$ or $s' = o$ together with the lifted parameters needed in the last-step congruence. In practice, we use Table 12 to select a branch consistent with the desired output family s' (and, in the solving regime, solvability), and then apply the one-shot inversion procedure to compute a compatible input class.

6. AFFINE WORD FORMS AND INDEX EVOLUTION

We now lift the single-step row calculus to sequences of tokens (words). We show that any admissible word W induces a strictly affine map on the internal index m , and we derive the exact recurrence relation that governs the evolution of m_t along the trajectory.

6.1. The Affine Word Form. Let $W = T_1 T_2 \dots T_n$ be a sequence of n tokens, where the t -th token uses column-lift p_t . Let $(\alpha_t, \beta_t, c_t, \delta_t)$ be the parameters of the row selected at step t .

Recall the single-step update from Section 3:

$$x_t = 6(2^{\alpha_t + 6p_t} m_{t-1} + k_t^{(p_t)}) + \delta_t,$$

where $k_t^{(p)} = (\beta_t 64^{p_t} + c_t)/9$.

By composing these linear maps, the action of the entire word W on the initial index m_0 takes a unified affine form.

Lemma 4 (Affine Word Form). *For any admissible word W of length n , there exist constants $A_W > 0$ and $B_W \in \mathbb{Z}$ such that the terminal value x_n is given by*

$$(6) \quad x_W(m_0) = 6(A_W m_0 + B_W) + \delta_W,$$

where:

•

$$A_W = \prod_{t=1}^n 2^{\alpha_t + 6p_t} = 2^{\sum_{t=1}^n (\alpha_t + 6p_t)} = 2^{\alpha(W)},$$

where

$$\alpha(W) := \sum_{t=1}^n (\alpha_t + 6p_t).$$

- B_W is an integer constant determined by the sequence of accumulated shifts.
- δ_W is the offset δ of the last token T_n .

Proof. By induction on n . For $n = 1$, the form matches the row definition with $A_W = 2^{\alpha+6p}$ and $B_W = k^{(p)}$. For the inductive step, assume $x_{n-1} = 6(A_{n-1}m_0 + B_{n-1}) + \delta_{n-1}$. The next index is $m_{n-1} = \lfloor x_{n-1}/18 \rfloor$. Substituting this into the linear form for step n preserves the affine structure (see Index Evolution below for the exact coefficients). $\square$

6.2. Index Evolution and the Router. The non-linearity of the Collatz map enters solely through the floor function $m_t = \lfloor x_t/18 \rfloor$. We can resolve this floor exactly by tracking the **router remainders**.

Proposition 2 (The Index Recurrence). *Let $a_t = 2^{\alpha_t + 6p_t}$ be the slope of step t . The internal index evolves according to:*

$$(7) \quad m_t = \frac{a_t m_{t-1} + k_t^{(p)} - j_t}{3},$$

where $j_t = (a_t m_{t-1} + k_t^{(p)}) \bmod 3$ is the **router index** required for the next step to be admissible.

Proof. Recall $x_t = 6(a_t m_{t-1} + k_t) + \delta_t$. Dividing by 18:

$$\frac{x_t}{18} = \frac{6(a_t m_{t-1} + k_t) + \delta_t}{18} = \frac{a_t m_{t-1} + k_t}{3} + \frac{\delta_t}{18}.$$

Let $N = a_t m_{t-1} + k_t$. We can write $N = 3q + r$, where $r \in \{0, 1, 2\}$. Then:

$$m_t = \left\lfloor \frac{x_t}{18} \right\rfloor = \left\lfloor q + \frac{r}{3} + \frac{\delta_t}{18} \right\rfloor.$$

Since $r \leq 2$ and $\delta_t \leq 5$, the fractional part is $\frac{r}{3} + \frac{\delta_t}{18} \leq \frac{2}{3} + \frac{5}{18} = \frac{17}{18} < 1$. Thus the floor is exactly q . Solving $N = 3m_t + r$ for m_t yields $m_t = (N - r)/3$. In our notation, the remainder r is exactly the router index j_t for the subsequent step. $\square$

Remark (Router digit as a remainder). Let an odd integer in the active residue classes be written in the form

$$x = 18m + 6j + \delta, \quad j \in \{0, 1, 2\}, \quad \delta \in \{1, 5\}.$$

Equivalently,

$$x = 6(3m + j) + \delta.$$

Thus, if

$$q := \frac{x - \delta}{6},$$

then

$$q = 3m + j.$$

Since $j \in \{0, 1, 2\}$, this is precisely the Euclidean division of q by 3. Therefore the router digit j is the remainder of q modulo 3:

$$j \equiv q \pmod{3}.$$

Consequently, whenever q is written as

$$q = 3M + r, \quad r \in \{0, 1, 2\},$$

we have

$$r = j.$$

This should be distinguished from the residue of the index variable m modulo 3. If one writes

$$m = 3M + r, \quad r \in \{0, 1, 2\},$$

then this r records $m \bmod 3$, not the current router digit j . It is the residue used to determine the next router value under the row transition.

Remark (Closed form for the index update and the router remainder). In the proof of Proposition (The Index Recurrence), we set

$$N := a_t m_{t-1} + k_t, \quad \text{so that} \quad N = 3q + r, \quad r \in \{0, 1, 2\}.$$

By Euclidean division, the quotient is

$$q = \left\lfloor \frac{N}{3} \right\rfloor,$$

and we identify this quotient with the next index m_t , i.e. $m_t = q$. Writing the remainder as the router output

$$j_t := N \bmod 3 \in \{0, 1, 2\},$$

we obtain the floor-free index recurrence

$$m_t = \frac{N - j_t}{3} = \frac{a_t m_{t-1} + k_t - j_t}{3}.$$

In our parameterization,

$$a_t = 2^{\alpha_t+6p_t}, \quad k_t = \frac{\beta_t 2^{6p_t} + c_t}{9} = \frac{\beta_t 64^{p_t} + c_t}{9},$$

so equivalently

$$\begin{aligned} m_t &= \frac{9(2^{\alpha_t+6p_t}m_{t-1} - j_t) + (\beta_t 64^{p_t} + c_t)}{27} \\ &= \frac{2^{\alpha_t+6p_t}m_{t-1} - j_t}{3} + \underbrace{\frac{\beta_t 64^{p_t} + c_t}{27}}_{=:G_t(p_t)}. \end{aligned}$$

Here $G_t(p) := (\beta_t 64^p + c_t)/27$ collects the p -dependence of the affine offset, while the router remainder j_t is chosen so that $a_t m_{t-1} + k_t - j_t$ is divisible by 3.

Remark (Index update and complementary $\frac{1}{3}$ -fractions (router)). Fix a step t and let

$$N := a_t m_{t-1} + k_t^{(p_t)} \in \mathbb{Z}, \quad j_t := N \bmod 3 \in \{0, 1, 2\}.$$

Then $N = 3m_t + j_t$, hence

$$m_t = \left\lfloor \frac{N}{3} \right\rfloor = \frac{N - j_t}{3} = \frac{a_t m_{t-1} + k_t^{(p_t)} - j_t}{3}.$$

In our parameterization,

$$a_t = 2^{\alpha_t+6p_t}, \quad k_t^{(p_t)} = \frac{\beta_t 64^{p_t} + c_t}{9},$$

so one may also write (split form)

$$m_t = \frac{a_t m_{t-1} - j_t}{3} + \frac{k_t^{(p_t)}}{3}.$$

Although $k_t^{(p_t)}/3$ need not be an integer, it is *complementary* to the first term: since $j_t \equiv a_t m_{t-1} + k_t^{(p_t)} \pmod{3}$, we have

$$a_t m_{t-1} - j_t \equiv -k_t^{(p_t)} \pmod{3},$$

so the two summands always carry opposite $\{0, \frac{1}{3}, \frac{2}{3}\}$ -fractional parts, and therefore their sum m_t is an integer.

Remark (Complementary $\frac{1}{3}$ -fractions and a convenient choice of h). Let

$$N := a_t m_{t-1} + k_t^{(p_t)} \in \mathbb{Z}, \quad j_t := N \bmod 3 \in \{0, 1, 2\},$$

so that $N = 3m_t + j_t$ and therefore

$$m_t = \frac{a_t m_{t-1} + k_t^{(p_t)} - j_t}{3} \in \mathbb{Z}.$$

Define

$$h_t := \left\{ \frac{a_t m_{t-1}}{3} \right\} = \frac{(a_t m_{t-1}) \bmod 3}{3} \in \left\{ 0, \frac{1}{3}, \frac{2}{3} \right\},$$

so that $\frac{a_t m_{t-1}}{3} - h_t \in \mathbb{Z}$. Writing

$$m_t = \frac{a_t m_{t-1}}{3} + \frac{k_t^{(p_t)} - j_t}{3},$$

and using $m_t \in \mathbb{Z}$, we obtain the complementary integrality relation

$$\frac{k_t^{(p_t)} - j_t}{3} + h_t \in \mathbb{Z}.$$

Equivalently, the two $/3$ terms have opposite fractional parts:

$$\frac{a_t m_{t-1}}{3} \equiv - \frac{k_t^{(p_t)} - j_t}{3} \pmod{1}.$$

6.3. Closed Form for m_n . Unrolling the recurrence yields a summation that describes the trajectory's "history."

Theorem 3 (Index Evolution Formula). *For a word of length n , the final index m_n is related to the start m_0 by:*

$$(8) \quad m_n = \frac{A_W m_0}{3^n} + \sum_{t=1}^n \frac{P_{n,t}}{3^{n-t+1}} (k_t^{(p)} - j_t),$$

where $P_{n,t} = \prod_{i=t+1}^n a_i$ is the product of slopes from step $t+1$ to n .

Remark (Zero-Start Independence). If we start from $x_0 = 1$ (or any minimal element where $m_0 = 0$), the first term vanishes. The entire trajectory is then determined solely by the structural constants of the word (the k values) and the routing choices (j). This confirms that the path from 1 is deterministic and algebraically fixed.

7. STEERING AND MONOTONE PADDING

The core of the inverse calculus is the ability to manipulate the affine parameters of a word to satisfy specific modular congruences. We achieve this via *padding*: appending short sequences of tokens to the end of a word.

7.1. Steering Gadgets.

Definition 2 (Steering Gadget). A **Steering Gadget** is a short admissible word S that begins and ends in the same family $f \in \{e, o\}$. Appending S to a prefix W ending in f preserves the terminal family but modifies the affine parameters:

- **Slope Boost:** It multiplies the slope A_W by $2^{\Delta_{v_2}}$, strictly increasing the 2-adic valuation.

- **Intercept Control:** It modifies the intercept B_W modulo 2 (and modulo 3).

Because these gadgets preserve the terminal family, they allow us to "steer" the values of A and B without altering the routing requirements for any subsequent steps.

7.2. The Finite Steering Menu. We fix a finite set of canonical gadgets $\mathcal{S}_p$ for each column $p \geq 0$. These are sufficient to generate any required 2-adic lift and parity.

Table 6. Canonical Steering Gadgets (Base $p = 0$).

Family	Block	Type Path	$\Delta v_2(A)$	Effect on B
Family e	Ψ_1	$e \rightarrow e$	+4	Preserves Parity ($k \equiv 0$)
	Ψ_2	$e \rightarrow e$	+6	Preserves Parity ($k \equiv 0$)
	$\psi_2 \circ \omega_1$	$e \rightarrow o \rightarrow e$	+3	Toggles Parity ($k \equiv 1$)
Family o	Ω_1	$o \rightarrow o$	+3	Preserves Parity ($k \equiv 0$)
	Ω_0	$o \rightarrow o$	+5	Preserves Parity ($k \equiv 0$)
	Ω_2	$o \rightarrow o$	+1	Toggles Parity ($k \equiv 1$)

Remark (Higher Columns). For $p \geq 1$, the structure remains identical, but the valuation lift increases by $6p$ per token. The parity effects depend on $k^{(p)} \bmod 2$, which is tabulated in the full reference implementation. The menu above is sufficient for $p = 0$.

7.3. Mod-3 Steering. In addition to parity, we must often control $B \pmod{3}$ to ensure the final congruence is solvable (removing factors of 3).

Lemma 5 (Mod-3 Control). *For each family $f \in \{e, o\}$, the affine maps $B \mapsto 2^\alpha B + k \pmod{3}$ induced by the available tokens generate the full affine group $\text{AGL}_1(\mathbb{F}_3)$. Consequently, from any current state B_W , there exists a steering gadget of length ≤ 2 that sets the new intercept $B' \equiv r \pmod{3}$ for any target $r \in \{0, 1, 2\}$.*

Proof. In family o , Ω_1 maps $B \mapsto 2B + 1$ and Ω_0 maps $B \mapsto 2B + 2$. These two operations generate all permutations of $\{0, 1, 2\}$. In family e , Ψ_0 maps $B \mapsto B$ (identity) and Ψ_2 maps $B \mapsto B + 1$ (shift). Iterating Ψ_2 reaches any residue. $\square$

7.4. The Monotone Padding Lemma. We combine these results into the primary tool used for inductive lifting.

Lemma 6 (Monotone Padding). *Let W be any admissible word ending in family f . For any target valuation K and any target parity $b \in \{0, 1\}$, there exists a padding string S such that the extended word $W' = W \cdot S$ satisfies:*

- (1) **Family Preservation:** W' ends in the same family f .
- (2) **Valuation Target:** $v_2(A_{W'}) \geq K$.
- (3) **Parity Control:** $B_{W'} \equiv b \pmod{2}$.

Proof. Since every gadget in Table 6 has $\Delta v_2 > 0$, we can repeat the "Preserves Parity" gadgets to raise $v_2(A)$ arbitrarily high (Monotone Lift). If the resulting B has the wrong parity, we append exactly one "Toggle Parity" gadget (e.g., Ω_2 or $\psi_2 \circ \omega_1$). This flips the bit $B \pmod{2}$ and adds positive valuation, satisfying all conditions. $\square$

Lemma 7 (Joint (mod2, mod3) steering with monotone padding). *Fix a family $f \in \{e, o\}$ and let W be any admissible word ending in f . Let $K \geq 1$ and let $r \in \{0, 1, 2, 3, 4, 5\}$ be any target residue class.*

Then there exists an admissible extension S such that the extended word $W' = W \cdot S$ satisfies:

- (1) **Family preservation:** W' ends in the same family f .
- (2) **Valuation target:** $v_2(A_{W'}) \geq K$.
- (3) **Joint steering:** $B_{W'} \equiv r \pmod{6}$.

Moreover, S can be chosen from a bounded-length concatenation of the canonical gadgets in Table 6.

Proof. Step 1 (raise $v_2(A)$ without leaving the family). By Lemma 6, for any prescribed parity class $b \in \{0, 1\}$, there exists an admissible padding word P_b such that $W_1 := W \cdot P_b$ ends in family f , satisfies $v_2(A_{W_1}) \geq K$, and has

$$B_{W_1} \equiv b \pmod{2}.$$

In particular, after this step we have achieved the valuation target and fixed $B \pmod{2}$.

Step 2 (finite controllability on $B \pmod{6}$ inside a fixed family). Consider the finite directed graph whose vertices are the residue classes

$$(f, \bar{B}) \in \{e, o\} \times (\mathbb{Z}/6\mathbb{Z}),$$

and whose directed edges are induced by appending one canonical gadget from Table 6 (i.e. $(f, \bar{B}) \rightarrow (f, \bar{B}')$ if appending that gadget is admissible from family f and maps $B \mapsto B'$ modulo 6 according to this paper's token update rules). Because the gadget set is finite and the state space has size 12, this is a finite graph.

By the Mod-3 control lemma (Lemma 5), within each family f the available steering moves act transitively on $B \pmod{3}$. Table 6 also contains at least one parity-preserving gadget (effect $B \equiv B' \pmod{2}$) and at least one parity-toggling gadget (effect $B' \equiv B+1 \pmod{2}$) within each family. Consequently,

the induced move set is transitive on $B \bmod 6$ within each family: from any $\bar{B} \in \mathbb{Z}/6\mathbb{Z}$ one can reach any other $\bar{r} \in \mathbb{Z}/6\mathbb{Z}$ by a word of bounded length (bounded by the diameter of this finite graph).

Let Q be such a steering word (a concatenation of canonical gadgets) that sends $B_{W_1} \bmod 6$ to the desired $r \bmod 6$ while remaining in family f . Set $W' := W_1 \cdot Q$. Since all gadgets in Q are admissible and family-preserving, W' ends in family f . Since $v_2(A)$ is nondecreasing under the padding/steering gadgets used in this paper (as recorded in Table 6), we still have $v_2(A_{W'}) \geq K$. Finally, by construction, $B_{W'} \equiv r \pmod{6}$.

The bounded-length claim follows because the steering word length can be chosen ≤ 11 by finiteness (of a 12-state directed graph), and in practice is much smaller (one can tabulate the shortest paths). $\square$

8. ROUTING COMPATIBILITY

A central challenge in the inverse calculus is that the "Router" $j_t = \lfloor x_t/6 \rfloor \bmod 3$ depends on the floor of the current value. If we adjust the initial input m to satisfy a condition at step n , we risk changing a router at step $t < n$, which would invalidate the chosen row sequence (a "branch flip").

We now prove that this can be prevented by imposing a sufficient 2-adic constraint on m .

8.1. The Stability Threshold. Let $W = T_1 T_2 \dots T_n$ be a fixed prefix. Let A_t denote the cumulative slope up to step t :

$$A_t = \prod_{i=1}^t 2^{\alpha_i + 6p_i} = 2^{S_t}, \quad \text{where } S_t = \sum_{i=1}^t (\alpha_i + 6p_i).$$

The value x_t depends on m via the term $A_t m$. To stabilize the floor functions, we must control the lower bits of m .

Definition 3 (Stability Threshold). The **Stability Threshold** S^* for a word W is the maximum accumulated exponent along the path plus one:

$$(9) \quad S^* := 1 + \max_{0 \leq t < n} S_t.$$

8.2. The Compatibility Lemma.

Lemma 8 (Routing Compatibility). *Let W be a fixed admissible prefix with planned routers $j_1, \dots, j_n$. If we restrict the input index m to a specific congruence class:*

$$m \equiv m^* \pmod{2^{S^*}},$$

(where m^ is compatible with the entry family), then the router remainders r_{t+1} computed along the trajectory are invariant for all m in that class. Specifically, if m^* generates the correct routers, then every $m \equiv m^* \pmod{2^{S^*}}$ generates the same routers.*

Proof. Recall the index recurrence (Section 6):

$$m_t = \frac{A_t m + B_t - r_{t+1}}{3}.$$

The router r_{t+1} is determined by $(A_t m + B_t) \bmod 3$. Since $A_t = 2^{S_t}$ is coprime to 3, fixing $m \bmod 3$ fixes the router sequence modulo 3. However, we must also ensure the integrality of the division (the floor). If we fix $m \pmod{2^{S^*}}$, we fix the lower S^* bits of m . Since $S_t < S^*$, the term $A_t m$ is congruent to $A_t m^* \pmod{2^{S^*+S_t}}$. The division by 3 (multiplication by 3^{-1} in the 2-adic integers) preserves this 2-adic precision. Thus, the "decisions" made by the floor function (which depend on lower bits) remain constant. $\square$

8.3. Application: Freezing the Prefix. This lemma provides a modular "Locking Mechanism."

- (1) **Construct a Prefix:** Choose a word W to reach a specific family or intermediate value.
- (2) **Calculate S^* :** Sum the exponents.
- (3) **Restrict m :** Solve the linear congruences for the routers modulo 3, then lift to modulo 2^{S^*} .

Once m is restricted to this class, we can append any number of steering gadgets to the *end* of W . As long as the final choice of m respects the constraint $m \equiv m^* \pmod{2^{S^*}}$, the prefix W will execute exactly as planned, with no branch flips.

Part 3. The Construction: Lifting and Witnesses

9. RESIDUE TARGETING VIA LAST-ROW CONGRUENCE

Once a prefix is stabilized via routing compatibility, the task of hitting a specific target residue $x_{\text{tar}} \pmod{M_K}$ (where $M_K = 3 \cdot 2^K$) falls entirely on the *last token* of the word.

We analyze the affine map of a single last token T chosen from column p .

9.1. The Last-Step Congruence. Let the last token have parameters $(\alpha, \beta, c, \delta)$ and column-lift p . Its unified form is:

$$x' = 6(2^{\alpha_p} u + k^{(p)}) + \delta_T, \quad \text{where } \alpha_p = \alpha + 6p, \quad u = \left\lfloor \frac{x}{18} \right\rfloor.$$

We wish to solve $x' \equiv x_{\text{tar}} \pmod{M_K}$. Rearranging terms, this is equivalent to the linear congruence:

$$(10) \quad a^{(p)} u \equiv r^{(p)} \pmod{M_K},$$

where the coefficient is $a^{(p)} = 6 \cdot 2^{\alpha_p} = 3 \cdot 2^{\alpha_p+1}$, and the target remainder is $r^{(p)} = x_{\text{tar}} - (6k^{(p)} + \delta_T)$.

Notation. We use m for the global integer input to a word evaluation $x_W(m) = 6(A_W m + B_W) + \delta_W$. In the residue-targeting (last-step) analysis we instead parametrize the current residue class by

$$u := \left\lfloor \frac{x}{18} \right\rfloor,$$

so that fixing the router state (s, j) determines x from u via $x = 18u + 6j + \delta_s$. Thus m and u play the same role (a free integer index), but are attached to different stages of the construction. We retain $k^{(p)}$ for the lifted constant in the last-step unified form to match Table 2.

Lemma 9 (Solvability Criterion). *The congruence (10) is solvable for u if and only if*

$$g^{(p)} := \gcd(a^{(p)}, M_K) = 3 \cdot 2^{\min(\alpha_p+1, K)}$$

divides the target remainder $r^{(p)}$.

9.2. The Two Regimes: Pinning vs. Solving. Depending on the magnitude of the 2-adic lift α_p relative to the target precision K , the behavior of the last step falls into one of two distinct regimes.

Proposition 3 (Pinning vs. Solving). *(1) **The Pinning Regime** ($\alpha_p + 1 \geq K$): In this case, M_K divides $a^{(p)}$. The term $a^{(p)}u$ vanishes modulo M_K (Indeed, $2^{\alpha_p+1} \equiv 0 \pmod{2^K}$ once $\alpha_p + 1 \geq K$, cf. Lemma 14.) Consequently, the output residue is **fixed** (pinned) by the token parameters alone:*

$$x' \equiv 6k^{(p)} + \delta_T \pmod{M_K},$$

independently of the input u . This token acts as a "constant function" modulo M_K .

*(2) **The Solving Regime** ($\alpha_p + 1 < K$): In this case, the coefficient $a^{(p)}$ retains information about u . The congruence has a unique solution class for u modulo $2^{K-(\alpha_p+1)}$:*

$$u \equiv \frac{r^{(p)}}{3 \cdot 2^{\alpha_p+1}} \pmod{2^{K-(\alpha_p+1)}}.$$

This is exactly the 2-adic cancellation phenomenon (Lemma 13).

Remark (Canonical Choice). This dichotomy gives us a robust algorithm:

- If we want to hit a target r' regardless of the history, we can try to find a **Pinning** row (high p) that hits it naturally.
- If we are constrained to a specific family, we use **Steering** (Section 7) to ensure the divisibility condition holds, then **Solve** for the required u .

9.3. Examples of Targeting. We illustrate these regimes with concrete examples from the unified table.

Example 4 (Pinning at $K = 5$ ($M_5 = 96$)). Target: $x_{\text{tar}} \equiv 53 \pmod{96}$. Choose the row Ω_0 (type ∞) at $p = 0$. Parameters: $\alpha = 5$, $k^{(0)} = 8$, $\delta = 5$. Check Pinning Threshold: $\alpha_p + 1 = 5 + 1 = 6 \geq 5$. The condition holds. The pinned value is:

$$x' \equiv 6(8) + 5 = 53 \pmod{96}.$$

Thus, Ω_0 pins the target 53 exactly, regardless of the input u .

Example 5 (Solving at $K = 10$ ($M_{10} = 3072$)). Target: $x_{\text{tar}} \equiv 3071 \pmod{3072}$. Choose the row Ω_2 (type ∞) at $p = 0$. Parameters: $\alpha = 1$, $k^{(0)} = 1$, $\delta = 5$. Check Pinning: $\alpha_p + 1 = 2 < 10$. We are in the **Solving** regime. We solve for u :

$$6(2^1)u \equiv 3071 - (6(1) + 5) \pmod{3072} \implies 12u \equiv 3060 \pmod{3072}.$$

Dividing by 12:

$$u \equiv \frac{3060}{12} = 255 \pmod{256}.$$

Thus, any input $u \equiv 255 \pmod{256}$ will map to the target.

Summary tables (for $p = 0$). To streamline the control arguments, we summarize the last-step data in two complementary tables. Table 7 is a branch-by-branch *output classifier* at $p = 0$: for each row (i.e. inverse branch) it records the parameters $(\alpha, \beta, c, \delta)$, the derived constant $k^{(0)} = (\beta + c)/9$, and the induced output classifiers (s', j') and m' as functions of the residue class $r \equiv m \pmod{3}$ (equivalently $u \pmod{3}$ in the control normal form). This makes it immediate, once a branch is chosen, to read off the output family s' (via δ), the router class j' , and the next-index update. Table 8 reorganizes the same information by *input state* (s, j) : each state offers exactly two branch choices corresponding to $s' = e$ or $s' = o$, and the table lists the associated (α, q, δ) together with the router transition triple $j'(0, 1, 2)$. In practice, Table 8 is used first to select a row consistent with the desired output family (and, in the solving regime, modular solvability), while Table 7 is used to record the resulting output classification and index update.

Write $m = 3M + r$ with $r \in \{0, 1, 2\}$, and $k^{(0)} = (\beta + c)/9$, $\kappa = k^{(0)} \pmod{3}$. At $p = 0$,

$$F(0, m) = 2^\alpha m + k^{(0)}.$$

The output router is determined by

$$j' \equiv 2^\alpha r + \kappa \pmod{3},$$

and the new internal index is

$$m' = 2^\alpha M + t_r, \quad t_r = \left\lfloor \frac{2^\alpha r + k^{(0)}}{3} \right\rfloor.$$

Table 7. $p = 0$ output classification derived from the row parameters.

Row	(s, j)	Type	α	β	c	δ	s'	$k^{(0)} = (\beta + c)/9$	$\kappa = k^{(0)} \bmod 3$	$j'(r = 0, 1, 2)$	$m' = 2^\alpha M + (t_0, t_1, t_2)$
Ψ_0	$(e, 0)$	ee	2	2	-2	1	e	0	0	$(0, 1, 2)$	$4M + (0, 1, 2)$
Ψ_1	$(e, 1)$	ee	4	56	-2	1	e	6	0	$(0, 1, 2)$	$16M + (2, 7, 12)$
Ψ_2	$(e, 2)$	ee	6	416	-2	1	e	46	1	$(1, 2, 0)$	$64M + (15, 36, 58)$
ω_0	$(o, 0)$	oe	3	20	-2	1	e	2	2	$(2, 0, 1)$	$8M + (0, 3, 6)$
ω_1	$(o, 1)$	oe	1	11	-2	1	e	1	1	$(1, 2, 0)$	$2M + (0, 1, 1)$
ω_2	$(o, 2)$	oe	5	272	-2	1	e	30	0	$(0, 1, 2)$	$32M + (10, 20, 31)$
ψ_0	$(e, 0)$	eo	4	8	-8	5	o	0	0	$(0, 1, 2)$	$16M + (0, 5, 10)$
ψ_1	$(e, 1)$	eo	6	224	-8	5	o	24	0	$(0, 1, 2)$	$64M + (8, 29, 50)$
ψ_2	$(e, 2)$	eo	2	26	-8	5	o	2	2	$(2, 0, 1)$	$4M + (0, 2, 3)$
Ω_0	$(o, 0)$	oo	5	80	-8	5	o	8	2	$(2, 0, 1)$	$32M + (2, 13, 24)$
Ω_1	$(o, 1)$	oo	3	44	-8	5	o	4	1	$(1, 2, 0)$	$8M + (1, 4, 6)$
Ω_2	$(o, 2)$	oo	1	17	-8	5	o	1	1	$(1, 2, 0)$	$2M + (0, 1, 1)$

Table 8. Control transition table at $p = 0$, grouped by input state (s, j) . Each input state admits two branch choices: $s' = e$ or $s' = o$ (equivalently Type $\ast e$ or $\ast o$). For each chosen row, we list $(\alpha, k^{(0)}, \delta)$ with $k^{(0)} = (\beta + c)/9$, and the router transition triple $j'(0, 1, 2)$, where $r := u \bmod 3$ and $j'(r) \equiv 2^\alpha u + k^{(0)} \pmod{3}$.

Input (s, j)	Choice	Row	Type	$(\alpha, k^{(0)}, \delta)$	$j'(0, 1, 2)$
$(e, 0)$	$s' = e$	Ψ_0	ee	$(2, 0, 1)$	$(0, 1, 2)$
$(e, 0)$	$s' = o$	ψ_0	eo	$(4, 0, 5)$	$(0, 1, 2)$
$(e, 1)$	$s' = e$	Ψ_1	ee	$(4, 6, 1)$	$(0, 1, 2)$
$(e, 1)$	$s' = o$	ψ_1	eo	$(6, 24, 5)$	$(0, 1, 2)$
$(e, 2)$	$s' = e$	Ψ_2	ee	$(6, 46, 1)$	$(1, 2, 0)$
$(e, 2)$	$s' = o$	ψ_2	eo	$(2, 2, 5)$	$(2, 0, 1)$
$(o, 0)$	$s' = e$	ω_0	oe	$(3, 2, 1)$	$(2, 1, 0)$
$(o, 0)$	$s' = o$	Ω_0	oo	$(5, 8, 5)$	$(2, 1, 0)$
$(o, 1)$	$s' = e$	ω_1	oe	$(1, 1, 1)$	$(1, 0, 2)$
$(o, 1)$	$s' = o$	Ω_1	oo	$(3, 4, 5)$	$(1, 0, 2)$
$(o, 2)$	$s' = e$	ω_2	oe	$(5, 30, 1)$	$(0, 2, 1)$
$(o, 2)$	$s' = o$	Ω_2	oo	$(1, 1, 5)$	$(1, 0, 2)$

Summary tables for $p = 0$. At $p = 0$, each row has the affine form

$$F(0, m) = 2^\alpha m + k^{(0)}, \quad k^{(0)} = \frac{\beta + c}{9}.$$

To classify the output state, write

$$m = 3M + r, \quad r \in \{0, 1, 2\}.$$

Then

$$F(0, m) = 2^\alpha(3M + r) + k^{(0)} = 3(2^\alpha M) + 2^\alpha r + k^{(0)}.$$

Thus

$$j' \equiv 2^\alpha r + k^{(0)} \pmod{3},$$

and

$$m' = 2^\alpha M + \left\lfloor \frac{2^\alpha r + k^{(0)}}{3} \right\rfloor.$$

Equivalently, with

$$\kappa = k^{(0)} \pmod{3},$$

we have

$$j' \equiv 2^\alpha r + \kappa \pmod{3}.$$

The output-classification table is split into three tables below. Table 9 gives the row constants, Table 10 gives the output router j' , and Table 11 gives the new internal index m' .

Table 9. Row parameters at $p = 0$. The constant $k^{(0)} = (\beta + c)/9$ is the additive term in $F(0, m) = 2^\alpha m + k^{(0)}$, and $\kappa = k^{(0)} \pmod{3}$.

Row	(s, j)	Type	α	β	c	δ	s'	$k^{(0)}$	κ
Ψ_0	$(e, 0)$	ee	2	2	-2	1	e	0	0
Ψ_1	$(e, 1)$	ee	4	56	-2	1	e	6	0
Ψ_2	$(e, 2)$	ee	6	416	-2	1	e	46	1
ω_0	$(o, 0)$	oe	3	20	-2	1	e	2	2
ω_1	$(o, 1)$	oe	1	11	-2	1	e	1	1
ω_2	$(o, 2)$	oe	5	272	-2	1	e	30	0
ψ_0	$(e, 0)$	eo	4	8	-8	5	o	0	0
ψ_1	$(e, 1)$	eo	6	224	-8	5	o	24	0
ψ_2	$(e, 2)$	eo	2	26	-8	5	o	2	2
Ω_0	$(o, 0)$	oo	5	80	-8	5	o	8	2
Ω_1	$(o, 1)$	oo	3	44	-8	5	o	4	1
Ω_2	$(o, 2)$	oo	1	17	-8	5	o	1	1

Table 10. Router update at $p = 0$. For each row and each residue $r = m \bmod 3$, the table gives the output router j' , computed from $j' \equiv 2^\alpha r + \kappa \pmod{3}$.

Row	j' when $r = 0$	j' when $r = 1$	j' when $r = 2$
Ψ_0	0	1	2
Ψ_1	0	1	2
Ψ_2	1	2	0
ω_0	2	1	0
ω_1	1	0	2
ω_2	0	2	1
ψ_0	0	1	2
ψ_1	0	1	2
ψ_2	2	0	1
Ω_0	2	1	0
Ω_1	1	0	2
Ω_2	1	0	2

Table 11. Internal-index update at $p = 0$. Writing $m = 3M + r$, the table lists the new internal index m' for each residue $r \in \{0, 1, 2\}$.

Row	m' when $r = 0$	m' when $r = 1$	m' when $r = 2$
Ψ_0	$4M$	$4M + 1$	$4M + 2$
Ψ_1	$16M + 2$	$16M + 7$	$16M + 12$
Ψ_2	$64M + 15$	$64M + 36$	$64M + 58$
ω_0	$8M$	$8M + 3$	$8M + 6$
ω_1	$2M$	$2M + 1$	$2M + 1$
ω_2	$32M + 10$	$32M + 20$	$32M + 31$
ψ_0	$16M$	$16M + 5$	$16M + 10$
ψ_1	$64M + 8$	$64M + 29$	$64M + 50$
ψ_2	$4M$	$4M + 2$	$4M + 3$
Ω_0	$32M + 2$	$32M + 13$	$32M + 24$
Ω_1	$8M + 1$	$8M + 4$	$8M + 6$
Ω_2	$2M$	$2M + 1$	$2M + 1$

Together, Tables 9–11 describe the full transition

$$(s, j, m) \mapsto (s', j', m')$$

at $p = 0$. First choose the row from Table 9. Then use $r = m \bmod 3$ to read off j' from Table 10 and m' from Table 11.

Example 6 (Solving at $K = 5$ ($M_5 = 96$) with row-selection via the control-transition table). Target: $x_{\text{tar}} \equiv 91 \pmod{96}$.

Step 1 (Infer the required output family s'). Since $96 \equiv 0 \pmod{6}$, reducing the target modulo 6 is immediate:

$$91 \equiv 1 \pmod{6}.$$

Hence the target lies in the e -family, so we must have $s' = e$, equivalently $\delta = 1$ in the final-step normal form

$$x' = 6(2^\alpha u + q) + \delta.$$

Step 2 (Use the control-transition table to select a candidate row).

In the $p = 0$ control-transition table, the constraint $s' = e$ means we must choose a row of type $*e$. Testing the six $*e$ options, the only row that admits solutions for this target is

$$\omega_1 \quad (\text{type } oe),$$

which occurs under input state $(s, j) = (o, 1)$ and has parameters

$$(\alpha, q, \delta) = (1, 1, 1).$$

How to test the six $*e$ options in Step 2:

Since the target value is 91, we first determine its terminal index by

$$m' = \left\lfloor \frac{91}{18} \right\rfloor = 5.$$

Since

$$6j' + \delta' = x' - 18m'$$

Then the remaining residue is

$$91 - 18m' = 91 - 18 \cdot 5 = 1.$$

Thus

$$6j' + \delta' = 1.$$

Since

$$j' \in \{0, 1, 2\}, \quad \delta' \in \{1, 5\},$$

the only possibility is

$$j' = 0, \quad \delta' = 1.$$

Therefore

$$91 = 18 \cdot 5 + 6 \cdot 0 + 1,$$

so

$$m' = 5, \quad j' = 0, \quad \delta' = 1.$$

we have

$$m' = 5, \quad j' = 0, \quad \delta' = 1.$$

Thus, in the classification table, a candidate row must have an m' -entry which can equal 5, and it must occur in the case where the corresponding router output is $j' = 0$.

For each of the six $*e$ -options, write the relevant table entry in the form

$$m' = 2^\alpha M + t_r, \quad r \in \{0, 1, 2\}.$$

Then test

$$2^\alpha M + t_r = 5.$$

Equivalently,

$$M = \frac{5 - t_r}{2^\alpha}.$$

The row passes precisely when this gives a nonnegative integer M , and the same case has router output $j' = 0$.

In the table, the successful case is the ω_1 -row. For that row, the relevant case gives

$$j' = 0$$

and

$$m' = 2M + 1.$$

Setting this equal to the target value $m' = 5$, we get

$$2M + 1 = 5.$$

Hence

$$2M = 4,$$

so

$$M = 2.$$

Therefore this case is admissible.

Since

$$m = 3M + r,$$

with

$$M = 2,$$

and with the residue r specified by the ω_1 -row case in the classification table, we obtain the corresponding input index m . Thus the successful tested option is

$$\boxed{\omega_1}.$$

Step 3 (Pinning vs. Solving). Here $K = 5$ and $\alpha + 1 = 2 < 5$, so we are in the **Solving** regime.

Step 4 (Solve the congruence for u). Impose $x' \equiv 91 \pmod{96}$ in the row formula $x' = 6(2^\alpha u + k^{(0)}) + \delta$:

$$6(2^1 u + 1) + 1 \equiv 91 \pmod{96}.$$

Rearranging gives

$$6(2u + 1) \equiv 90 \pmod{96}.$$

Divide by 6 (valid since $6 \mid 96$):

$$2u + 1 \equiv 15 \pmod{16}$$

$$\implies 2u \equiv 14 \pmod{16}$$

$$\implies u \equiv 7 \pmod{8}.$$

Thus, any input with $u \equiv 7 \pmod{8}$ maps to the target residue class.

Step 5 (Extrapolate a compatible starting input x). The input decomposition is

$$x = 18u + 6j + \delta_s, \quad \delta_s = \begin{cases} 1, & s = e, \\ 5, & s = o. \end{cases}$$

From the chosen row we have the required input state $(s, j) = (o, 1)$, hence $\delta_s = 5$ and $j = 1$. Therefore

$$x = 18u + 6 \cdot 1 + 5 = 18u + 11.$$

Taking the smallest solution $u = 7$ yields the explicit starting value

$$x = 18 \cdot 7 + 11 = 137.$$

More generally, writing $u = 7 + 8t$ gives the full congruence class

$$x = 18(7 + 8t) + 11 = 137 + 144t, \quad \text{i.e.} \quad x \equiv 137 \pmod{144}.$$

Step 6 (Verification). For row ω_1 at $p = 0$ we have

$$x' = 6(2u + 1) + 1 = 12u + 7.$$

With $u = 7$ this gives $x' = 12 \cdot 7 + 7 = 91$, and if $u \equiv 7 \pmod{8}$ then

$$x' = 12u + 7 \equiv 12 \cdot 7 + 7 = 91 \pmod{96},$$

since $12 \cdot 8 = 96$.

Example 7 (Solving at $K = 10$ ($M_{10} = 3072$) via the one-shot reduction). Target: $x_{\text{tar}} \equiv 3071 \pmod{3072}$. Choose the row Ω_2 (type ∞) at $p = 0$, with parameters

$$(\alpha, q, \delta) = (1, 1, 5), \quad k^{(0)} = \frac{\beta + c}{9}.$$

Step 1 (Consistency and one-shot reduction). Since $3071 \equiv 5 \pmod{6}$, the target is compatible with $\delta = 5$. Define the reduced target

$$T := \frac{x_{\text{tar}} - \delta}{6} = \frac{3071 - 5}{6} = 511, \quad v := T - k^{(0)} = 511 - 1 = 510.$$

The row congruence

$$x' \equiv 6(2^\alpha u + k^{(0)}) + \delta \pmod{M_{10}}$$

is equivalent to the single 2-adic congruence

$$2^\alpha u \equiv v \pmod{2^{10}}, \quad \text{i.e.} \quad 2u \equiv 510 \pmod{1024}.$$

Step 2 (Solving regime and maximal division). Here $\alpha < 10$, so we are in the **Solving** regime. Since 510 is divisible by $2^\alpha = 2$, we may divide by 2:

$$u \equiv 255 \pmod{512}.$$

To present the *coarser* (but often more convenient) family obtained by dividing the original congruence by the full factor $6 \cdot 2^\alpha = 12$, note that the original condition is

$$6 \cdot 2^\alpha u \equiv x_{\text{tar}} - (6q + \delta) \pmod{3 \cdot 2^{10}} \iff 12u \equiv 3060 \pmod{3072},$$

and since $\gcd(12, 3072) = 12$ we can divide by 12 to obtain

$$u \equiv 255 \pmod{256}.$$

Thus, any input $u \equiv 255 \pmod{256}$ maps to x_{tar} modulo M_{10} under this row.

(Optional) Reconstruct a compatible starting input x . For the row Ω_2 the input state is $(s, j) = (o, 2)$, hence

$$x = 18u + 6j + \delta_s = 18u + 12 + 5 = 18u + 17.$$

Taking the smallest representative $u = 255$ gives $x = 18 \cdot 255 + 17 = 4607$, and in general

$$u \equiv 255 \pmod{256} \implies x \equiv 4607 \pmod{4608}.$$

Example 8 (Pinning at $K = 10$ ($M_{10} = 3072$) by lifting). Fix $K = 10$, so $M_{10} = 3 \cdot 2^{10} = 3072$. Choose the row Ω_2 (type oo) and lift to $p = 2$. For this row one has $(\alpha, \beta, c, \delta) = (1, 17, -8, 5)$, hence

$$\alpha_p = \alpha + 6p = 1 + 12 = 13 \geq 10, \quad k^{(p)} = \frac{\beta 64^p + c}{9} = \frac{17 \cdot 64^2 - 8}{9} = 7736.$$

Therefore we are in the **Pinning** regime: since $\alpha_p \geq K$, the term $2^{\alpha_p u}$ vanishes modulo 2^K and the output is independent of the input index u . Indeed, the lifted row map is

$$x' \equiv 6(2^{\alpha_p u} + k^{(p)}) + \delta \pmod{M_{10}},$$

and reducing modulo M_{10} gives

$$x' \equiv 6k^{(p)} + \delta \pmod{3072}.$$

Substituting $k^{(2)} = 7736$ and $\delta = 5$ yields

$$x' \equiv 6 \cdot 7736 + 5 = 46421 \equiv 341 \pmod{3072}.$$

Thus, *every* input in the state $(s, j) = (o, 2)$ (i.e. every $x = 18u + 6 \cdot 2 + 5$ with arbitrary u) is mapped, under the $(\Omega_2, p = 2)$ branch, to the *same* residue class

$$x' \equiv 341 \pmod{3072}.$$

In particular, at $K = 10$ the branch Ω_2 becomes pinned as soon as $p \geq 2$, since $\alpha + 6p \geq K$.

9.4. Structural consequences of the $p = 0$ tables. The $p = 0$ tables above are used in two ways. First, they give a controlled transition system: from each input state (s, j) , there are exactly two available exits, one landing in the e -family and one landing in the o -family. Choosing an exit therefore fixes both the output family and the local scale of the inverse step through the exponent α .

Second, for each fixed row the router update is a permutation of the three residue classes $r = m \pmod{3}$. Indeed, writing

$$F(0, m) = 2^\alpha m + k^{(0)}, \quad m = 3M + r,$$

we have

$$j' \equiv 2^\alpha r + k^{(0)} \pmod{3}.$$

Since 2^α is invertible modulo 3, each chosen row gives a bijection

$$r \mapsto j'$$

of $\{0, 1, 2\}$. Thus, once a row and a desired output router are specified, there is a unique residue class $r = m \bmod 3$ that realizes that output router.

This is the mechanism behind residue targeting. A row may either *pin* a target residue automatically when its fixed part already contains enough 2-adic depth, or it may require *solving* for the input residue class u . The examples above illustrate both regimes: Ω_0 pins the target $53 \pmod{96}$, while Ω_2 at modulus 3072 requires solving

$$12u \equiv 3060 \pmod{3072}, \quad u \equiv 255 \pmod{256}.$$

The remaining formulas in this section give the same targeting mechanism symbolically, first for $p = 0$ and then for general $p \geq 0$. The large diagnostic tables formerly separated into control cost, router compression, pinning strength, and higher- p comparison are all summarized by this single principle: row choice fixes the output family, the router update is invertible modulo 3, and the lift parameter p increases the available pinning depth.

9.5. General symbolic form for j' and m' . The table entries for j' and m' can be replaced by a single general quotient–remainder calculation. This is useful because it gives one formula valid for all rows and all lifts $p \geq 0$.

For a row with parameters α, β, c, δ , define

$$\alpha_p = \alpha + 6p,$$

and

$$k^{(p)} = \frac{\beta 64^p + c}{9}.$$

Then the row has the affine form

$$F(p, m) = 2^{\alpha_p} m + k^{(p)}.$$

Now write

$$m = 3M + r, \quad r \in \{0, 1, 2\}.$$

Substituting this into the affine formula gives

$$F(p, m) = 2^{\alpha_p} (3M + r) + k^{(p)}.$$

Hence

$$F(p, m) = 3(2^{\alpha_p} M) + (2^{\alpha_p} r + k^{(p)}).$$

The second term is now divided by 3. Define $j' \in \{0, 1, 2\}$ and $t_r^{(p)}$ by

$$2^{\alpha_p} r + k^{(p)} = 3t_r^{(p)} + j'.$$

Equivalently,

$$j' = (2^{\alpha_p} r + k^{(p)}) \bmod 3,$$

and

$$t_r^{(p)} = \left\lfloor \frac{2^{\alpha_p} r + k^{(p)}}{3} \right\rfloor.$$

Therefore

$$F(p, m) = 3(2^{\alpha_p} M + t_r^{(p)}) + j'.$$

So the general output formulas are

$$j' = (2^{\alpha+6p} r + k^{(p)}) \bmod 3$$

and

$$m' = 2^{\alpha+6p} M + \left\lfloor \frac{2^{\alpha+6p} r + k^{(p)}}{3} \right\rfloor.$$

Thus the whole j' -table and m' -table are encoded by the single division identity

$$2^{\alpha+6p} r + k^{(p)} = 3t_r^{(p)} + j'.$$

The router formula and the period-three shift. The formula for j' can be simplified further. Since

$$2^{6p} \equiv 1 \pmod{3},$$

we have

$$2^{\alpha+6p} \equiv 2^\alpha \pmod{3}.$$

It remains to understand $k^{(p)}$ modulo 3. We have

$$k^{(p)} - k^{(0)} = \frac{\beta 64^p + c}{9} - \frac{\beta + c}{9} = \frac{\beta(64^p - 1)}{9}.$$

Since

$$64 = 1 + 63,$$

we get, modulo 27,

$$64^p = (1 + 63)^p \equiv 1 + 63p \pmod{27}.$$

Because $63 \equiv 9 \pmod{27}$, this gives

$$64^p \equiv 1 + 9p \pmod{27}.$$

Hence

$$\frac{64^p - 1}{9} \equiv p \pmod{3}.$$

Therefore

$$k^{(p)} - k^{(0)} \equiv \beta p \pmod{3}.$$

So in general,

$$k^{(p)} \equiv k^{(0)} + \beta p \pmod{3}.$$

For the rows in the $p = 0$ table, one has

$$\beta \equiv 2 \pmod{3}.$$

Thus, for these rows,

$$k^{(p)} \equiv k^{(0)} + 2p \pmod{3}.$$

Writing

$$\kappa = k^{(0)} \pmod{3},$$

we obtain the useful router formula

$$j' \equiv 2^\alpha r + \kappa + 2p \pmod{3}.$$

Equivalently, if

$$j'_0(r) \equiv 2^\alpha r + \kappa \pmod{3}$$

is the router at $p = 0$, then

$$j'_p(r) \equiv j'_0(r) + 2p \pmod{3}.$$

Thus increasing p does not change the basic form of the router rule; it rotates the output router by $2p$ modulo 3. The router pattern therefore cycles with period 3 in p .

Using p to target a desired router. The formula

$$j' \equiv 2^\alpha r + \kappa + 2p \pmod{3}$$

can be solved for p . Suppose the desired output router is j'_{tar} . Then p must satisfy

$$2p \equiv j'_{\text{tar}} - 2^\alpha r - \kappa \pmod{3}.$$

Since $2^{-1} \equiv 2 \pmod{3}$, this is equivalent to

$$p \equiv 2(j'_{\text{tar}} - 2^\alpha r - \kappa) \pmod{3}.$$

Thus the lift parameter p can be chosen modulo 3 to force a desired output router.

Example 1: shifting the router for ω_1 . Consider the row ω_1 . From the table,

$$\alpha = 1, \quad \beta = 11, \quad c = -2, \quad k^{(0)} = 1, \quad \kappa = 1.$$

At $p = 0$, the router rule is

$$j'_0(r) \equiv 2r + 1 \pmod{3}.$$

Thus

$$(j'_0(0), j'_0(1), j'_0(2)) = (1, 0, 2).$$

At general p , the router rule becomes

$$j'_p(r) \equiv 2r + 1 + 2p \pmod{3}.$$

Therefore:

$$p = 0 : \quad (1, 0, 2),$$

$$p = 1 : \quad (0, 2, 1),$$

and

$$p = 2 : \quad (2, 1, 0).$$

At $p = 3$, the pattern returns to

$$(1, 0, 2).$$

So the router pattern for ω_1 cycles with period 3.

Let us also compute a concrete lifted output. Take

$$m = 1 = 3(0) + 1,$$

so

$$M = 0, \quad r = 1.$$

For $p = 1$,

$$\alpha_p = \alpha + 6p = 1 + 6 = 7,$$

and

$$k^{(1)} = \frac{11 \cdot 64 - 2}{9} = 78.$$

Thus

$$F(1, 1) = 2^7(1) + 78 = 206.$$

Now decompose

$$206 = 3(68) + 2.$$

Therefore

$$m' = 68, \quad j' = 2.$$

This agrees with the router formula:

$$j'_1(1) \equiv 2(1) + 1 + 2(1) = 5 \equiv 2 \pmod{3}.$$

Example 2: choosing p to hit a target router. Consider the row Ψ_2 . From the table,

$$\alpha = 6, \quad k^{(0)} = 46, \quad \kappa = 46 \bmod 3 = 1.$$

Since α is even,

$$2^\alpha \equiv 1 \pmod{3}.$$

Hence the general router rule is

$$j'_p(r) \equiv r + 1 + 2p \pmod{3}.$$

Suppose we are in the residue class

$$r = 2$$

and we want the output router to be

$$j'_{\text{tar}} = 1.$$

Then we solve

$$1 \equiv 2 + 1 + 2p \pmod{3}.$$

This gives

$$1 \equiv 3 + 2p \pmod{3},$$

or

$$2p \equiv 1 \pmod{3}.$$

Multiplying by 2, since $2^{-1} \equiv 2 \pmod{3}$, gives

$$p \equiv 2 \pmod{3}.$$

Thus any lift satisfying

$$p \equiv 2 \pmod{3}$$

will send the residue class $r = 2$ to the target router $j' = 1$ for the row Ψ_2 .

For example, at $p = 2$,

$$j'_2(2) \equiv 2 + 1 + 4 = 7 \equiv 1 \pmod{3},$$

as required.

Interpretation. The general formula is useful in two different ways. For the router j' , it gives a genuine simplification: the effect of p is only a period-three shift,

$$j'_p(r) \equiv j'_0(r) + 2p \pmod{3}.$$

For the index m' , the formula does not remove the large powers of 2, but it organizes the calculation:

$$m' = 2^{\alpha+6p}M + \left\lfloor \frac{2^{\alpha+6p}r + k(p)}{3} \right\rfloor.$$

Thus p has two roles. Modulo 3, it rotates the router. In the integer index m' , it lifts the affine scale by replacing 2^α with $2^{\alpha+6p}$.

9.6. One-shot inversion at $p = 0$: solving for a compatible input x from a prescribed output x' . Fix $K \geq 1$ and write

$$M_K := 3 \cdot 2^K.$$

Assume we are given a target output residue class

$$x' \equiv x_{\text{tar}} \pmod{M_K},$$

and we choose a specific row (branch) with parameters (s, j) , type $s \rightarrow s'$, and

$$(\alpha, \beta, c, \delta) \quad (\text{at } p = 0),$$

so that s' is determined by $\delta \in \{1, 5\}$ and

$$q := k^{(0)} := \frac{\beta + c}{9} \in \mathbb{Z}.$$

In the $p = 0$ normal form, this row maps an input index u to an output x' by

$$(11) \quad x' \equiv 6(2^\alpha u + q) + \delta \pmod{M_K}.$$

We now describe a direct procedure to compute a compatible input x (if one exists).

Step 1 (Consistency with the output family). First check that the target is compatible with the chosen row's output family:

$$x_{\text{tar}} \equiv \delta \pmod{6}.$$

If not, then (11) has no solutions for this row.

Step 2 (Reduce to a congruence for u). Define the reduced target

$$T := \frac{x_{\text{tar}} - \delta}{6} \in \mathbb{Z},$$

and set

$$v := T - q.$$

Reducing (11) and dividing by 6 yields the equivalent congruence

$$(12) \quad 2^\alpha u \equiv v \pmod{2^K}.$$

Step 3 (Pinning vs. solving).

(*Pinning*) If $\alpha \geq K$, then $2^\alpha u \equiv 0 \pmod{2^K}$ for every u , and (12) is solvable iff

$$v \equiv 0 \pmod{2^K}.$$

In this case, (12) places no restriction on u modulo 2^K (any u works).

(*Solving*) If $\alpha < K$, then (12) is solvable iff

$$v \equiv 0 \pmod{2^\alpha}.$$

When solvable, divide (12) by 2^α to obtain the explicit solution class

$$(13) \quad u \equiv \frac{v}{2^\alpha} \pmod{2^{K-\alpha}}.$$

Step 4 (Reconstruct the input x). Once a solution class for u is obtained (either arbitrary u in the pinning case, or the class (13) in the solving case), reconstruct a compatible input integer x in the chosen input state (s, j) via

$$(14) \quad x = 18u + 6j + \delta_s, \quad \delta_s := \begin{cases} 1, & s = e, \\ 5, & s = o. \end{cases}$$

In particular, if (13) yields $u \equiv u_0 \pmod{2^{K-\alpha}}$, then (14) gives the congruence class

$$x \equiv 18u_0 + 6j + \delta_s \pmod{18 \cdot 2^{K-\alpha}}.$$

Step 5 (Optional: router class of the target). Since δ fixes s' , the output router class is read off directly from the target by

$$j' \equiv \left\lfloor \frac{x'}{6} \right\rfloor \equiv \frac{x_{\text{tar}} - \delta}{6} = T \pmod{3},$$

so j' can be computed without solving for u .

Table 12. General $p \geq 0$ control-transition table (lifted parameters). Each input state (s, j) admits two choices corresponding to $s' = e$ or $s' = o$ (Type $*e$ or $*o$). For each branch (row) we list the lifted exponent $\alpha_p = \alpha + 6p$ and the lifted constant $k^{(p)} = (\beta 64^p + c)/9$ so that $x' \equiv 6(2^{\alpha_p} u + k^{(p)}) + \delta \pmod{M_K}$ with $M_K = 3 \cdot 2^K$. The router transition triple $j'(0, 1, 2)$ (where $r := u \bmod 3$) is independent of p since $64^p \equiv 1 \pmod{3}$ and $6p$ is even.

Input (s, j)	Choice	Row	Type	δ	$(\alpha_p, k^{(p)})$	$j'(0, 1, 2)$
$(e, 0)$	$s' = e$	Ψ_0	ee	1	$(2 + 6p, (2 \cdot 64^p - 2)/9)$	$(0, 1, 2)$
$(e, 0)$	$s' = o$	ψ_0	eo	5	$(4 + 6p, (8 \cdot 64^p - 8)/9)$	$(0, 1, 2)$
$(e, 1)$	$s' = e$	Ψ_1	ee	1	$(4 + 6p, (56 \cdot 64^p - 2)/9)$	$(0, 1, 2)$
$(e, 1)$	$s' = o$	ψ_1	eo	5	$(6 + 6p, (224 \cdot 64^p - 8)/9)$	$(0, 1, 2)$
$(e, 2)$	$s' = e$	Ψ_2	ee	1	$(6 + 6p, (416 \cdot 64^p - 2)/9)$	$(1, 2, 0)$
$(e, 2)$	$s' = o$	ψ_2	eo	5	$(2 + 6p, (26 \cdot 64^p - 8)/9)$	$(2, 0, 1)$
$(o, 0)$	$s' = e$	ω_0	oe	1	$(3 + 6p, (20 \cdot 64^p - 2)/9)$	$(2, 1, 0)$
$(o, 0)$	$s' = o$	Ω_0	oo	5	$(5 + 6p, (80 \cdot 64^p - 8)/9)$	$(2, 1, 0)$
$(o, 1)$	$s' = e$	ω_1	oe	1	$(1 + 6p, (11 \cdot 64^p - 2)/9)$	$(1, 0, 2)$
$(o, 1)$	$s' = o$	Ω_1	oo	5	$(3 + 6p, (44 \cdot 64^p - 8)/9)$	$(1, 0, 2)$
$(o, 2)$	$s' = e$	ω_2	oe	1	$(5 + 6p, (272 \cdot 64^p - 2)/9)$	$(0, 2, 1)$
$(o, 2)$	$s' = o$	Ω_2	oo	5	$(1 + 6p, (17 \cdot 64^p - 8)/9)$	$(1, 0, 2)$

9.7. One-shot inversion for general $p \geq 0$: solving for a compatible input x from a prescribed output x' . Fix $K \geq 1$ and set

$$M_K := 3 \cdot 2^K.$$

Assume we are given a target output residue class

$$x' \equiv x_{\text{tar}} \pmod{M_K},$$

and we choose a specific row (branch) with input state (s, j) , type $s \rightarrow s'$, and parameters

$$(\alpha, \beta, c, \delta), \quad \delta \in \{1, 5\}.$$

For each $p \geq 0$, recall

$$F(p, u) := \frac{(9u2^\alpha + \beta) 64^p + c}{9}, \quad x' = 6F(p, u) + \delta,$$

and define

$$\alpha_p := \alpha + 6p, \quad k^{(p)} := \frac{\beta 64^p + c}{9}.$$

(For the rows in the parameter table, $k^{(p)} \in \mathbb{Z}$ for all $p \geq 0$.) Using $64^p = 2^{6p}$, one has the affine form

$$(15) \quad F(p, u) = u 2^{\alpha_p} + k^{(p)}, \quad x' \equiv 6(u 2^{\alpha_p} + k^{(p)}) + \delta \pmod{M_K}.$$

We now describe a direct procedure to compute a compatible input x (if one exists), at a fixed choice of (row, p, K) .

Step 1 (Consistency with the output family). First check that the target is compatible with the chosen row's output family:

$$x_{\text{tar}} \equiv \delta \pmod{6}.$$

If not, then (15) has no solutions for this row.

Step 2 (Reduce to a congruence for u). Define the reduced target

$$T := \frac{x_{\text{tar}} - \delta}{6} \in \mathbb{Z},$$

and set

$$v := T - k^{(p)}.$$

Dividing (15) by 6 and reducing modulo 2^K yields the equivalent congruence

$$(16) \quad 2^{\alpha_p} u \equiv v \pmod{2^K}.$$

Step 3 (Pinning vs. solving).

(Pinning) If $\alpha_p \geq K$, then $2^{\alpha_p} u \equiv 0 \pmod{2^K}$ for every u , and (16) is solvable iff

$$v \equiv 0 \pmod{2^K}.$$

In this case, (16) places no restriction on u modulo 2^K (any u works).

(Solving) If $\alpha_p < K$, then (16) is solvable iff

$$v \equiv 0 \pmod{2^{\alpha_p}}.$$

When solvable, divide (16) by 2^{α_p} to obtain the explicit solution class

$$(17) \quad u \equiv \frac{v}{2^{\alpha_p}} \pmod{2^{K-\alpha_p}}.$$

Step 4 (Reconstruct the input x). Once a solution class for u is obtained (either arbitrary u in the pinning case, or the class (17) in the solving case), reconstruct a compatible input integer x in the chosen input state (s, j) via

$$(18) \quad x = 18u + 6j + \delta_s, \quad \delta_s := \begin{cases} 1, & s = e, \\ 5, & s = o. \end{cases}$$

In particular, if (17) yields $u \equiv u_0 \pmod{2^{K-\alpha_p}}$, then (18) gives the congruence class

$$x \equiv 18u_0 + 6j + \delta_s \pmod{18 \cdot 2^{K-\alpha_p}}.$$

Step 5 (Optional: router class of the target). Since δ fixes s' , the output router class is read off directly from the target by

$$j' \equiv \left\lfloor \frac{x'}{6} \right\rfloor \equiv \frac{x_{\text{tar}} - \delta}{6} = T \pmod{3},$$

so j' can be computed without solving for u .

10. BASE WITNESSES (MOD 24)

To initialize the inductive lifting procedure, we must establish that every odd residue class modulo $M_3 = 24$ is reachable.

Theorem 4 (Uniform Base Coverage). *For each odd residue $r \in \{1, 5, 7, 11, 13, 17, 19, 23\}$ modulo 24, there exists a certified inverse word W_r and an admissible choice of the internal index m such that*

$$x_{W_r}(m) \equiv r \pmod{24}.$$

Moreover, W_r can be chosen to end in the correct family determined by $r \bmod 6$: family e if $r \equiv 1 \pmod{6}$, and family o if $r \equiv 5 \pmod{6}$.

10.1. Witnesses from $x_0 = 1$. We demonstrate existence by providing explicit words that generate these residues starting from the seed 1 (where $m_0 = 0$). Table 13 lists a specific word W_r for each target r .

Table 13. Base witnesses mod 24 from $x_0 = 1$. Each step obeys routing and type navigation.

Target r	Family	Word W_r	Step Trace from 1
1	e	Ψ	1
5	o	ψ	$1 \xrightarrow{\psi} 5$
13	e	$\psi \omega$	$1 \xrightarrow{\psi} 5 \xrightarrow{\omega} 13$
17	o	$\Psi \psi \omega \psi$	$1 \xrightarrow{\Psi} 1 \xrightarrow{\psi} 5 \xrightarrow{\omega} 13 \xrightarrow{\psi} 17$
11	o	$\psi \omega \psi \Omega$	$1 \xrightarrow{\psi} 5 \xrightarrow{\omega} 13 \xrightarrow{\psi} 17 \xrightarrow{\Omega} 11$
7	e	$\psi \omega \psi \Omega \omega$	$1 \rightarrow 5 \rightarrow 13 \rightarrow 17 \rightarrow 11 \rightarrow 7$
19	e	$\psi \omega \psi \Omega \Omega \omega$	$1 \rightarrow 5 \rightarrow 13 \rightarrow 17 \rightarrow 11 \rightarrow 29 \rightarrow 19$
23	o	$\psi \Omega \Omega \Omega$	$1 \xrightarrow{\psi} 5 \xrightarrow{\Omega} 53 \xrightarrow{\Omega} 35 \xrightarrow{\Omega} 23$

Proposition 4 (Verification). *For each row in Table 13, the step trace is router-admissible at every token, terminates in the stated family, and yields a final value $x \equiv r \pmod{24}$. Furthermore, the forward check $U(x_{t+1}) = x_t$ holds for every step.*

Lemma 10 (Prefix routing compatibility under joint congruence control). *Fix a certified program (word) prefix $W = t_1 t_2 \cdots t_L$ of length L . Let the router at step ℓ be*

$$j_\ell(m) := (a_\ell m + k_\ell) \bmod 3,$$

where $a_\ell, k_\ell \in \mathbb{Z}$ are the (prefix-determined) coefficients appearing in the index recurrence of this paper, so that j_ℓ depends on m only through $(a_\ell m + k_\ell) \bmod 3$.

Then there exists an integer $S^* = S^*(W) \geq 0$ such that for any two indices m, m' satisfying

$$m \equiv m' \pmod{3 \cdot 2^{S^*}},$$

we have

$$j_\ell(m) = j_\ell(m') \quad \text{for all } 1 \leq \ell \leq L.$$

In particular, the row selections determined by the router sequence are identical on the first L steps for m and m' .

Proof. Fix $\ell \leq L$. Since the router depends on $(a_\ell m + k_\ell) \bmod 3$, it suffices to show $a_\ell(m - m') \equiv 0 \pmod{3}$. By hypothesis, $m - m'$ is divisible by 3, hence $a_\ell(m - m') \equiv 0 \pmod{3}$ for any integer a_ℓ . Therefore $(a_\ell m + k_\ell) \equiv (a_\ell m' + k_\ell) \pmod{3}$ and $j_\ell(m) = j_\ell(m')$. The 2^{S^*} factor is included because later steps in the construction impose 2-adic integrality/floor constraints; choosing S^* large enough to satisfy those constraints for the prefix W yields the stated joint modulus. $\square$

11. INDUCTIVE LIFTING ($M_K \rightarrow M_{K+1}$)

Having established the base case at $K = 3$ (Section 10), we now prove the inductive step: given full reachability of odd residues modulo $M_K = 3 \cdot 2^K$, we can extend reachability to $M_{K+1} = 3 \cdot 2^{K+1}$.

11.1. The Lifting Lemma. The lifting mechanism relies on the fact that any target residue $r' \in M_{K+1}$ projects down to a known residue $r \in M_K$. We use the existing witness for r as a template, then apply steering gadgets to refine the precision.

Lemma 11 (Lifting $K \rightarrow K + 1$). *Fix $K \geq 3$. Assume the inductive hypothesis that for every odd residue $r \pmod{M_K}$ there exists a certified word W and an integer input m such that*

$$x_W(m) \equiv r \pmod{M_K}.$$

Then for every odd target residue $r' \pmod{M_{K+1}}$ there exists a certified word W' and an integer input m' such that

$$x_{W'}(m') \equiv r' \pmod{M_{K+1}}.$$

Proof. Let $r' \in \{0, \dots, M_{K+1} - 1\}$ be the target odd residue.

- (1) **Project Down:** Let $r = r' \bmod M_K$. By the induction hypothesis, there exists a word W (ending in the correct family for r') such that $x_W(m) \equiv r \pmod{M_K}$.
- (2) **Mod-3 Alignment:** The affine form is $x_W(m) = 6(A_W m + B_W) + \delta_W$. To ensure the final congruence is solvable, we must remove the factor of 3 from the obstruction. Use Lemma 5 to append a short same-family steering gadget so that the new intercept satisfies:

$$B_{W'} \equiv \frac{r' - \delta_W}{6} \pmod{3}.$$

- (3) **2-adic Steering:** Use Lemma 6 to append same-family gadgets until $v_2(A_{W'}) \geq K$. Simultaneously, toggle the parity of $B_{W'}$ to ensure that the term $\frac{r' - \delta_W}{6} - B_{W'}$ is divisible by $2^{\min(\alpha_p+1, K)}$. This guarantees that the solvability criterion $g^{(p)} \mid r^{(p)}$ from Lemma 9 is met.
- (4) **Routing Stability:** Apply Lemma 8 to restrict the input m to a class modulo 2^{S^*} that freezes all prefix routers. This ensures that the steering in steps 2 and 3 does not invalidate the original path W .
- (5) **Solve:** With the algebra aligned, the last-row congruence

$$A_{W'}m \equiv \frac{r' - \delta_W}{6} - B_{W'} \pmod{2^K}$$

is solvable. Choose m' to be the *least nonnegative* solution in its congruence class. More precisely, writing $\sigma := v_2(A_{W'})$ and $A_{W'} = 2^\sigma v$ with v odd, the congruence determines m' uniquely modulo $2^{K-\sigma}$; take the representative

$$0 \leq m' < 2^{K-\sigma}.$$

By Lemma 10, this canonical choice lies in the stable routing class ensured by step 4.

Thus, the constructed word and input satisfy $x_{W'}(m') \equiv r' \pmod{M_{K+1}}$. $\square$

11.2. Global Reachability Theorem.

Theorem 5 (Reachability for all K). *Let $M_K := 3 \cdot 2^K$ and let*

$$\mathcal{R}_K := \{ r \in \mathbb{Z}/M_K\mathbb{Z} : r \text{ is odd and } r \not\equiv 0 \pmod{3} \}$$

be the set of active odd residue classes modulo M_K . For every $K \geq 3$ and every $r \in \mathcal{R}_K$, there exists a certified word W and an integer $m \geq 0$ such that

$$x_W(m) \equiv r \pmod{M_K}.$$

Moreover, in the terminal solve step one may choose m canonically (least nonnegative representative in its congruence class), so that if $\sigma := v_2(A_W)$ then

$$0 \leq m < 2^{K-\sigma}.$$

In addition, when the theorem is applied to a fixed odd target x , the inductive construction may be carried out inside the router class

$$j(x) := \left\lfloor \frac{x}{6} \right\rfloor \pmod{3},$$

so that the resulting witness value $y := x_W(m)$ satisfies

$$y \equiv x \pmod{M_K}, \quad y \bmod 6 = x \bmod 6, \quad \left\lfloor \frac{y}{6} \right\rfloor \pmod{3} = \left\lfloor \frac{x}{6} \right\rfloor \pmod{3}.$$

Proof. Base case ($K = 3$). By Section 10 (Table 13), every active odd residue class modulo $M_3 = 24$ has an explicit certified witness. This establishes the claim for $K = 3$.

Inductive step. Assume the claim holds for some $K \geq 3$, and fix any $r' \in \mathcal{R}_{K+1}$. Let r be the reduction of r' modulo M_K . By the inductive hypothesis there exist a certified word W and $m \geq 0$ such that

$$x_W(m) \equiv r \pmod{M_K}.$$

Apply Lemma 11 (the $K \rightarrow K+1$ lifting lemma) to lift this witness from modulus M_K to modulus M_{K+1} , producing an extended certified word W' (and a terminal input m') such that

$$x_{W'}(m') \equiv r' \pmod{M_{K+1}}.$$

When the target is a fixed odd integer x , we may execute the lifting step inside the router class $j(x) = \lfloor x/6 \rfloor \bmod 3$ (Lemma 12). In the terminal congruence solve step inside Lemma 11, choose m' as the least nonnegative solution in its congruence class. Writing $\sigma := v_2(A_{W'})$, this yields the canonical bound $0 \leq m' < 2^{K+1-\sigma}$.

Thus every $r' \in \mathcal{R}_{K+1}$ is reachable. By induction, reachability holds for all $K \geq 3$. $\square$

Example 9 (Lifting Trace). Suppose we have a witness for $13 \pmod{24}$ ($K = 3$), which is $W = \psi\omega$. To hit $13 \pmod{48}$ ($K = 4$):

- (1) Check the current value. If $x_W(m) \equiv 13 \pmod{48}$, we are done.
- (2) If $x_W(m) \equiv 13 + 24 = 37 \pmod{48}$, the value is "off" by the half-modulus.
- (3) We append a steering gadget (e.g., Ψ_1 at $p = 0$, which adds drift $+24$ modulo 48, or a parity toggler) to shift the residue by exactly the required amount.
- (4) The new word $W' = \psi\omega \dots$ hits $13 \pmod{48}$ exactly.

Corollary 6 (Nested lifting of witnesses). *In the inductive construction in the proof of Theorem 5, the witnesses may be chosen nested in the sense that for each $K \geq 3$ one can select a certified word W_K and input m_K such that*

$$x_{W_K}(m_K) \equiv r \pmod{M_K},$$

and the next-stage witness satisfies

$$W_{K+1} = W_K \cdot G_K$$

for some (bounded-length) gadget word G_K produced by the lifting procedure (Lemma 11), i.e. each witness word is a prefix of the next. Moreover, the terminal input can be chosen canonically as in Theorem 5.

Proof. In the inductive step of Theorem 5, the witness at level $K+1$ is constructed by taking an existing witness word W_K and appending a finite concatenation of the steering/padding gadgets used in Lemma 11. Denote this concatenation by G_K . Then the lifted witness word is exactly $W_{K+1} = W_K \cdot G_K$, proving nesting. The canonical choice of the terminal input is the one specified in the solve step of Lemma 11 (least nonnegative representative). $\square$

Lemma 12 (Router-locking within reachability). *Fix $K \geq 3$ and an odd target x . Let*

$$\delta_6 := x \bmod 6 \in \{1, 5\}, \quad j := \left\lfloor \frac{x}{6} \right\rfloor \bmod 3 \in \{0, 1, 2\}.$$

Then Theorem 5 can be applied so that the produced witness value $y = x_W(m)$ satisfies in addition:

$$y \bmod 6 = \delta_6, \quad \left\lfloor \frac{y}{6} \right\rfloor \bmod 3 = j.$$

Proof. In the lifting procedure (Lemma 11), Step 4 enforces *routing stability*: the lifted word is constructed inside a stable routing class determined by the router digit. Choosing the stable routing class indexed by j forces the terminal witness value y to lie in the same router class, i.e. $\lfloor y/6 \rfloor \bmod 3 = j$. Meanwhile $y \equiv x \pmod{M_{K+1}}$ implies $y \equiv x \pmod{6}$, hence $y \bmod 6 = \delta_6$. $\square$

12. FROM RESIDUES TO EXACT INTEGERS

The previous sections established that for any K , we can find a word W and input m such that $x_W(m) \equiv x_{\text{tar}} \pmod{3 \cdot 2^K}$. We now show that by taking the limit as $K \rightarrow \infty$, we obtain an exact solution in the integers.

12.1. Linear 2-adic Lifting. The affine form of a word is $x_W(m) = 6(A_W m + B_W) + \delta_W$. Solving $x_W(m) = x_{\text{tar}}$ is equivalent to solving the linear equation:

$$(19) \quad A_W m = \frac{x_{\text{tar}} - \delta_W}{6} - B_W.$$

Since $A_W = 3 \cdot 2^{\alpha(W)}$, this is a linear equation over the 2-adic integers.

Reusable 2-adic tools. We will repeatedly reduce control and lifting statements to elementary congruences modulo powers of 2. For convenience, we isolate three standard facts that will be used throughout without further comment: Lemma 13 formalizes cancellation of an odd factor and yields the resulting solution class modulo a power of two; Lemma 14 records the “pinning” phenomenon that occurs once the 2-power exceeds the modulus; and Lemma 15 observes that, once the family digit $\delta \in \{1, 5\}$ is fixed, the router digit j' is read off directly from the target output via $\lfloor x'/6 \rfloor \bmod 3$. These lemmas are the basic arithmetic engine behind both the pinning/solving dichotomy and the congruence-class descriptions of solution sets.

Lemma 13 (2-adic cancellation and classes). *Let $K \geq 1$, let $\sigma \geq 0$, and write $A = 2^\sigma v$ with v odd. For any integers m, m' , if $K \geq \sigma$ then*

$$Am \equiv Am' \pmod{2^K} \iff m \equiv m' \pmod{2^{K-\sigma}}.$$

Equivalently, for fixed c the congruence

$$Am \equiv Ac \pmod{2^K}$$

has solution set the unique residue class $m \equiv c \pmod{2^{K-\sigma}}$.

Proof. Assume $K \geq \sigma$. From $2^\sigma vm \equiv 2^\sigma vm' \pmod{2^K}$ we may cancel 2^σ to get $vm \equiv vm' \pmod{2^{K-\sigma}}$. Since v is odd, it is invertible modulo $2^{K-\sigma}$, hence $m \equiv m' \pmod{2^{K-\sigma}}$. The converse is immediate by multiplying a congruence $m \equiv m' \pmod{2^{K-\sigma}}$ by $2^\sigma v$. $\square$

Lemma 14 (2-adic pinning). *Let $K \geq 1$ and $\sigma \geq K$. Then for every integer m ,*

$$2^\sigma m \equiv 0 \pmod{2^K}.$$

More generally, if $A = 2^\sigma v$ with v odd and $\sigma \geq K$, then $Am \equiv 0 \pmod{2^K}$ for all m .

Proof. If $\sigma \geq K$ then $2^K \mid 2^\sigma$, hence $2^K \mid 2^\sigma m$ for all m . The general case follows since v is an integer. $\square$

Lemma 15 (Router digit from the target). *Let $\delta \in \{1, 5\}$ and let x' be an integer with $x' \equiv \delta \pmod{6}$. Define*

$$F := \frac{x' - \delta}{6} \in \mathbb{Z}.$$

Then the router digit of x' satisfies

$$\left\lfloor \frac{x'}{6} \right\rfloor \equiv F \pmod{3}.$$

In particular, the router class $j' = \lfloor x'/6 \rfloor \pmod{3}$ is determined by $x' \pmod{18}$ once δ is fixed.

Proof. Since $x' = 6F + \delta$ with $0 < \delta < 6$, we have $\lfloor x'/6 \rfloor = F$. Reducing modulo 3 gives the claim. $\square$

Lemma 16 (2-adic Completeness). *Let W be a fixed certified word, with affine evaluation*

$$x_W(m) = 6(A_W m + B_W) + \delta_W,$$

where $A_W, B_W \in \mathbb{Z}$ and $\delta_W \in \{1, 5\}$ are fixed by W . Fix an integer target x_{tar} .

Assume that for every $K \geq K_0$ there exists an integer m_K such that

$$x_W(m_K) \equiv x_{\text{tar}} \pmod{3 \cdot 2^K}.$$

Equivalently (after dividing by 6 and using that the resulting right-hand side is an integer in the application), there is a fixed integer RHS such that

$$(20) \quad A_W m_K \equiv \text{RHS} \pmod{2^K} \quad \text{for all } K \geq K_0.$$

Let $\sigma := v_2(A_W)$ and write $A_W = 2^\sigma v$ with v odd.

If the solutions are chosen compatibly, i.e.

$$m_{K+1} \equiv m_K \pmod{2^{K-\sigma}} \quad (K \geq \max\{K_0, \sigma + 1\}),$$

then $(m_K)_K$ is a Cauchy sequence in $\mathbb{Z}_2$ and converges to a unique $m \in \mathbb{Z}_2$ satisfying $A_W m = \text{RHS}$ in $\mathbb{Z}_2$, hence $x_W(m) = x_{\text{tar}}$ in $\mathbb{Z}_2$.

Moreover, if u divides $\text{RHS}/2^\sigma$ as an ordinary integer (in particular, if $u = 1$, i.e. A_W is a power of two), then $m \in \mathbb{Z}$ and the equality is exact in $\mathbb{Z}$: $x_W(m) = x_{\text{tar}}$.

Proof. From (20) and $v_2(A_W) = \sigma$, it follows that for each $K > \sigma$ the congruence class of m_K is unique modulo $2^{K-\sigma}$: if $A_W m \equiv A_W m' \pmod{2^K}$, then $2^\sigma v(m - m') \equiv 0 \pmod{2^K}$ and since v is odd (hence invertible modulo $2^{K-\sigma}$), we get $m \equiv m' \pmod{2^{K-\sigma}}$. Thus compatibility means precisely that each m_{K+1} refines m_K modulo $2^{K-\sigma}$, so $(m_K)_K$ defines an element of the inverse limit $\varprojlim \mathbb{Z}/2^n \mathbb{Z} = \mathbb{Z}_2$. Hence $(m_K)_K$ is Cauchy and converges to some $m \in \mathbb{Z}_2$.

Passing to the limit in (20) (multiplication by the fixed integer A_W is continuous in the 2-adic metric) yields $A_W m = \text{RHS}$ in $\mathbb{Z}_2$. Therefore $x_W(m) = x_{\text{tar}}$ in $\mathbb{Z}_2$.

To obtain an ordinary integer, reduce (20) modulo 2^σ . Since $A_W m_K \equiv 0 \pmod{2^\sigma}$ for every K , we obtain $\text{RHS} \equiv 0 \pmod{2^\sigma}$, so $\text{RHS} = 2^\sigma r$ for some $r \in \mathbb{Z}$. Then the 2-adic equation becomes

$$u m = r \quad \text{in } \mathbb{Z}_2,$$

which has the unique solution $m = u^{-1}r \in \mathbb{Z}_2$. If in addition $u \mid r$ in $\mathbb{Z}$ (in particular if $u = 1$), then $m = r/u \in \mathbb{Z}$, and substituting back into $x_W(m)$ gives $x_W(m) = x_{\text{tar}}$ as an equality of integers. $\square$

Lemma 17 (Power-of-two linear lifting yields an integer solution). *Let $\sigma \geq 0$ and let $\text{RHS} \in \mathbb{Z}$. If there exists $m \in \mathbb{Z}_2$ such that*

$$2^\sigma m \equiv \text{RHS} \pmod{2^K} \quad \text{for all } K > \sigma,$$

then $v_2(\text{RHS}) \geq \sigma$ and $m = \text{RHS}/2^\sigma \in \mathbb{Z}$.

Proof. Reducing the congruence modulo 2^σ gives $\text{RHS} \equiv 0 \pmod{2^\sigma}$, hence $v_2(\text{RHS}) \geq \sigma$. In $\mathbb{Z}_2$, division by 2^σ is well-defined, so $m = \text{RHS}/2^\sigma$. Since RHS is an integer divisible by 2^σ , the quotient $\text{RHS}/2^\sigma$ is an ordinary integer. $\square$

Lemma 18 (Finite hit from canonical representatives). *Fix $x \in \mathbb{N}$. Choose K so that $M_K = 3 \cdot 2^K > x$. Assume the reachability procedure returns a certified witness (W_K, m_K) such that $x_{W_K}(m_K) \equiv x \pmod{M_K}$ and $0 \leq x_{W_K}(m_K) < M_K$. Then $x_{W_K}(m_K) = x$.*

Proof. Since $0 \leq x_{W_K}(m_K), x < M_K$ and $x_{W_K}(m_K) \equiv x \pmod{M_K}$, we must have $x_{W_K}(m_K) = x$. $\square$

Lemma 19 (Finite hit in the quotient coordinate). *Let $m \in \mathbb{Z}_{\geq 0}$ be fixed. Suppose that for each $K \geq 1$ there exists an integer m_K such that*

- (1) $m_K \equiv m \pmod{2^K}$,
- (2) $0 \leq m_K < 2^K$ (canonical representative modulo 2^K).

Then there exists a finite K_0 such that for all $K \geq K_0$,

$$m_K = m.$$

In fact one may take

$$K_0 = \min\{K \geq 1 : 2^K > m\} = \lceil \log_2(m+1) \rceil.$$

Proof. Choose K_0 such that $2^{K_0} > m$. For any $K \geq K_0$ we have $0 \leq m < 2^K$, and by assumption also $0 \leq m_K < 2^K$. Since $m_K \equiv m \pmod{2^K}$ and both integers lie in $[0, 2^K)$, it follows that $m_K = m$. $\square$

Corollary 7 (Finite hit for x once (δ_6, j) are fixed). *Let x be an odd integer, and define*

$$\begin{aligned} \delta_6 &:= x \bmod 6 \in \{1, 5\}, \\ j &:= \left\lfloor \frac{x}{6} \right\rfloor \bmod 3 \in \{0, 1, 2\}, \\ m &:= \left\lfloor \frac{x}{18} \right\rfloor \in \mathbb{Z}_{\geq 0}. \end{aligned}$$

Suppose that for each $K \geq 1$ this construction produces an odd integer of the form

$$x_K := 18m_K + 6j + \delta_6,$$

where $(m_K)_{K \geq 1}$ satisfies the hypotheses of Lemma 19. Then there exists a finite K_0 such that for all $K \geq K_0$,

$$x_K = x.$$

Proof. By Lemma 19, there is K_0 such that $m_K = m$ for all $K \geq K_0$. Substituting into $x_K = 18m_K + 6j + \delta_6$ yields $x_K = 18m + 6j + \delta_6 = x$ for all $K \geq K_0$. $\square$

Lemma 20 (Quotient congruence extraction). *Let x and y be odd integers with the same*

$$\begin{aligned}\delta_6 &:= x \bmod 6 = y \bmod 6 \in \{1, 5\}, \\ j &:= \left\lfloor \frac{x}{6} \right\rfloor \bmod 3 = \left\lfloor \frac{y}{6} \right\rfloor \bmod 3 \in \{0, 1, 2\}.\end{aligned}$$

Write

$$x = 18m + 6j + \delta_6, \quad y = 18m' + 6j + \delta_6,$$

with $m, m' \in \mathbb{Z}_{\geq 0}$. Then for every $K \geq 1$,

$$y \equiv x \pmod{3 \cdot 2^{K+1}} \implies m' \equiv m \pmod{2^K}.$$

Proof. If $y \equiv x \pmod{3 \cdot 2^{K+1}}$, then $3 \cdot 2^{K+1}$ divides $y - x = 18(m' - m)$. Dividing by $18 = 3 \cdot 2 \cdot 3$ gives $2^K \mid (m' - m)$, i.e. $m' \equiv m \pmod{2^K}$. $\square$

12.2. The Global Existence Theorem. We can now state the final result of the constructive calculus. By combining the topological reachability of residues (Theorem 5) with the algebraic completeness of the 2-adic integers (Lemma 16), we obtain a result for exact integers.

Theorem 8 (Exact Reachability). *For every odd integer $x \geq 1$, there exists a finite certified word W and an integer $m \in \mathbb{Z}_{\geq 0}$ such that*

$$x_W(m) = x.$$

Consequently, the forward orbit of x under the accelerated map U traverses the path W in reverse and terminates at 1.

Proof. Fix an odd $x \geq 1$ and define

$$\begin{aligned}\delta_6 &:= x \bmod 6 \in \{1, 5\}, \\ j &:= \left\lfloor \frac{x}{6} \right\rfloor \bmod 3 \in \{0, 1, 2\}, \\ m &:= \left\lfloor \frac{x}{18} \right\rfloor \in \mathbb{Z}_{\geq 0}.\end{aligned}$$

Choose K_0 such that $2^{K_0} > m$.

Apply Theorem 5 at some precision $K \geq K_0 + 1$ together with Lemma 12 to obtain a certified witness (word and input) producing an odd integer x_K such that

$$x_K \equiv x \pmod{M_K}, \quad x_K \bmod 6 = \delta_6, \quad \left\lfloor \frac{x_K}{6} \right\rfloor \bmod 3 = j.$$

Write

$$x_K = 18m_K + 6j + \delta_6$$

for the (necessarily integral) quotient $m_K \in \mathbb{Z}_{\geq 0}$.

Let $\sigma := v_2(A_W)$ for the witness word W producing x_K . By the canonical choice in the terminal solve step (Lemma 11, Step 5), m_K is determined modulo $2^{K-\sigma}$ and we may choose it as the least nonnegative representative, so that

$$0 \leq m_K < 2^{K-\sigma}.$$

Choose K large enough so that $K - \sigma \geq K_0$ (equivalently $2^{K-\sigma} \geq 2^{K_0}$). Then the congruence $x_K \equiv x \pmod{M_K}$ implies $x_K \equiv x \pmod{M_{K_0+1}}$, and Lemma 20 (with $y = x_K$ and $K - 1 = K_0$) yields

$$m_K \equiv m \pmod{2^{K_0}}.$$

Now replace m_K by its reduction modulo 2^{K_0} (which does not change x_K modulo M_{K_0+1} and preserves the router class): namely, set

$$\tilde{m}_K := m_K \bmod 2^{K_0}, \quad 0 \leq \tilde{m}_K < 2^{K_0}.$$

Then $\tilde{m}_K \equiv m_K \equiv m \pmod{2^{K_0}}$, and since also $0 \leq m < 2^{K_0}$ (by $2^{K_0} > m$), Lemma 19 gives $\tilde{m}_K = m$. Substituting into $x_K = 18\tilde{m}_K + 6j + \delta_6$ yields $x_K = 18m + 6j + \delta_6 = x$.

Therefore the corresponding witness word W and input m satisfy $x_W(m) = x$ exactly. Certification of W implies the reverse traversal property, hence $U^{|W|}(x) = 1$. $\square$

Corollary 9 (Exclusion of Disconnected Cycles). *Since every odd integer x is the root of a finite inverse chain terminating at 1, no integer can belong to a disconnected cycle (a cycle that does not contain 1). Furthermore, no integer can belong to a divergent trajectory that does not eventually enter the tree of 1. The Collatz graph on the odd integers is therefore a single connected tree rooted at 1.*

Example 10 (Execution of Exact Reachability for $x = 497$). To target $x = 497$, the algorithm constructs a certified program W .

- **Targeting:** $497 \equiv 17 \pmod{24}$. We start with the base witness for 17.
- **Lifting:** We append steering tokens to satisfy the 2-adic linear constraints.
- **Resulting Program:** The certified path is $W = (\psi, \Omega, \Omega, \omega, \psi)$.
- **Instantiation:** The template equation $A_W m + B_W = (497 - \delta)/6$ yields the unique integer solution $m = 1$.
- **Verification:** $x_W(1) = 497$. The forward orbit traverses the path in reverse: $497 \rightarrow 373 \rightarrow 35 \rightarrow 53 \rightarrow 5 \rightarrow 1$.

13. RESIDUE OBSTRUCTIONS, DESCENT, AND EXCEPTIONAL ANTI-GROWTH

This section studies two obstructions using the same residue-lift mechanism. The first is a descent obstruction: a proposed infinite descent would force

the starting index into deeper and deeper residue classes modulo powers of 3. The second is an exceptional anti-growth obstruction arising from the $p = 0$ table: repeated passage through the $(o, 2)$ -row creates inverse debt, and limited $(o, 1)$ -runs provide the least compensation. In both cases, the essential question is whether an infinite obstruction can be carried by a finite positive integer, or whether the associated residue constraints either stabilize or define only a 3-adic object.

The purpose of this section is to analyze the residue obstructions that could prevent forward descent of an odd Collatz orbit. In the forward direction, a successful descent argument must show that every odd input eventually reaches a smaller odd value. In the inverse calculus, this same question appears as a growth problem: ordinary inverse branches expand, while exceptional anti-growth branches represent the possible failure modes of descent when read forward.

The section has two parts. First, we analyze descent obstructions at the level of words in the row alphabet. Second, we isolate the exceptional $(o, 2)$ -mechanism in the $p = 0$ table, where inverse anti-growth appears. Both analyses use the same residue-lift mechanism: an attempted infinite obstruction forces the starting index into deeper and deeper congruence classes modulo powers of 3.

The guiding question is whether such an obstruction can be carried by a finite positive integer. The recurring conclusion is that an attempted infinite obstruction either stabilizes, exhausting the finite residue construction, or defines only a 3-adic object rather than an ordinary positive integer.

13.1. Inverse growth and forward descent. The inverse calculus is used here because forward descent is easier to detect after reversing the direction of comparison. In the forward odd Collatz dynamics, the goal is to show that every odd input eventually reaches a smaller odd value. In the inverse calculus, this becomes a growth question: a branch which grows in the inverse direction corresponds to descent in the forward direction.

More precisely, suppose an inverse word carries an odd value x' back to a predecessor x . If

$$x > x',$$

then reading the same word forward carries x down to x' . Thus inverse growth,

$$x' \mapsto x > x',$$

is equivalent to forward descent,

$$x \mapsto x' < x.$$

Therefore, throughout this section, inverse growth is not an obstruction. It is the desired behavior. The obstruction is inverse anti-growth: an inverse branch, or a contiguous portion of an inverse branch, which fails to grow enough. In the $p = 0$ table, the exceptional source of such anti-growth is the

$(o, 2)$ -row. The $(o, 1)$ -row supplies the least compensation available within the same base routing. This is why the local obstruction studied below has the form

$$(o, 2)^{k_0}(o, 1)^{r_0}(o, 2)^{k_1}(o, 1)^{r_1} \dots$$

The purpose of the section is therefore to rule out the possibility that such anti-growth can persist indefinitely for a finite positive segment-start. If every attempted infinite anti-growth obstruction either stabilizes or becomes only 3-adic, then it cannot prevent forward descent in the ordinary positive integer domain.

Proposition 5 (Non-exceptional rows give forward descent). *In the inverse calculus, every non-exceptional row is strictly inverse-expansive away from the terminal fixed point $1 \mapsto 1$. The only equality case is this fixed point, which is represented in the $p = 0$ table and is excluded when studying descent from odd $x > 1$. Consequently, every non-exceptional non-terminal row gives strict descent when read in the forward orbital direction.*

Proof. This is Corollary 2, together with the $p = 0$ terminal case in the base table. For inputs in the e-family, the inverse drift is positive away from the terminal fixed point. For inputs in the o-family, negative drift occurs only when $p = 0$ and the token is one of

$$\omega_1, \quad \Omega_2.$$

All rows with $p \geq 1$ are strongly expansive in the inverse direction. Thus, apart from the fixed point $1 \mapsto 1$, every non-exceptional row is strictly inverse-expansive. Reading such a row in the forward orbital direction gives strict descent. $\square$

Thus the only local rows requiring special analysis are the exceptional anti-growth rows. The $(o, 2)$ -mechanism is the most dangerous because repeated passage through it creates inverse debt. Since, as we will show, a pure $(o, 2)$ -run is finite, the next possible obstruction is a compensated exceptional segment in which finite $(o, 2)$ -runs are followed by limited $(o, 1)$ -runs, the least compensating mechanism visible in the $p = 0$ table.

Corollary 10 (All higher- p rows descend). *For every row with $p \geq 1$, the inverse drift is positive. Hence every $p \geq 1$ row is expansive in the inverse direction and descending in the forward orbital direction.*

Proof. This is the $p \geq 1$ case of Corollary 2. The drift bounds show that all $p \geq 1$ rows are strongly expansive in the inverse calculus. Therefore, if an inverse step carries x' to a larger predecessor x , then reading the same step forward carries x to the smaller value x' . Thus every $p \geq 1$ row gives forward descent. $\square$

The row calculus isolates the only possible local obstruction to immediate forward descent. Let a row/lift produce an inverse step

$$3x' + 1 = 2^\sigma x, \quad \sigma = \alpha + 6p.$$

Equivalently, the accelerated odd Collatz map sends

$$x' \mapsto x.$$

Thus the forward step is descending precisely when

$$x < x'.$$

Theorem 11 (Local descent and exceptional rows). *Let x' be a positive odd integer produced by one of the row/lift inverse branches, and suppose*

$$3x' + 1 = 2^\sigma x, \quad \sigma = \alpha + 6p.$$

If

$$\sigma \geq 2,$$

then the corresponding forward odd step is strictly descending:

$$x < x'.$$

Consequently, every row/lift with $p > 0$ is forward-descending. At $p = 0$, the only non-descending rows are the two rows with $\alpha = 1$, namely

$$\omega_1 \quad \text{and} \quad \Omega_2.$$

Equivalently, all row/lift steps are forward-descending except the two exceptional cases

$$(\omega_1, p = 0) \quad \text{and} \quad (\Omega_2, p = 0).$$

Proof. From

$$3x' + 1 = 2^\sigma x,$$

we have

$$x = \frac{3x' + 1}{2^\sigma}.$$

Therefore $x < x'$ is equivalent to

$$\frac{3x' + 1}{2^\sigma} < x',$$

or

$$3x' + 1 < 2^\sigma x'.$$

If $\sigma \geq 2$, then $2^\sigma \geq 4$, and hence it is enough to check

$$3x' + 1 < 4x'.$$

This is equivalent to

$$1 < x',$$

which holds for every positive odd $x' > 1$. Thus every step with $\sigma \geq 2$ is strictly descending.

Now

$$\sigma = \alpha + 6p.$$

If $p > 0$, then

$$\sigma \geq \alpha + 6 \geq 7,$$

since the row table has $\alpha \geq 1$. Hence every row/lift with $p > 0$ is descending.

At $p = 0$, one has $\sigma = \alpha$. The only rows in the table with $\alpha = 1$ are

$$\omega_1 \quad \text{and} \quad \Omega_2.$$

All other $p = 0$ rows have $\alpha \geq 2$, and are therefore descending by the first part of the proof. $\square$

Theorem 12 (No infinite exceptional run). *There is no infinite positive-integer forward orbit consisting only of the exceptional $\sigma = 1$ steps. Equivalently, no positive integer can remain forever in the two exceptional cases*

$$\omega_1 \quad \text{and} \quad \Omega_2$$

at $p = 0$.

Proof. A $\sigma = 1$ odd step has the form

$$U(x) = \frac{3x+1}{2}.$$

For this step to occur, one must have

$$v_2(3x+1) = 1,$$

which, for odd x , is equivalent to

$$x \equiv 3 \pmod{4}.$$

Since for an odd integer x , the condition

$$v_2(3x+1) = 1$$

is equivalent to

$$x \equiv 3 \pmod{4}.$$

Indeed, $v_2(3x+1) = 1$ means that $3x+1$ is divisible by 2, but not divisible by 4. Equivalently,

$$3x+1 \equiv 2 \pmod{4}.$$

Since x is odd, there are only two cases:

$$x \equiv 1 \pmod{4} \quad \text{or} \quad x \equiv 3 \pmod{4}.$$

If

$$x \equiv 1 \pmod{4},$$

then

$$3x+1 \equiv 3(1)+1 \equiv 4 \equiv 0 \pmod{4},$$

so

$$v_2(3x+1) \geq 2.$$

Thus this case does not give a $\sigma = 1$ step.

On the other hand, if

$$x \equiv 3 \pmod{4},$$

then

$$3x + 1 \equiv 3(3) + 1 \equiv 10 \equiv 2 \pmod{4}.$$

Hence $3x + 1$ is divisible by 2, but not by 4, and therefore

$$v_2(3x + 1) = 1.$$

Thus, for odd x ,

$$v_2(3x + 1) = 1 \iff x \equiv 3 \pmod{4}.$$

More generally, n consecutive $\sigma = 1$ steps force the initial value to satisfy

$$x \equiv -1 \pmod{2^{n+1}}.$$

Equivalently,

$$2^{n+1} \mid x + 1.$$

Thus an infinite run of $\sigma = 1$ steps would require

$$2^N \mid x + 1$$

for every $N \geq 1$.

This is impossible for a fixed finite positive integer x . Indeed, once

$$2^N > x + 1,$$

the divisibility relation

$$2^N \mid x + 1$$

can hold only if

$$x + 1 = 0.$$

That would force

$$x = -1,$$

which is not a positive integer.

Equivalently, the infinite system of congruences

$$x \equiv -1 \pmod{2^N} \quad (N \geq 1)$$

has the 2-adic solution

$$x = -1,$$

whose 2-adic expansion is

$$\cdots 11111_2.$$

But no ordinary positive integer has infinitely many trailing binary 1's. Therefore no positive-integer orbit can consist indefinitely only of exceptional $\sigma = 1$ steps. The only object satisfying these congruences in the 2-adic integers is the 2-adic value -1 , which is not a positive integer. Therefore no positive-integer orbit can consist indefinitely only of exceptional steps. $\square$

13.2. From One-Step Rows to Word-Level Reverse Growth. The one-step row calculus gives a precise inverse transition

$$(s, j, m) \mapsto (s', j', m').$$

Although the Collatz conjecture is a statement about forward descent, the row calculus is naturally written in the reverse direction. Thus, rather than trying to prove descent directly at each forward step, we study when an inverse word produces growth.

This change of viewpoint is harmless, and in fact useful. Suppose an inverse word sends

$$x_0 \mapsto x_n.$$

Then the corresponding forward orbit travels in the opposite direction:

$$x_n \mapsto x_0.$$

Therefore, if the inverse word satisfies

$$x_n > x_0,$$

then the corresponding forward block satisfies

$$x_0 < x_n.$$

In other words, reverse growth is precisely forward descent after reversing the direction of the block.

Thus the main goal of the word calculus is to prove inequalities of the form

$$x_n > x_0$$

for inverse words. Once such an inequality is proved, it immediately becomes a descent statement for the corresponding forward orbit.

In the state coordinates, we usually work with the internal indices rather than the odd values themselves. Since

$$x = 18m + 6j + \delta_s,$$

the internal index m is a safe size proxy for x : strict growth in m implies strict growth in x . Therefore, the central task becomes to show that suitable inverse words satisfy

$$m_n > m_0.$$

This is the reverse-growth form of the forward-descent problem.

13.2.1. Reverse-growth criterion for a word. A word grows in the inverse direction at the m -level when

$$m_n > m_0.$$

Using the word affine formula

$$m_0 = \frac{3^n m_n + C_W}{2^{\Sigma_W}},$$

the inequality $m_n > m_0$ becomes

$$m_n > \frac{3^n m_n + C_W}{2^{\Sigma_W}}.$$

Equivalently,

$$2^{\Sigma_W} m_n > 3^n m_n + C_W,$$

or

$$\boxed{C_W < (2^{\Sigma_W} - 3^n) m_n.}$$

Thus the quantity

$$2^{\Sigma_W} - 3^n$$

controls the leading reverse-growth behavior of the word. If

$$2^{\Sigma_W} > 3^n,$$

then the inverse word has positive multiplicative growth, and the remaining question is whether the accumulated carry term C_W is small enough for the strict inequality above to hold.

Theorem 13 (Reverse-growth criterion). *Let W be an admissible inverse word of length n , with total exponent Σ_W and accumulated carry C_W . Then*

$$m_n > m_0$$

if and only if

$$C_W < (2^{\Sigma_W} - 3^n) m_n.$$

Consequently, the inverse word grows at the internal-index level exactly when this carry inequality holds. Reversing the word, the corresponding forward block descends exactly in this case.

Proof. From

$$m_0 = \frac{3^n m_n + C_W}{2^{\Sigma_W}},$$

we have

$$m_n > m_0$$

if and only if

$$m_n > \frac{3^n m_n + C_W}{2^{\Sigma_W}}.$$

Multiplying by 2^{Σ_W} gives

$$2^{\Sigma_W} m_n > 3^n m_n + C_W,$$

equivalently

$$C_W < (2^{\Sigma_W} - 3^n) m_n.$$

This proves the criterion. $\square$

13.2.2. *Return segments and the exceptional obstruction.* The reverse-growth criterion can be tested especially cleanly on *return segments*. By a return segment we mean an admissible word whose initial and terminal symbols agree:

$$(s_0, j_0) = (s_n, j_n).$$

In that case the initial and terminal odd values have the same residue offset:

$$x_0 = 18m_0 + 6j_0 + \delta_{s_0},$$

and

$$x_n = 18m_n + 6j_n + \delta_{s_0}.$$

Hence

$$x_n > x_0 \iff m_n > m_0.$$

Thus, for return segments, reverse growth in the internal index is exactly the same as reverse growth in the odd value.

One-step carry form. We begin with the one-step relation at the m -index level. For one row step, write

$$F_i = 3m_{i+1} + j_{i+1},$$

because the next odd value has the decomposition

$$x_{i+1} = 6F_i + \delta_{i+1} = 18m_{i+1} + 6j_{i+1} + \delta_{i+1}.$$

For a row with 2-adic exponent

$$\sigma_i = \alpha_i + 6p_i,$$

the row map is affine in m_i , so we may write

$$F_i = 2^{\sigma_i} m_i + k_i,$$

where k_i is the row-dependent offset. Hence

$$3m_{i+1} + j_{i+1} = 2^{\sigma_i} m_i + k_i.$$

Solving for m_{i+1} , we obtain

$$3m_{i+1} = 2^{\sigma_i} m_i + k_i - j_{i+1},$$

and therefore

$$m_{i+1} = \frac{2^{\sigma_i} m_i + k_i - j_{i+1}}{3}.$$

If we define the one-step carry by

$$c_i = j_{i+1} - k_i,$$

then this becomes

$$m_{i+1} = \frac{2^{\sigma_i} m_i - c_i}{3}.$$

Equivalently,

$$2^{\sigma_i} m_i = 3m_{i+1} + c_i,$$

and solving backward gives

$$m_i = \frac{3m_{i+1} + c_i}{2^{\sigma_i}}.$$

Composition of carries along a word. For one step, we have

$$m_0 = \frac{3m_1 + c_0}{2^{\sigma_0}}.$$

For two steps, substitute

$$m_1 = \frac{3m_2 + c_1}{2^{\sigma_1}}$$

into the preceding expression:

$$m_0 = \frac{3 \left(\frac{3m_2 + c_1}{2^{\sigma_1}} \right) + c_0}{2^{\sigma_0}}.$$

Thus

$$m_0 = \frac{9m_2 + 3c_1 + c_0 2^{\sigma_1}}{2^{\sigma_0 + \sigma_1}}.$$

So, for two steps,

$$m_0 = \frac{3^2 m_2 + C_2}{2^{\sigma_0 + \sigma_1}},$$

where

$$C_2 = 3c_1 + c_0 2^{\sigma_1}.$$

For three steps, substitute

$$m_2 = \frac{3m_3 + c_2}{2^{\sigma_2}}.$$

Then

$$m_0 = \frac{9 \left(\frac{3m_3 + c_2}{2^{\sigma_2}} \right) + 3c_1 + c_0 2^{\sigma_1}}{2^{\sigma_0 + \sigma_1}}.$$

Multiplying through by 2^{σ_2} , we get

$$m_0 = \frac{27m_3 + 9c_2 + 3c_1 2^{\sigma_2} + c_0 2^{\sigma_1 + \sigma_2}}{2^{\sigma_0 + \sigma_1 + \sigma_2}}.$$

Thus, for three steps,

$$m_0 = \frac{3^3 m_3 + C_3}{2^{\sigma_0 + \sigma_1 + \sigma_2}},$$

where

$$C_3 = 9c_2 + 3c_1 2^{\sigma_2} + c_0 2^{\sigma_1 + \sigma_2}.$$

In general, for a word W of length n , define

$$\Sigma_W = \sigma_0 + \sigma_1 + \cdots + \sigma_{n-1}.$$

Then the composed relation has the form

$$m_0 = \frac{3^n m_n + C_W}{2^{\Sigma_W}},$$

where the accumulated carry is

$$C_W = 3^{n-1} c_{n-1} + 3^{n-2} c_{n-2} 2^{\sigma_{n-1}} + 3^{n-3} c_{n-3} 2^{\sigma_{n-2} + \sigma_{n-1}} + \cdots + c_0 2^{\sigma_1 + \cdots + \sigma_{n-1}}.$$

Equivalently,

$$C_W = \sum_{i=0}^{n-1} 3^i c_{n-1-i} 2^{\sigma_{n-i} + \sigma_{n-i+1} + \dots + \sigma_{n-1}},$$

with the convention that the empty exponent sum is 0.

Thus, as the word length increases, the leading term becomes

$$3^n m_n,$$

the denominator becomes

$$2^{\Sigma_W},$$

and the carry terms accumulate with mixed powers of 3 and 2, depending on where they occur in the word.

Let S be a return segment of length n . At $p = 0$, write

$$A_S = \sum_{i=0}^{n-1} \alpha_i,$$

and

$$C_S^{(0)} = \sum_{i=0}^{n-1} 3^i 2^{\alpha_{i+1} + \dots + \alpha_{n-1}} (j_{i+1} - k_i^{(0)}).$$

Then the segment formula is

$$m_0 = \frac{3^n m_n + C_S^{(0)}}{2^{A_S}}.$$

Therefore the return segment grows in reverse precisely when

$$m_n > m_0,$$

or equivalently,

$$C_S^{(0)} < (2^{A_S} - 3^n) m_n.$$

This gives a natural way to test the row system: for each symbol

$$(e, 0), (e, 1), (e, 2), (o, 0), (o, 1), (o, 2),$$

one may examine admissible return segments beginning and ending at that symbol, and ask whether the segment has reverse growth.

The first evaluation shows that arbitrary return segments do not all grow. The obstruction comes from the two exceptional rows

$$\omega_1 \quad \text{and} \quad \Omega_2,$$

which are exactly the $p = 0$ rows with

$$\alpha = 1.$$

For such a row, the one-step exponent is too small:

$$2^\alpha = 2 < 3.$$

Thus a segment consisting only of such exceptional behavior can have negative reverse-growth tendency.

For example, the symbol $(o, 2)$ has the exceptional self-return

$$(o, 2) \xrightarrow{\Omega_2} (o, 2).$$

This segment has

$$A_S = 1, \quad n = 1,$$

and therefore

$$2^{A_S} - 3^n = 2 - 3 = -1.$$

So this one-step return segment does not grow in reverse. Symbolically, repeating this loop gives

$$\Omega_2^n,$$

with

$$A_S = n, \quad |S| = n,$$

and hence

$$2^{A_S} - 3^{|S|} = 2^n - 3^n < 0.$$

Thus the symbolic return-segment test immediately isolates the exceptional obstruction.

Other short return segments also inherit this obstruction. For example, there is a return pattern based at $(e, 2)$,

$$(e, 2) \rightarrow (o, 2) \rightarrow (o, 1) \rightarrow (e, 2),$$

using the rows

$$\psi_2, \Omega_2, \omega_1.$$

The corresponding exponents are

$$2, 1, 1.$$

Thus

$$A_S = 4, \quad |S| = 3,$$

and

$$2^{A_S} - 3^{|S|} = 2^4 - 3^3 = 16 - 27 = -11.$$

This return segment also fails the leading reverse-growth test.

Similarly, the symbol $(o, 1)$ has a corresponding short return pattern

$$(o, 1) \rightarrow (e, 2) \rightarrow (o, 2) \rightarrow (o, 1),$$

again involving the exponent pattern

$$1, 2, 1,$$

so that

$$A_S = 4, \quad |S| = 3,$$

and again

$$2^{A_S} - 3^{|S|} < 0.$$

By contrast, some symbols already have short return segments with positive exponent surplus. For instance, one return pattern based at $(e, 1)$ is

$$(e, 1) \rightarrow (e, 2) \rightarrow (o, 2) \rightarrow (o, 1) \rightarrow (e, 1),$$

with row pattern

$$\Psi_1, \psi_2, \Omega_2, \omega_1.$$

The exponents are

$$4, 2, 1, 1,$$

so

$$A_S = 8, \quad |S| = 4.$$

Hence

$$2^{A_S} - 3^{|S|} = 2^8 - 3^4 = 256 - 81 = 175 > 0.$$

This segment has positive leading reverse-growth tendency.

These examples show that the return-segment approach is useful, but the notion of segment must be refined. Arbitrary return segments are not sufficient, because the exceptional rows can produce non-growing symbolic returns. The remaining task is therefore to analyze return segments after the exceptional behavior has been properly accounted for.

A more precise target is to study one of the following refined objects:

- (1) maximal exceptional blocks together with their forced exits;
- (2) first return segments with positive exponent surplus;
- (3) return segments after excluding extendable exceptional self-loops;
- (4) return segments constrained by the actual integer compatibility conditions, rather than by the symbol graph alone.

Thus the segment analysis reduces the global obstruction to a sharper question: exceptional symbolic returns must either terminate or be absorbed into longer admissible segments whose accumulated exponent and carry term force reverse growth. In the forward direction, such reverse growth is exactly descent below the value at the beginning of the segment.

13.2.3. Shortest growing return segments by symbol. We now use the word-level reverse-growth criterion to test short return patterns in the $p = 0$ symbol system. The purpose of this subsection is not yet to prove global descent, but to identify which symbols already admit short reverse-growing return segments, and where the exceptional obstruction first appears.

A *symbol* is one of the six state labels

$$(e, 0), (e, 1), (e, 2), (o, 0), (o, 1), (o, 2).$$

A *word* is a finite admissible sequence of rows

$$W = R_0 R_1 \cdots R_{n-1}.$$

The length of the word is

$$|W| = n.$$

The word is admissible if each row begins at the symbol produced by the previous row. Thus it determines a symbol path

$$(s_0, j_0) \longrightarrow (s_1, j_1) \longrightarrow \cdots \longrightarrow (s_n, j_n).$$

A *segment* is such an admissible word together with its induced symbol path. A *return segment* is a segment whose initial and terminal symbols agree:

$$(s_0, j_0) = (s_n, j_n).$$

Return segments are useful because the initial and terminal odd numbers then have the same residue offset. Indeed, if

$$x_0 = 18m_0 + 6j_0 + \delta_{s_0}$$

and

$$x_n = 18m_n + 6j_n + \delta_{s_n},$$

then for a return segment,

$$j_n = j_0, \quad \delta_{s_n} = \delta_{s_0},$$

so

$$x_n > x_0 \iff m_n > m_0.$$

Thus, for return segments, reverse growth in the internal index is exactly the same as reverse growth in the odd value.

We work here at $p = 0$. Therefore each row contributes its base exponent α and its base constant

$$k^{(0)} = \frac{\beta + c}{9}.$$

For a return segment S of length n , define the total base exponent

$$A_S = \sum_{i=0}^{n-1} \alpha_i.$$

Also define the leading reverse-growth surplus

$$D_S = 2^{A_S} - 3^n.$$

The quantity D_S measures the multiplicative part of the reverse growth. If

$$D_S > 0,$$

then the segment has positive leading reverse-growth tendency. If

$$D_S < 0,$$

then the segment has negative leading reverse-growth tendency.

The accumulated carry term of the segment is

$$C_S^{(0)} = \sum_{i=0}^{n-1} 3^i 2^{\alpha_{i+1} + \cdots + \alpha_{n-1}} (j_{i+1} - k_i^{(0)}).$$

The word affine formula gives

$$m_0 = \frac{3^n m_n + C_S^{(0)}}{2^{A_S}}.$$

Therefore the segment grows in reverse precisely when

$$m_n > m_0.$$

Using the affine formula, this is equivalent to

$$C_S^{(0)} < (2^{A_S} - 3^n) m_n.$$

That is,

$$\boxed{C_S^{(0)} < D_S m_n.}$$

This is the proof criterion used in the table below. In particular, if

$$D_S > 0$$

and

$$C_S^{(0)} \leq 0,$$

then the return segment grows in reverse for every nonnegative terminal index m_n .

The following table lists, for each symbol, a shortest same-symbol return segment satisfying this reverse-growth criterion.

Table 14. Shortest $p = 0$ same-symbol return segments with reverse growth. Here n is the segment length, A_S is the sum of the base exponents, $D_S = 2^{A_S} - 3^n$, and $C_S^{(0)}$ is the accumulated carry term.

Start symbol	Shortest growing return segment	n	A_S	D_S	$C_S^{(0)}$
$(e, 0)$	Ψ_0	1	2	1	0
$(e, 1)$	Ψ_1	1	4	13	-5
$(e, 2)$	Ψ_2	1	6	61	-44
$(o, 0)$	Ω_0	1	5	29	-8
$(o, 1)$	Ω_1	1	3	5	-3
$(o, 2)$	Ω_2, Ω_1	2	4	7	-6

The table shows that five of the six symbols have a one-step growing return segment. For example,

$$(e, 2) \xrightarrow{\Psi_2} (e, 2)$$

has

$$A_S = 6, \quad n = 1,$$

so

$$D_S = 2^6 - 3 = 61 > 0.$$

Moreover,

$$C_S^{(0)} = -44,$$

so the reverse-growth inequality

$$C_S^{(0)} < D_S m_n$$

holds for every $m_n \geq 0$.

The exceptional symbol is $(o, 2)$. It has the one-step return

$$(o, 2) \xrightarrow{\Omega_2} (o, 2),$$

but this segment is not reverse-growing, because Ω_2 is one of the two exceptional $\alpha = 1$ rows. In this case

$$A_S = 1, \quad n = 1,$$

and therefore

$$D_S = 2^1 - 3^1 = -1.$$

Thus the one-step self-return at $(o, 2)$ has negative leading reverse-growth tendency.

However, $(o, 2)$ has a two-step growing return segment:

$$(o, 2) \xrightarrow{\Omega_2} (o, 1) \xrightarrow{\Omega_1} (o, 2).$$

The exponents are

$$1, 3,$$

so

$$A_S = 4, \quad n = 2.$$

Hence

$$D_S = 2^4 - 3^2 = 16 - 9 = 7 > 0.$$

The accumulated carry is

$$C_S^{(0)} = -6.$$

Therefore the growth inequality becomes

$$-6 < 7m_n,$$

which holds for every $m_n \geq 0$.

Disposal of the remaining ω -rows. We record why the lower-case ω -rows do not create a second persistent obstruction.

For $p \geq 1$, every row is strictly inverse-expansive, since the inverse multiplier is

$$\frac{2^{\alpha+6p}}{3},$$

and the factor 2^{6p} makes the multiplier larger than the corresponding $p = 0$ multiplier. Thus no lower-case ω -row with $p \geq 1$ can contribute to an anti-growth obstruction.

At $p = 0$, the rows ω_0 and ω_2 are already non-exceptional in the drift table. Hence they are handled by the ordinary inverse-growth argument.

The only lower-case row requiring comment is ω_1 . Its $p = 0$ inverse multiplier is

$$\frac{2}{3}.$$

Thus a single ω_1 -step may create local inverse anti-growth. Equivalently, if debt is measured as the reciprocal of inverse growth, then one such step contributes the debt factor

$$\left(\frac{2}{3}\right)^{-1} = \frac{3}{2}.$$

This is the same one-step debt factor as an exceptional $\sigma = 1$ step.

However, the important point is that ω_1 does not support an arbitrarily long self-loop analogous to the Ω_2 -run. The output classification table shows that a $p = 0$ ω_1 -transition does not generate a persistent block

$$\omega_1^k$$

of arbitrary length especially since it exits to family e. Thus ω_1 can contribute only a bounded local debt term, not a persistent exceptional obstruction.

Consequently, the only row requiring a separate run-length analysis is Ω_2 . The Ω_2 -row is exceptional not merely because a single step has inverse multiplier $2/3$, but because it can support long blocks

$$(o, 2)^k,$$

whose accumulated debt is

$$\left(\frac{3}{2}\right)^k.$$

The remaining analysis therefore focuses on the Ω_2 -self-loop and on its compensation by $(o, 1)$ -growth blocks.

We conclude that every symbol admits a same-symbol return segment with reverse growth. This does not yet prove global descent, because one must still prove that every actual orbit can be decomposed into, or routed into, such growing return segments. Nevertheless, the table gives a useful local certificate: at the symbol level, each state has a short return word whose inverse action grows.

13.2.4. The exceptional $(o, 2)$ obstruction. We now isolate the only remaining local obstruction in the $p = 0$ symbol system. The preceding return-segment analysis showed that every symbol has a same-symbol return segment with reverse growth. Five of the six symbols have a one-step growing return segment. The only symbol requiring further analysis is

$$(o, 2).$$

This subsection explains why the obstruction is confined to the residue class

$$x \equiv 17 \pmod{18},$$

and why the bad self-loop inside this class cannot persist indefinitely.

Recall that the state coordinates write odd integers in the form

$$x = 18m + 6j + \delta_s,$$

where

$$s \in \{e, o\}, \quad j \in \{0, 1, 2\},$$

and

$$\delta_e = 1, \quad \delta_o = 5.$$

Thus the symbol $(o, 2)$ corresponds to

$$s = o, \quad j = 2.$$

Therefore

$$x = 18m + 6(2) + 5.$$

Hence

$$x = 18m + 17.$$

So

$$(o, 2) \iff x \equiv 17 \pmod{18}.$$

Equivalently, this class may be described in two stages. Since $s = o$, we have

$$x \equiv 5 \pmod{6}.$$

Write

$$x = 6q + 5.$$

The router condition $j = 2$ says

$$q \equiv 2 \pmod{3}.$$

Thus

$$q = 3m + 2,$$

and therefore

$$x = 6(3m + 2) + 5 = 18m + 17.$$

So the two conditions

$$x \equiv 5 \pmod{6}$$

and

$$\left\lfloor \frac{x}{6} \right\rfloor \equiv 2 \pmod{3}$$

are exactly equivalent to

$$x \equiv 17 \pmod{18}.$$

The exceptional row at this symbol is

$$\Omega_2.$$

At $p = 0$, the row Ω_2 has

$$\alpha = 1, \quad k^{(0)} = 1.$$

Thus, at the level of the internal index,

$$F(0, m) = 2m + 1.$$

The output state is obtained by decomposing

$$F(0, m) = 3m' + j', \quad j' \in \{0, 1, 2\}.$$

For Ω_2 , the bad self-return is the case

$$(o, 2) \xrightarrow{\Omega_2} (o, 2).$$

This occurs precisely when

$$j' = 2.$$

Since

$$2m + 1 \equiv 2 \pmod{3}$$

is equivalent to

$$2m \equiv 1 \pmod{3},$$

and since $2^{-1} \equiv 2 \pmod{3}$, this is equivalent to

$$m \equiv 2 \pmod{3}.$$

Because in modular arithmetic,

$$2^{-1} \pmod{3}$$

means the multiplicative inverse of 2 modulo 3. That is, it is the residue a satisfying

$$2a \equiv 1 \pmod{3}.$$

Taking $a = 2$, we have

$$2 \cdot 2 = 4 \equiv 1 \pmod{3}.$$

Therefore

$$2^{-1} \equiv 2 \pmod{3}.$$

Thus 2 is its own inverse modulo 3. Consequently, dividing by 2 modulo 3 is the same as multiplying by 2. For example, if

$$2q \equiv 0 \pmod{3},$$

then multiplying both sides by $2^{-1} \equiv 2 \pmod{3}$ gives

$$q \equiv 0 \pmod{3}.$$

Similarly, if

$$2q \equiv 1 \pmod{3},$$

then

$$q \equiv 2^{-1} \equiv 2 \pmod{3}.$$

So, modulo 3,

$$\frac{1}{2} \equiv 2.$$

Therefore the Ω_2 self-loop occurs exactly for indices of the form

$$m = 3M + 2.$$

Substituting this into $F(0, m) = 2m + 1$, we get

$$F(0, m) = 2(3M + 2) + 1 = 6M + 5.$$

Thus

$$3m' + 2 = 6M + 5,$$

so

$$m' = 2M + 1.$$

Since

$$m = 3M + 2,$$

we may write this directly as

$$m' = \frac{2m - 1}{3}.$$

Hence the exceptional self-loop acts on the internal index by

$$\boxed{m \mapsto \frac{2m - 1}{3}}$$

whenever

$$m \equiv 2 \pmod{3}.$$

This recurrence has a simple stopping criterion. Set

$$y = m + 1.$$

Then

$$m' + 1 = \frac{2m - 1}{3} + 1 = \frac{2m + 2}{3} = \frac{2(m + 1)}{3}.$$

Therefore

$$\boxed{m' + 1 = \frac{2}{3}(m + 1)}.$$

After k consecutive Ω_2 self-loops, we obtain

$$m_k + 1 = \left(\frac{2}{3}\right)^k (m_0 + 1).$$

For this to remain an integer through k consecutive self-loops, it is necessary and sufficient that

$$3^k \mid m_0 + 1.$$

Thus

$$k \text{ consecutive } \Omega_2 \text{ self-loops} \iff 3^k \mid m_0 + 1.$$

Consequently, the exact number of consecutive Ω_2 self-loops is

$$\boxed{v_3(m_0 + 1)}.$$

In particular, the bad self-loop cannot persist indefinitely for any finite positive integer m_0 .

In terms of the original odd value, the $(o, 2)$ class is

$$x = 18m + 17.$$

Hence

$$x + 1 = 18m + 18 = 18(m + 1).$$

Since

$$18 = 2 \cdot 3^2,$$

we have

$$v_3(x+1) = 2 + v_3(m+1).$$

Therefore the number of possible consecutive Ω_2 self-loops can also be written as

$$\boxed{v_3(x+1) - 2}$$

for odd integers in the class

$$x \equiv 17 \pmod{18}.$$

This proves that the only local obstruction is sharply confined. The bad self-loop can occur only for numbers of the form

$$x = 18m + 17,$$

and, even there, it can occur only finitely many times. More precisely, it stops after

$$v_3(m+1)$$

iterations, or equivalently after

$$v_3(x+1) - 2$$

iterations in the original odd coordinate.

The corresponding growing return segment at $(o, 2)$ is

$$(o, 2) \xrightarrow{\Omega_2} (o, 1) \xrightarrow{\Omega_1} (o, 2).$$

At $p = 0$, the exponents are

$$1, 3.$$

Thus

$$A_S = 1 + 3 = 4, \quad |S| = 2.$$

The leading reverse-growth surplus is

$$D_S = 2^{A_S} - 3^{|S|} = 2^4 - 3^2 = 16 - 9 = 7.$$

The accumulated carry term is

$$C_S^{(0)} = -6.$$

Therefore the reverse-growth criterion

$$C_S^{(0)} < D_S m_n$$

becomes

$$-6 < 7m_n.$$

This holds for every nonnegative terminal index m_n . Hence the two-step segment

$$\Omega_2, \Omega_1$$

is unconditionally reverse-growing.

The situation is therefore as follows. The one-step Ω_2 self-loop is the only non-growing return at the symbol level, and it occurs only in the class

$$x \equiv 17 \pmod{18}.$$

Moreover, repeated use of this self-loop is controlled exactly by the finite valuation

$$v_3(m+1).$$

Once the orbit exits the self-loop, the symbol $(o, 2)$ has a short return segment with unconditional reverse growth. Thus the remaining global issue is not the existence of a growing return segment at $(o, 2)$, but the decomposition of actual orbits into such growing segments after any finite exceptional self-loop run has terminated.

13.2.5. *The output constraint from the $(o, 2)$ input class.* The preceding discussion showed that the exceptional symbol $(o, 2)$ corresponds to the residue class

$$x \equiv 17 \pmod{18}.$$

We now refine this further by asking how the Ω_2 row constrains the output symbol.

Let the input be in the $(o, 2)$ class. Then

$$x = 18m + 17.$$

For the row Ω_2 at $p = 0$, we have

$$\alpha = 1, \quad k^{(0)} = 1.$$

Therefore

$$F(0, m) = 2m + 1.$$

Since Ω_2 lands in the o -family, the output odd number is

$$x' = 6F(0, m) + 5.$$

Substituting the formula for $F(0, m)$, we get

$$x' = 6(2m + 1) + 5.$$

Hence

$$x' = 12m + 11.$$

Now decompose m modulo 3:

$$m = 3M + r, \quad r \in \{0, 1, 2\}.$$

Then

$$x' = 12(3M + r) + 11 = 36M + 12r + 11.$$

Thus the output residue modulo 18 depends on $r = m \bmod 3$.

If $r = 0$, then

$$x' = 36M + 11 \equiv 11 \pmod{18}.$$

If $r = 1$, then

$$x' = 36M + 23 \equiv 5 \pmod{18}.$$

If $r = 2$, then

$$x' = 36M + 35 \equiv 17 \pmod{18}.$$

Therefore

$$x' \equiv \begin{cases} 11 \pmod{18}, & m \equiv 0 \pmod{3}, \\ 5 \pmod{18}, & m \equiv 1 \pmod{3}, \\ 17 \pmod{18}, & m \equiv 2 \pmod{3}. \end{cases}$$

In symbol language, this says

$$(o, 2) \longrightarrow \begin{cases} (o, 1), & m \equiv 0 \pmod{3}, \\ (o, 0), & m \equiv 1 \pmod{3}, \\ (o, 2), & m \equiv 2 \pmod{3}. \end{cases}$$

Thus the exceptional self-loop

$$(o, 2) \xrightarrow{\Omega_2} (o, 2)$$

does not occur for every input in the class

$$x \equiv 17 \pmod{18}.$$

It occurs only in the subcase

$$m \equiv 2 \pmod{3}.$$

Since

$$x = 18m + 17,$$

the condition

$$m \equiv 2 \pmod{3}$$

is equivalent to

$$x \equiv 18(2) + 17 \pmod{54}.$$

Hence

$$x \equiv 53 \pmod{54}.$$

Therefore the actual self-loop obstruction is narrower than the full $(o, 2)$ class:

$$(o, 2) \text{ input class: } x \equiv 17 \pmod{18},$$

but

$$\Omega_2 \text{ self-loop subcase: } x \equiv 53 \pmod{54}.$$

This refinement is important. The class

$$x \equiv 17 \pmod{18}$$

is the only symbol class in which the exceptional Ω_2 row can appear. But the genuinely bad self-return

$$(o, 2) \rightarrow (o, 2)$$

requires the stronger congruence

$$x \equiv 53 \pmod{54}.$$

If this stronger congruence fails, then the Ω_2 row exits the $(o, 2)$ symbol immediately, landing instead in either $(o, 1)$ or $(o, 0)$, both of which have one-step growing return segments.

13.2.6. *Refining the exceptional $(o, 2)$ class.* The previous subsection isolated $(o, 2)$ as the only symbol requiring further analysis. We now refine this obstruction. The main point is that the exceptional behavior is not attached to all odd integers, nor even to all odd integers in the $(o, 2)$ class. Each additional exceptional self-loop imposes a stronger congruence condition, narrowing the possible input class by another factor of 3.

Recall that the symbol $(o, 2)$ corresponds to

$$x = 18m + 17.$$

Equivalently,

$$x \equiv 17 \pmod{18}.$$

This is the only symbol class in which the exceptional Ω_2 self-loop can appear.

At $p = 0$, the row Ω_2 has

$$\alpha = 1, \quad k^{(0)} = 1.$$

Thus

$$F(0, m) = 2m + 1.$$

The output state is determined by decomposing

$$F(0, m) = 3m' + j', \quad j' \in \{0, 1, 2\}.$$

Since

$$F(0, m) = 2m + 1,$$

we have

$$j' \equiv 2m + 1 \pmod{3}.$$

Now write

$$m = 3M + r, \quad r \in \{0, 1, 2\}.$$

Then

$$j' \equiv 2r + 1 \pmod{3}.$$

Therefore

$$r = 0 \implies j' = 1,$$

$$r = 1 \implies j' = 0,$$

and

$$r = 2 \implies j' = 2.$$

So the row Ω_2 sends

$$(o, 2) \longrightarrow \begin{cases} (o, 1), & m \equiv 0 \pmod{3}, \\ (o, 0), & m \equiv 1 \pmod{3}, \\ (o, 2), & m \equiv 2 \pmod{3}. \end{cases}$$

Thus the exceptional self-loop

$$(o, 2) \xrightarrow{\Omega_2} (o, 2)$$

occurs only in the subcase

$$m \equiv 2 \pmod{3}.$$

Since

$$x = 18m + 17,$$

the condition $m \equiv 2 \pmod{3}$ is equivalent to

$$x \equiv 18(2) + 17 \pmod{54}.$$

Hence

$$x \equiv 53 \pmod{54}.$$

Thus a single Ω_2 self-loop requires the sharper congruence

$$\boxed{x \equiv 53 \pmod{54}}.$$

The broader condition

$$x \equiv 17 \pmod{18}$$

places the number in the $(o, 2)$ class, but the actual self-loop requires the narrower condition

$$x \equiv 53 \pmod{54}.$$

13.2.7. Several consecutive Ω_2 self-loops. The congruence condition becomes stronger with each additional Ω_2 self-loop. In the self-loop case, we have

$$m = 3M + 2,$$

and the output index is

$$m' = 2M + 1.$$

Equivalently,

$$m' = \frac{2m - 1}{3}.$$

Adding 1 to both sides gives

$$m' + 1 = \frac{2m - 1}{3} + 1 = \frac{2m + 2}{3} = \frac{2(m + 1)}{3}.$$

Thus

$$\boxed{m' + 1 = \frac{2}{3}(m + 1)}.$$

After k consecutive Ω_2 self-loops, this gives

$$m_k + 1 = \left(\frac{2}{3}\right)^k (m_0 + 1).$$

For this to remain integral through k consecutive self-loops, one must have

$$3^k \mid m_0 + 1.$$

Conversely, if

$$3^k \mid m_0 + 1,$$

then the first k such divisions are possible. Hence

$$\boxed{k \text{ consecutive } \Omega_2 \text{ self-loops require } 3^k \mid m_0 + 1.}$$

In terms of the original odd value, since

$$x = 18m + 17,$$

we have

$$x + 1 = 18(m + 1).$$

Therefore

$$3^k \mid m + 1$$

is equivalent to

$$18 \cdot 3^k \mid x + 1.$$

So k consecutive Ω_2 self-loops require

$$\boxed{x \equiv -1 \pmod{18 \cdot 3^k}.$$

For the first few values of k , this gives

Number of consecutive Ω_2 self-loops	Required congruence for x
1	$x \equiv 53 \pmod{54}$
2	$x \equiv 161 \pmod{162}$
3	$x \equiv 485 \pmod{486}$
k	$x \equiv -1 \pmod{18 \cdot 3^k}.$

Thus each additional exceptional self-loop narrows the pool of possible odd integers by another factor of 3.

Equivalently, the exact number of consecutive Ω_2 self-loops is controlled by the 3-adic valuation:

$$\boxed{\#\{\text{consecutive } \Omega_2 \text{ self-loops}\} = v_3(m + 1).}$$

Since

$$x + 1 = 18(m + 1) = 2 \cdot 3^2(m + 1),$$

we also have

$$v_3(x + 1) = 2 + v_3(m + 1).$$

Therefore the same number may be written as

$$\boxed{v_3(x + 1) - 2.}$$

This proves that the exceptional self-loop cannot persist indefinitely for any finite positive integer. An infinite run would require

$$x \equiv -1 \pmod{18 \cdot 3^k}$$

for every k , which is impossible for an ordinary positive integer.

13.2.8. *Effect on the proof strategy.* The preceding refinement has an important consequence for the proof strategy. The local obstruction is not distributed across all odd integers. It is concentrated in the nested family of congruence classes

$$\begin{aligned} 17 \pmod{18} \supset 53 \pmod{54} \supset 161 \pmod{162} \\ \supset 485 \pmod{486} \supset \dots \end{aligned}$$

The first class

$$x \equiv 17 \pmod{18}$$

is the full $(o, 2)$ class. The next class

$$x \equiv 53 \pmod{54}$$

is the subclass that loops once under Ω_2 . The next class

$$x \equiv 161 \pmod{162}$$

is the subclass that loops twice, and so on.

Thus a long exceptional run can occur only for increasingly special numbers. The exceptional pool shrinks by a factor of 3 at each additional loop. In particular, the limiting exceptional set is empty among finite positive integers.

This does not by itself prove global descent, but it sharply narrows the remaining obstruction. The remaining task is no longer to control all odd integers uniformly. Rather, one must control finite exceptional runs of the form

$$(o, 2) \xrightarrow{\Omega_2} (o, 2) \xrightarrow{\Omega_2} \dots \xrightarrow{\Omega_2} (o, 2),$$

and then show that, once the run terminates, the subsequent admissible segment gives enough reverse growth to compensate for the finite exceptional shrinkage.

13.2.9. *Growth rate of ordinary and exceptional segments.* For a general word W of length n , the word formula is

$$m_0 = \frac{3^n m_n + C_W}{2^{\Sigma_W}}.$$

Solving for m_n , we get

$$m_n = \frac{2^{\Sigma_W}}{3^n} m_0 - \frac{C_W}{3^n}.$$

Thus the leading reverse-growth multiplier of W is

$$G_W = \frac{2^{\Sigma_W}}{3^n}.$$

At $p = 0$, this becomes

$$G_S^{(0)} = \frac{2^{A_S}}{3^{|S|}},$$

where

$$A_S = \sum_i \alpha_i.$$

For the shortest growing return segments listed above, the leading reverse-growth multipliers are

$$(e, 0) : \quad G = \frac{2^2}{3} = \frac{4}{3},$$

$$(e, 1) : \quad G = \frac{2^4}{3} = \frac{16}{3},$$

$$(e, 2) : \quad G = \frac{2^6}{3} = \frac{64}{3},$$

$$(o, 0) : \quad G = \frac{2^5}{3} = \frac{32}{3},$$

$$(o, 1) : \quad G = \frac{2^3}{3} = \frac{8}{3},$$

and

$$(o, 2) : \quad G = \frac{2^4}{3^2} = \frac{16}{9}.$$

The weakest of these growing return segments is the two-step $(o, 2)$ segment, with multiplier

$$\frac{16}{9} > 1.$$

By contrast, the exceptional Ω_2 self-loop has

$$A = 1, \quad n = 1,$$

and therefore

$$G_{\Omega_2} = \frac{2}{3} < 1.$$

So in the inverse direction, the exceptional self-loop shrinks. After k consecutive exceptional self-loops, the leading multiplier is

$$G_{\Omega_2^k} = \left(\frac{2}{3}\right)^k.$$

Equivalently, in the forward direction, such a run grows by the reciprocal factor

$$\left(\frac{3}{2}\right)^k.$$

If a run of k exceptional loops is followed by the two-step growing return segment

$$(o, 2) \xrightarrow{\Omega_2} (o, 1) \xrightarrow{\Omega_1} (o, 2),$$

then the combined leading reverse multiplier is

$$\left(\frac{2}{3}\right)^k \cdot \frac{16}{9}.$$

This is greater than 1 for $k = 0$ and $k = 1$, but not for all larger k . Thus a long exceptional run may require further compensating growth after it exits.

This gives the current precise form of the remaining problem. The exceptional run is finite and occurs only in increasingly thin congruence classes, but the proof must still show that the finite reverse shrinkage

$$\left(\frac{2}{3}\right)^k$$

is eventually compensated by subsequent reverse-growing segments. Once that compensation is proved, the corresponding forward orbit descends below the value at the beginning of the segment.

13.2.10. *Compensating exceptional shrinkage.* The row $(o, 1)$ is not itself an anti-growth obstruction. Its role is conservative: it is the weakest compensating o -family row available after a finite $(o, 2)$ -run. Thus the alternating

$$(o, 2)^{k_0}(o, 1)^{r_0}(o, 2)^{k_1}(o, 1)^{r_1} \dots$$

model is studied as the compensated exceptional segment. In this segment, the $(o, 2)$ -blocks create the exceptional debt, while the intervening $(o, 1)$ -blocks provide the weakest available compensation before a possible return to the exceptional $(o, 2)$ -state.

No assumption is made at this stage that the compensation supplied by the $(o, 1)$ -blocks is sufficient to repay the accumulated $(o, 2)$ -debt. The purpose of the compensated exceptional analysis is more basic: to show that an ordinary positive integer cannot support an infinite $(o, 2)/(o, 1)$ -segment. Thus the analysis rules out the only remaining mechanism by which the $(o, 2)$ -obstruction could persist indefinitely.

The exceptional Ω_2 self-loop has reverse multiplier

$$G_{\Omega_2} = \frac{2}{3}.$$

Thus a block of k consecutive exceptional self-loops has reverse multiplier

$$G_{\Omega_2^k} = \left(\frac{2}{3}\right)^k.$$

This is reverse shrinkage, not reverse growth. In the forward direction, the same block has growth factor

$$\left(\frac{3}{2}\right)^k.$$

Therefore a finite exceptional run of length k creates a growth debt: any subsequent reverse-growing segment must compensate for the factor

$$\left(\frac{2}{3}\right)^k$$

lost during the exceptional run.

Suppose that after the exceptional run exits the $(o, 2)$ self-loop, the orbit enters a compensating segment T . Let

$$G_T = \frac{2^{A_T}}{3^{|T|}}$$

be the leading reverse-growth multiplier of T . Then the combined leading multiplier of the exceptional run followed by T is

$$G_{\Omega_2^k T} = \left(\frac{2}{3}\right)^k G_T.$$

For this combined block to have positive leading reverse growth, we need

$$\left(\frac{2}{3}\right)^k G_T > 1.$$

Equivalently,

$$\boxed{G_T > \left(\frac{3}{2}\right)^k}.$$

Thus the compensating segment must have enough reverse growth to overcome the forward growth accumulated during the exceptional run.

One natural conservative choice is to use the guaranteed growing return at the exit symbol. If the exceptional run exits into $(o, 1)$, then the one-step return

$$(o, 1) \xrightarrow{\Omega_1} (o, 1)$$

has reverse multiplier

$$G_{\Omega_1} = \frac{2^3}{3} = \frac{8}{3}.$$

This is the weaker of the two immediate o -family exit returns, since

$$(o, 0) \xrightarrow{\Omega_0} (o, 0)$$

has multiplier

$$G_{\Omega_0} = \frac{2^5}{3} = \frac{32}{3}.$$

Therefore $(o, 1)$ gives the conservative exit-growth estimate.

If r such $(o, 1)$ -strength returns are used to compensate k exceptional self-loops, the leading multiplier is

$$\left(\frac{2}{3}\right)^k \left(\frac{8}{3}\right)^r.$$

The compensation condition is

$$\left(\frac{2}{3}\right)^k \left(\frac{8}{3}\right)^r > 1.$$

Equivalently,

$$\left(\frac{8}{3}\right)^r > \left(\frac{3}{2}\right)^k.$$

Taking logarithms gives

$$r \log \left(\frac{8}{3} \right) > k \log \left(\frac{3}{2} \right),$$

so it is enough that

$$r > k \frac{\log(3/2)}{\log(8/3)}.$$

Numerically,

$$\frac{\log(3/2)}{\log(8/3)} \approx 0.414.$$

Thus, at the level of leading multipliers, approximately

$$r \geq \lceil 0.414 k \rceil$$

$(o, 1)$ -strength growing returns are enough to compensate k exceptional Ω_2 self-loops.

For example, one $(o, 1)$ -return compensates one or two exceptional self-loops:

$$\left(\frac{2}{3} \right) \left(\frac{8}{3} \right) = \frac{16}{9} > 1,$$

and

$$\left(\frac{2}{3} \right)^2 \left(\frac{8}{3} \right) = \frac{32}{27} > 1.$$

However, one such return does not compensate three exceptional loops:

$$\left(\frac{2}{3} \right)^3 \left(\frac{8}{3} \right) = \frac{64}{81} < 1.$$

For $k = 3$, two $(o, 1)$ -strength returns suffice:

$$\left(\frac{2}{3} \right)^3 \left(\frac{8}{3} \right)^2 = \frac{512}{243} > 1.$$

This gives a quantitative form of the compensation problem. A longer exceptional run requires a longer compensating stretch afterward, but not one-for-one: because the $(o, 1)$ return grows by a factor $8/3$, one such return compensates more than two exceptional Ω_2 self-loops at the leading-multiplier level.

The exact word-level proof must still include the accumulated carry terms. The multiplier calculation gives the leading criterion, while the full reverse-growth criterion remains

$$C_W^{(0)} < \left(2^{A_W} - 3^{|W|} \right) m_n.$$

Since the known growing return segments have nonpositive accumulated carry, their carry terms help rather than obstruct reverse growth. Thus the leading compensation estimate above gives a conservative guide for how much post-exceptional growth is needed to overcome a finite exceptional run.

13.2.11. *Exit conditions and compensation after an exceptional $(o, 2)$ run.* We now summarize the exceptional $(o, 2)$ run in a form useful for the growth argument. The exceptional self-loop is the row

$$\Omega_2 : (o, 2) \rightarrow (o, 2)$$

at $p = 0$. In the $(o, 2)$ class we have

$$x = 18m + 17.$$

For Ω_2 , the base parameters are

$$\alpha = 1, \quad k^{(0)} = 1,$$

so

$$F(0, m) = 2m + 1.$$

The self-loop occurs when the output router is again $j' = 2$. Since

$$j' \equiv 2m + 1 \pmod{3},$$

the self-loop condition is

$$2m + 1 \equiv 2 \pmod{3}.$$

Equivalently,

$$m \equiv 2 \pmod{3}.$$

Thus

$$(o, 2) \xrightarrow{\Omega_2} (o, 2)$$

occurs exactly when

$$m \equiv 2 \pmod{3}.$$

In this case, write

$$m = 3M + 2.$$

Then

$$F(0, m) = 2(3M + 2) + 1 = 6M + 5.$$

Since the self-loop has $j' = 2$, we write

$$F(0, m) = 3m' + 2.$$

Therefore

$$3m' + 2 = 6M + 5,$$

so

$$m' = 2M + 1.$$

In terms of m , this is

$$m' = \frac{2m - 1}{3}.$$

Adding 1 gives the simpler recurrence

$$\boxed{m' + 1 = \frac{2}{3}(m + 1).}$$

After k consecutive Ω_2 self-loops, we therefore have

$$m_k + 1 = \left(\frac{2}{3}\right)^k (m_0 + 1).$$

For the expression to remain integral through k self-loops, it is necessary and sufficient that

$$3^k \mid m_0 + 1.$$

Hence

$$k \text{ consecutive } \Omega_2 \text{ self-loops require } m_0 \equiv -1 \pmod{3^k}.$$

An exact run of length k satisfies

$$v_3(m_0 + 1) = k.$$

Equivalently, there is an integer q with

$$3 \nmid q$$

such that

$$m_0 + 1 = 3^k q.$$

Thus

$$m_0 = 3^k q - 1, \quad 3 \nmid q.$$

After the k exceptional self-loops,

$$m_k + 1 = 2^k q,$$

so

$$m_k = 2^k q - 1.$$

The variable q is introduced to measure the exact 3-adic depth of the pure $(o, 2)$ -run.

A pure $(o, 2)$ -run of length k means that the state remains in the $(o, 2)$ -self-loop for k consecutive steps. The local $(o, 2)$ -condition is

$$m_i \equiv -1 \pmod{3},$$

or equivalently,

$$3 \mid m_i + 1.$$

Thus, for one $(o, 2)$ -step, we need

$$3 \mid m_0 + 1.$$

For k consecutive $(o, 2)$ -steps, this divisibility must persist through k successive updates. This forces

$$3^k \mid m_0 + 1.$$

Therefore we write

$$m_0 + 1 = 3^k q.$$

The condition

$$3 \nmid q$$

means that the run has exact length k . If q were still divisible by 3, then

$$3^{k+1} \mid m_0 + 1,$$

and the pure $(o, 2)$ -run would continue for at least one more step.

Now we derive the exit value

$$m_k = 2^k q - 1.$$

The $(o, 2)$ -self-loop updates the m -index by

$$m_{i+1} = \frac{2(m_i + 1)}{3} - 1.$$

Equivalently,

$$m_{i+1} + 1 = \frac{2(m_i + 1)}{3}.$$

Apply this repeatedly. After one step,

$$m_1 + 1 = \frac{2(m_0 + 1)}{3}.$$

Using

$$m_0 + 1 = 3^k q,$$

we obtain

$$m_1 + 1 = \frac{2 \cdot 3^k q}{3} = 2 \cdot 3^{k-1} q.$$

After two steps,

$$m_2 + 1 = \frac{2(m_1 + 1)}{3} = \frac{2 \cdot 2 \cdot 3^{k-1} q}{3} = 2^2 3^{k-2} q.$$

Continuing inductively, after i steps,

$$m_i + 1 = 2^i 3^{k-i} q.$$

Therefore, after k steps,

$$m_k + 1 = 2^k 3^{k-k} q = 2^k q.$$

Hence

$$m_k = 2^k q - 1.$$

Thus each $(o, 2)$ -step removes one factor of 3 from $m_i + 1$ and introduces one factor of 2. After k such steps, all k factors of 3 have been removed and replaced by k factors of 2:

$$3^k q \quad \mapsto \quad 2^k q.$$

This is the reason for the formula

$$m_k = 2^k q - 1.$$

In terms of the odd value

$$x_0 = 18m_0 + 17,$$

the condition becomes

$$x_0 + 1 = 18(m_0 + 1) = 18 \cdot 3^k q.$$

Therefore k consecutive exceptional self-loops require

$$\boxed{x_0 \equiv -1 \pmod{18 \cdot 3^k}.$$

For the first few values this gives

Number of consecutive Ω_2 loops	Required congruence for x_0
1	$x_0 \equiv 53 \pmod{54}$
2	$x_0 \equiv 161 \pmod{162}$
3	$x_0 \equiv 485 \pmod{486}$
k	$x_0 \equiv -1 \pmod{18 \cdot 3^k}.$

Thus each additional exceptional loop narrows the possible input class by another factor of 3.

Examples of exact exceptional runs. For an exact k -loop example, the simplest choice is

$$q = 1,$$

so

$$m_0 = 3^k - 1.$$

Then

$$m_i + 1 = 2^i 3^{k-i},$$

and hence

$$\boxed{m_i = 2^i 3^{k-i} - 1.}$$

The corresponding odd values are

$$x_i = 18m_i + 17 = 18(2^i 3^{k-i} - 1) + 17,$$

so

$$\boxed{x_i = 18 \cdot 2^i 3^{k-i} - 1.}$$

For $k = 1$,

$$m_0 = 3^1 - 1 = 2.$$

The exceptional segment is

$$2 \mapsto 1.$$

In odd values,

$$x_0 = 18(2) + 17 = 53, \quad x_1 = 18(1) + 17 = 35.$$

Thus

$$\boxed{53 \mapsto 35.}$$

For $k = 5$,

$$m_0 = 3^5 - 1 = 242.$$

The exceptional segment is

$$242 \mapsto 161 \mapsto 107 \mapsto 71 \mapsto 47 \mapsto 31.$$

In odd values this is

$$4373 \mapsto 2915 \mapsto 1943 \mapsto 1295 \mapsto 863 \mapsto 575.$$

For $k = 7$,

$$m_0 = 3^7 - 1 = 2186.$$

The exceptional segment is

$$2186 \mapsto 1457 \mapsto 971 \mapsto 647 \mapsto 431 \mapsto 287 \mapsto 191 \mapsto 127.$$

In odd values this is

$$\begin{aligned} 39365 &\mapsto 26243 \mapsto 17495 \mapsto 11663 \\ &\mapsto 7775 \mapsto 5183 \mapsto 3455 \mapsto 2303. \end{aligned}$$

These examples all use $q = 1$. In general, the terminal residue after an exact k -loop run is

$$m_k \equiv 2^k q - 1 \pmod{3}.$$

Since $3 \nmid q$, this terminal residue is either 0 or 1, but never 2. Therefore the exceptional run must exit the $(o, 2)$ self-loop. More precisely,

$$m_k \equiv 0 \pmod{3} \implies (o, 2) \rightarrow (o, 1),$$

whereas

$$m_k \equiv 1 \pmod{3} \implies (o, 2) \rightarrow (o, 0).$$

Thus every finite exceptional run exits to either $(o, 1)$ or $(o, 0)$.

Worst-case exit compensation. Both exit symbols have unconditional same-symbol growing returns. The $(o, 0)$ return

$$(o, 0) \xrightarrow{\Omega_0} (o, 0)$$

has leading reverse multiplier

$$G_{\Omega_0} = \frac{2^5}{3} = \frac{32}{3}.$$

The $(o, 1)$ return

$$(o, 1) \xrightarrow{\Omega_1} (o, 1)$$

has leading reverse multiplier

$$G_{\Omega_1} = \frac{2^3}{3} = \frac{8}{3}.$$

Since

$$\frac{8}{3} < \frac{32}{3},$$

the $(o, 1)$ exit gives the conservative, worst-case compensation estimate.

The exceptional block Ω_2^k has leading reverse multiplier

$$G_{\Omega_2^k} = \left(\frac{2}{3}\right)^k.$$

This is reverse shrinkage. In the forward direction, it corresponds to growth by

$$\left(\frac{3}{2}\right)^k.$$

If r $(o, 1)$ -strength growing returns are used to compensate k exceptional loops, the combined leading reverse multiplier is

$$\left(\frac{2}{3}\right)^k \left(\frac{8}{3}\right)^r.$$

For this to exceed 1, we need

$$\left(\frac{2}{3}\right)^k \left(\frac{8}{3}\right)^r > 1.$$

Equivalently,

$$\left(\frac{8}{3}\right)^r > \left(\frac{3}{2}\right)^k.$$

Taking logarithms gives

$$r \log\left(\frac{8}{3}\right) > k \log\left(\frac{3}{2}\right).$$

Therefore it is enough that

$$r > k \frac{\log(3/2)}{\log(8/3)}.$$

Numerically,

$$\frac{\log(3/2)}{\log(8/3)} \approx 0.414.$$

Thus, at the leading-multiplier level,

$$r \geq \lceil 0.414 k \rceil$$

$(o, 1)$ -strength growing returns compensate k exceptional $(o, 2)$ self-loops.

Equivalently, fewer than half as many $(o, 1)$ -strength growing returns are needed to compensate a run of exceptional $(o, 2)$ loops. For example,

$$k = 2 \implies r = 1 \text{ is enough,}$$

since

$$\left(\frac{2}{3}\right)^2 \left(\frac{8}{3}\right) = \frac{32}{27} > 1.$$

For $k = 3$, one return is not enough:

$$\left(\frac{2}{3}\right)^3 \left(\frac{8}{3}\right) = \frac{64}{81} < 1,$$

but two returns are enough:

$$\left(\frac{2}{3}\right)^3 \left(\frac{8}{3}\right)^2 = \frac{512}{243} > 1.$$

Similarly,

$$k = 5 \implies r = 3 \text{ is enough,}$$

and

$$k = 10 \implies r = 5 \text{ is enough.}$$

This gives a quantitative compensation principle. A long exceptional $(o, 2)$ run creates reverse shrinkage by the factor

$$\left(\frac{2}{3}\right)^k,$$

but the conservative $(o, 1)$ exit-growth multiplier

$$\frac{8}{3}$$

is strong enough that approximately 41.4% as many such growing returns compensate the loss at the leading-multiplier level. The exact word-level verification must still include the carry term, using

$$C_W^{(0)} < (2^{A_W} - 3^{|W|}) m_n,$$

but the known $(o, 1)$ and $(o, 0)$ returns have nonpositive carry, so their carry contributions assist rather than obstruct reverse growth.

Lemma 21 (Exit from a finite Ω_2 run). *Let an orbit remain in the exceptional self-loop*

$$(o, 2) \xrightarrow{\Omega_2} (o, 2)$$

for exactly $k \geq 1$ consecutive steps. Then, after the k -th self-loop, the next Ω_2 -transition cannot return to $(o, 2)$. Instead, it exits to either $(o, 0)$ or $(o, 1)$.

Proof. An exact k -step self-loop run satisfies

$$m_0 + 1 = 3^k q, \quad 3 \nmid q.$$

After k self-loops,

$$m_k + 1 = 2^k q,$$

so

$$m_k = 2^k q - 1.$$

Since $3 \nmid q$, the residue $m_k \bmod 3$ is either 0 or 1, but not 2. The self-loop $(o, 2) \rightarrow (o, 2)$ occurs exactly when

$$m \equiv 2 \pmod{3}.$$

Therefore the next Ω_2 -transition cannot be another self-loop. For Ω_2 , the routing is

$$m \equiv 0 \pmod{3} \implies (o, 2) \rightarrow (o, 1),$$

and

$$m \equiv 1 \pmod{3} \implies (o, 2) \rightarrow (o, 0).$$

Hence the exact finite run exits to either $(o, 1)$ or $(o, 0)$. $\square$

13.2.12. *No infinite positive Ω_2 self-loop.* We now record the finiteness of the exceptional Ω_2 self-loop. This is the key point that prevents the exceptional $(o, 2)$ behavior from persisting forever in the positive integers.

Recall that the exceptional self-loop is

$$(o, 2) \xrightarrow{\Omega_2} (o, 2).$$

At $p = 0$, the row Ω_2 has

$$\alpha = 1, \quad k^{(0)} = 1,$$

so

$$F(0, m) = 2m + 1.$$

The self-loop occurs precisely when

$$m \equiv 2 \pmod{3}.$$

In that case, writing

$$m = 3M + 2,$$

we get

$$F(0, m) = 2(3M + 2) + 1 = 6M + 5.$$

Since the output router is again $j' = 2$, we write

$$F(0, m) = 3m' + 2.$$

Thus

$$3m' + 2 = 6M + 5,$$

and hence

$$m' = 2M + 1.$$

In terms of m , this is

$$m' = \frac{2m - 1}{3}.$$

Adding 1 to both sides gives the simpler recurrence

$$\boxed{m' + 1 = \frac{2}{3}(m + 1)}.$$

Therefore, after k consecutive Ω_2 self-loops,

$$\boxed{m_k + 1 = \left(\frac{2}{3}\right)^k (m_0 + 1)}.$$

For the first k self-loops to be possible in the ordinary integers, one must have

$$3^k \mid m_0 + 1.$$

Equivalently,

$$m_0 \equiv -1 \pmod{3^k}.$$

Lemma 22 (No infinite positive Ω_2 self-loop). *No positive integer state can remain forever in the exceptional Ω_2 self-loop*

$$(o, 2) \xrightarrow{\Omega_2} (o, 2).$$

Equivalently, every positive-integer Ω_2 self-loop run is finite.

Proof. Suppose, for contradiction, that a positive integer state remains in the Ω_2 self-loop forever. Then for every $k \geq 1$, the first k self-loops are possible. Hence

$$3^k \mid m_0 + 1$$

for every $k \geq 1$.

The only ordinary integer divisible by every power of 3 is 0. Therefore

$$m_0 + 1 = 0,$$

so

$$m_0 = -1.$$

But $m_0 = -1$ does not correspond to a positive $(o, 2)$ state. Indeed, in the $(o, 2)$ class,

$$x_0 = 18m_0 + 17,$$

so

$$x_0 = 18(-1) + 17 = -1.$$

This is not a positive integer. Hence no positive integer can remain in the Ω_2 self-loop indefinitely. $\square$

Thus an infinite exceptional run is possible only as the formal fixed point

$$m = -1, \quad x = -1,$$

outside the positive Collatz domain. Indeed,

$$\frac{2(-1) - 1}{3} = -1.$$

So -1 is the non-positive limiting object associated with the exceptional self-loop, but it is not a positive integer state.

For every positive integer in the $(o, 2)$ class, the exceptional run therefore terminates after finitely many steps. More precisely, if

$$v_3(m_0 + 1) = k,$$

then the self-loop occurs exactly k times. After those k loops, we may write

$$m_0 + 1 = 3^k q, \quad 3 \nmid q.$$

Then

$$m_k + 1 = 2^k q,$$

so

$$m_k = 2^k q - 1.$$

Since $3 \nmid q$, we cannot have

$$m_k \equiv 2 \pmod{3}.$$

Indeed, if

$$m_k \equiv 2 \pmod{3},$$

then

$$2^k q - 1 \equiv 2 \pmod{3},$$

so

$$2^k q \equiv 0 \pmod{3},$$

forcing

$$q \equiv 0 \pmod{3},$$

a contradiction.

Hence

$$m_k \equiv 0 \pmod{3} \quad \text{or} \quad m_k \equiv 1 \pmod{3}.$$

Under the Ω_2 routing,

$$m_k \equiv 0 \pmod{3} \implies (o, 2) \rightarrow (o, 1),$$

and

$$m_k \equiv 1 \pmod{3} \implies (o, 2) \rightarrow (o, 0).$$

Therefore every finite positive Ω_2 self-loop run exits to one of the two good symbols:

$$\boxed{(o, 0) \quad \text{or} \quad (o, 1)}.$$

Combining these observations, the exceptional obstruction has the following exact form:

$$\boxed{\text{A positive } (o, 2) \text{ state may loop under } \Omega_2 \text{ only finitely many times,}}$$

and after the finite run terminates, it must exit to either $(o, 0)$ or $(o, 1)$. Thus the exceptional self-loop cannot produce an infinite positive-integer shrink path to 1, to 0, or to $-\infty$; within the positive-integer system, the infinite exceptional path does not exist.

Lemma 23 (Logarithmic bound on exceptional $(o, 2)$ runs). *Let*

$$x_0 = 18m_0 + 17$$

be a positive odd integer in the $(o, 2)$ class. Suppose that the Ω_2 self-loop

$$(o, 2) \xrightarrow{\Omega_2} (o, 2)$$

occurs for k consecutive steps. Then

$$3^k \mid m_0 + 1.$$

Consequently,

$$k \leq \lfloor \log_3(m_0 + 1) \rfloor.$$

Equivalently, since

$$m_0 + 1 = \frac{x_0 + 1}{18},$$

we have

$$k \leq \left\lfloor \log_3 \left(\frac{x_0 + 1}{18} \right) \right\rfloor.$$

Thus the length of an exceptional $(o, 2)$ run is bounded logarithmically in the starting value.

Proof. For the exceptional Ω_2 self-loop at $p = 0$, the internal index satisfies

$$m_{i+1} + 1 = \frac{2}{3}(m_i + 1).$$

After k consecutive self-loops, this gives

$$m_k + 1 = \left(\frac{2}{3} \right)^k (m_0 + 1).$$

For m_k to be an integer after k such steps, it is necessary that

$$3^k \mid m_0 + 1.$$

Therefore

$$3^k \leq m_0 + 1.$$

Taking logarithms base 3, we get

$$k \leq \log_3(m_0 + 1).$$

Since k is an integer,

$$k \leq \lfloor \log_3(m_0 + 1) \rfloor.$$

Finally, because the $(o, 2)$ class has

$$x_0 = 18m_0 + 17,$$

we have

$$x_0 + 1 = 18(m_0 + 1),$$

and hence

$$m_0 + 1 = \frac{x_0 + 1}{18}.$$

Substituting this into the previous bound gives

$$k \leq \left\lfloor \log_3 \left(\frac{x_0 + 1}{18} \right) \right\rfloor.$$

□

The bound is sharp in the following sense. For any $k \geq 1$, choose

$$m_0 = 3^k - 1.$$

Then

$$m_0 + 1 = 3^k,$$

so

$$v_3(m_0 + 1) = k.$$

Thus this choice gives an exact k -step exceptional $(o, 2)$ run. The corresponding odd value is

$$x_0 = 18(3^k - 1) + 17 = 18 \cdot 3^k - 1.$$

Therefore there is no universal finite maximum length for exceptional $(o, 2)$ runs over all positive integers. However, for any fixed size bound $x_0 \leq X$, the length satisfies

$$k \leq \left\lfloor \log_3 \left(\frac{X+1}{18} \right) \right\rfloor.$$

Numerical size bounds. The logarithmic bound also shows that very large numerical ranges can contain only modest-length exceptional runs. If

$$x_0 \leq X,$$

then

$$k \leq \left\lfloor \log_3 \left(\frac{X+1}{18} \right) \right\rfloor.$$

For example,

$$X = 10^6 \implies k \leq \left\lfloor \log_3 \left(\frac{10^6 + 1}{18} \right) \right\rfloor = 9.$$

Similarly,

$$X = 10^{12} \implies k \leq \left\lfloor \log_3 \left(\frac{10^{12} + 1}{18} \right) \right\rfloor = 22,$$

and

$$X = 10^{18} \implies k \leq \left\lfloor \log_3 \left(\frac{10^{18} + 1}{18} \right) \right\rfloor = 35.$$

Thus even below 10^{18} , an exceptional $(o, 2)$ run can have length at most 35. The possible length grows only logarithmically with the starting value. So exceptional runs can be arbitrarily long in principle, but their length grows only logarithmically with the starting value.

13.2.13. *A growth-debt budget for exceptional runs.* The exceptional Ω_2 self-loop creates reverse shrinkage. A run of k consecutive exceptional loops has leading reverse multiplier

$$G_{\Omega_2^k} = \left(\frac{2}{3} \right)^k.$$

Equivalently, in the forward direction, the same block has growth factor

$$\left(\frac{3}{2} \right)^k.$$

It is therefore useful to regard a k -step exceptional run as creating a *growth debt*

$$B_k = \left(\frac{3}{2} \right)^k.$$

This debt measures the amount of subsequent reverse growth required to compensate for the shrinkage caused by the exceptional block.

If a later segment has leading reverse multiplier $G > 1$, then it reduces the outstanding debt by a factor of G :

$$B \mapsto \frac{B}{G}.$$

The debt is cleared once

$$B < 1.$$

After a finite Ω_2 self-loop run, the orbit exits to either $(o, 0)$ or $(o, 1)$. The corresponding one-step growing return multipliers are

$$(o, 0) : \quad G_{\Omega_0} = \frac{32}{3},$$

and

$$(o, 1) : \quad G_{\Omega_1} = \frac{8}{3}.$$

Since

$$\frac{8}{3} < \frac{32}{3},$$

the $(o, 1)$ exit gives the conservative worst-case compensation rate.

Suppose therefore that we compensate a k -step exceptional run using r worst-case $(o, 1)$ -strength returns. The remaining debt is

$$B_{k,r} = \frac{(3/2)^k}{(8/3)^r}.$$

The debt is cleared when

$$B_{k,r} < 1.$$

Equivalently,

$$\frac{(3/2)^k}{(8/3)^r} < 1,$$

or

$$\left(\frac{8}{3}\right)^r > \left(\frac{3}{2}\right)^k.$$

Taking logarithms gives

$$r \log \left(\frac{8}{3}\right) > k \log \left(\frac{3}{2}\right),$$

so it is enough that

$$\boxed{r > k \frac{\log(3/2)}{\log(8/3)}}.$$

Numerically,

$$\frac{\log(3/2)}{\log(8/3)} \approx 0.414.$$

Thus, at the level of leading multipliers, approximately 41.4% as many $(o, 1)$ -strength returns as exceptional $(o, 2)$ loops are enough to compensate the exceptional shrinkage.

For a cleaner sufficient bound, we may use

$$r = \left\lceil \frac{k}{2} \right\rceil.$$

Since

$$\frac{1}{2} > \frac{\log(3/2)}{\log(8/3)},$$

this choice is safely sufficient. Therefore

$$\left(\frac{8}{3}\right)^{\lceil k/2 \rceil} > \left(\frac{3}{2}\right)^k,$$

and hence

$$\frac{(3/2)^k}{(8/3)^{\lceil k/2 \rceil}} < 1.$$

Lemma 24 (Worst-case compensation budget). *Let a positive orbit contain a finite exceptional block*

$$\Omega_2^k.$$

At the leading-multiplier level, this block creates growth debt

$$B_k = \left(\frac{3}{2}\right)^k.$$

Then

$$\left\lceil \frac{k}{2} \right\rceil$$

worst-case $(o, 1)$ -strength growing returns are sufficient to clear this debt.

Proof. Each exceptional Ω_2 self-loop has reverse multiplier

$$\frac{2}{3},$$

so k such loops have reverse multiplier

$$\left(\frac{2}{3}\right)^k.$$

Equivalently, they create forward growth debt

$$B_k = \left(\frac{3}{2}\right)^k.$$

The conservative post-exception return multiplier is the $(o, 1)$ value

$$G_{\Omega_1} = \frac{8}{3}.$$

After r such returns, the remaining debt is

$$B_{k,r} = \frac{(3/2)^k}{(8/3)^r}.$$

It is enough to show that this is < 1 for

$$r = \left\lceil \frac{k}{2} \right\rceil.$$

Since

$$\left\lceil \frac{k}{2} \right\rceil \geq \frac{k}{2},$$

we have

$$\left(\frac{8}{3}\right)^{\lceil k/2 \rceil} \geq \left(\frac{8}{3}\right)^{k/2}.$$

Thus it is enough to check

$$\left(\frac{8}{3}\right)^{k/2} > \left(\frac{3}{2}\right)^k.$$

Taking the k -th root, this is equivalent to

$$\sqrt{\frac{8}{3}} > \frac{3}{2}.$$

Squaring both sides gives

$$\frac{8}{3} > \frac{9}{4}.$$

Multiplying by 12, this becomes

$$32 > 27,$$

which is true. Therefore

$$\left(\frac{8}{3}\right)^{\lceil k/2 \rceil} > \left(\frac{3}{2}\right)^k,$$

and so

$$B_{k,\lceil k/2 \rceil} < 1.$$

Hence $\lceil k/2 \rceil$ worst-case $(o, 1)$ -strength returns clear the growth debt created by k exceptional Ω_2 loops. $\square$

For example,

$$k = 2 \implies \left\lceil \frac{k}{2} \right\rceil = 1,$$

and indeed

$$\left(\frac{2}{3}\right)^2 \left(\frac{8}{3}\right) = \frac{32}{27} > 1.$$

For

$$k = 3,$$

one $(o, 1)$ -strength return is not enough:

$$\left(\frac{2}{3}\right)^3 \left(\frac{8}{3}\right) = \frac{64}{81} < 1.$$

But

$$\left\lceil \frac{3}{2} \right\rceil = 2,$$

and two such returns give

$$\left(\frac{2}{3}\right)^3 \left(\frac{8}{3}\right)^2 = \frac{512}{243} > 1.$$

Similarly,

$$k = 5 \implies \left\lceil \frac{k}{2} \right\rceil = 3,$$

and

$$k = 10 \implies \left\lceil \frac{k}{2} \right\rceil = 5.$$

This debt-budget lemma is still a leading-multiplier result. The exact word-level inequality remains

$$C_W^{(0)} < (2^{A_W} - 3^{|W|}) m_n.$$

However, the relevant $(o, 1)$ and $(o, 0)$ return segments have nonpositive carry terms, so their carry contributions assist the reverse-growth inequality rather than obstructing it.

13.2.14. Debt compensation after an exceptional run. We now formalize the compensation principle suggested by the exceptional $(o, 2)$ analysis. A run of k consecutive exceptional Ω_2 self-loops has leading reverse multiplier

$$G_{\Omega_2^k} = \left(\frac{2}{3}\right)^k.$$

Thus, in reverse, the exceptional block shrinks by the factor

$$\left(\frac{2}{3}\right)^k.$$

Equivalently, it creates a forward-growth debt of

$$B_k = \left(\frac{3}{2}\right)^k.$$

Once the exceptional run exits, it enters either $(o, 0)$ or $(o, 1)$. The corresponding same-symbol growing returns have leading reverse multipliers

$$G_{\Omega_0} = \frac{32}{3}, \quad G_{\Omega_1} = \frac{8}{3}.$$

Since

$$\frac{8}{3} < \frac{32}{3},$$

the $(o, 1)$ return gives the conservative worst-case compensation rate. Therefore, to prove a worst-case compensation estimate, it is enough to show that sufficiently many $(o, 1)$ -strength returns overcome the debt created by the exceptional Ω_2 -run.

Lemma 25 (Debt compensation). *Let $k \geq 1$. A block of k exceptional Ω_2 self-loops creates growth debt*

$$B_k = \left(\frac{3}{2}\right)^k.$$

Then

$$r = \left\lceil \frac{k}{2} \right\rceil$$

worst-case $(o, 1)$ -strength growing returns are sufficient to clear this debt. Equivalently,

$$\left(\frac{2}{3}\right)^k \left(\frac{8}{3}\right)^{\lceil k/2 \rceil} > 1.$$

Proof. The exceptional block Ω_2^k has leading reverse multiplier

$$\left(\frac{2}{3}\right)^k.$$

Each $(o, 1)$ -strength compensating return has leading reverse multiplier

$$\frac{8}{3}.$$

Thus r such compensating returns produce combined multiplier

$$\left(\frac{2}{3}\right)^k \left(\frac{8}{3}\right)^r.$$

We want this to be greater than 1.

Set

$$r = \left\lceil \frac{k}{2} \right\rceil.$$

Since

$$\left\lceil \frac{k}{2} \right\rceil \geq \frac{k}{2},$$

we have

$$\left(\frac{8}{3}\right)^{\lceil k/2 \rceil} \geq \left(\frac{8}{3}\right)^{k/2}.$$

Therefore it is enough to prove

$$\left(\frac{2}{3}\right)^k \left(\frac{8}{3}\right)^{k/2} > 1.$$

Rewrite the left-hand side as

$$\left[\frac{2}{3} \left(\frac{8}{3} \right)^{1/2} \right]^k.$$

Thus it is enough to show

$$\frac{2}{3} \sqrt{\frac{8}{3}} > 1.$$

Equivalently,

$$\sqrt{\frac{8}{3}} > \frac{3}{2}.$$

Squaring both sides gives

$$\frac{8}{3} > \frac{9}{4}.$$

Multiplying by 12, this becomes

$$32 > 27,$$

which is true. Hence

$$\left(\frac{2}{3} \right)^k \left(\frac{8}{3} \right)^{\lceil k/2 \rceil} > 1.$$

Therefore $\lceil k/2 \rceil$ worst-case $(o, 1)$ -strength returns clear the leading growth debt created by k exceptional Ω_2 self-loops. $\square$

Equivalently, one may express the exact threshold by solving

$$\left(\frac{8}{3} \right)^r > \left(\frac{3}{2} \right)^k.$$

Taking logarithms gives

$$r > k \frac{\log(3/2)}{\log(8/3)}.$$

Since

$$\frac{\log(3/2)}{\log(8/3)} \approx 0.414,$$

approximately 41.4% as many $(o, 1)$ -strength returns as exceptional $(o, 2)$ loops are enough at the leading-multiplier level. The simpler bound

$$r = \left\lceil \frac{k}{2} \right\rceil$$

is slightly stronger but easier to use.

This lemma is a leading-multiplier statement. The exact word-level criterion remains

$$C_W^{(0)} < (2^{A_W} - 3^{|W|}) m_n.$$

However, the relevant $(o, 1)$ and $(o, 0)$ return segments have nonpositive carry terms, so their accumulated carry contributions assist rather than obstruct reverse growth.

13.2.15. *Residue-level transitions from the exit symbols.* After a finite exceptional Ω_2 self-loop run, the orbit exits from $(o, 2)$ to either $(o, 0)$ or $(o, 1)$. We therefore examine the exact residue-level behavior of these two exit symbols.

Throughout this subsection we work at $p = 0$. Write

$$m = 3M + r, \quad r \in \{0, 1, 2\}.$$

For a row with base exponent α and base constant $k^{(0)}$, the output router is determined by

$$j' \equiv 2^\alpha r + k^{(0)} \pmod{3}.$$

The output index has the form

$$m' = 2^\alpha M + t_r,$$

where t_r is the carry determined by the corresponding row.

For the exit symbol $(o, 0)$, the relevant $p = 0$ rows are

$$\omega_0 : (o, 0) \rightarrow e, \quad \Omega_0 : (o, 0) \rightarrow o.$$

For the exit symbol $(o, 1)$, the relevant $p = 0$ rows are

$$\omega_1 : (o, 1) \rightarrow e, \quad \Omega_1 : (o, 1) \rightarrow o.$$

The resulting residue-level transitions are shown in Table 15.

Table 15. Residue-level $p = 0$ transitions from the exit symbols $(o, 0)$ and $(o, 1)$, where $r = m \bmod 3$.

Input	Row	r	Output
$(o, 0)$	ω_0	0	$(e, 2)$
$(o, 0)$	ω_0	1	$(e, 1)$
$(o, 0)$	ω_0	2	$(e, 0)$
$(o, 0)$	Ω_0	0	$(o, 2)$
$(o, 0)$	Ω_0	1	$(o, 1)$
$(o, 0)$	Ω_0	2	$(o, 0)$
$(o, 1)$	ω_1	0	$(e, 1)$
$(o, 1)$	ω_1	1	$(e, 0)$
$(o, 1)$	ω_1	2	$(e, 2)$
$(o, 1)$	Ω_1	0	$(o, 1)$
$(o, 1)$	Ω_1	1	$(o, 0)$
$(o, 1)$	Ω_1	2	$(o, 2)$

For $(o, 0)$, the same-symbol growing return is

$$(o, 0) \xrightarrow{\Omega_0} (o, 0).$$

From the table, this occurs when

$$r = 2, \quad m = 3M + 2.$$

For Ω_0 , the base exponent is

$$\alpha = 5,$$

and the output index in the $r = 2$ case is

$$m' = 32M + 24.$$

Therefore

$$m' - m = (32M + 24) - (3M + 2) = 29M + 22 > 0.$$

Thus this same-symbol return is strongly reverse-growing.

For $(o, 1)$, the same-symbol growing return is

$$(o, 1) \xrightarrow{\Omega_1} (o, 1).$$

From the table, this occurs when

$$r = 0, \quad m = 3M.$$

For Ω_1 , the base exponent is

$$\alpha = 3,$$

and the output index in the $r = 0$ case is

$$m' = 8M + 1.$$

Therefore

$$m' - m = (8M + 1) - 3M = 5M + 1 > 0.$$

Thus this same-symbol return is also reverse-growing.

The table has two important consequences. First, after a finite exceptional Ω_2 run exits to $(o, 0)$ or $(o, 1)$, both exit symbols have available same-symbol reverse-growing returns. Second, these same-symbol returns are residue-specific:

$$(o, 0) \xrightarrow{\Omega_0} (o, 0) \quad \text{requires} \quad m \equiv 2 \pmod{3},$$

whereas

$$(o, 1) \xrightarrow{\Omega_1} (o, 1) \quad \text{requires} \quad m \equiv 0 \pmod{3}.$$

Thus the next coverage problem is residue-level: after exiting an exceptional $(o, 2)$ run, one must determine which residues occur and how quickly they route into one of the available growing returns.

13.2.16. *Separate compensation bounds for the two exit symbols.* After a finite exceptional Ω_2 self-loop run, the orbit exits from $(o, 2)$ to either $(o, 0)$ or $(o, 1)$. These two exits have different compensating strengths. The $(o, 0)$ exit is stronger, while the $(o, 1)$ exit is the conservative worst case.

A block of k exceptional Ω_2 self-loops creates growth debt

$$B_k = \left(\frac{3}{2}\right)^k.$$

A compensating return segment with reverse multiplier $G > 1$ reduces this debt by a factor of G . Thus r such returns clear the debt provided

$$G^r > \left(\frac{3}{2}\right)^k.$$

The $(o, 0)$ exit. The same-symbol return

$$(o, 0) \xrightarrow{\Omega_0} (o, 0)$$

has leading reverse multiplier

$$G_{\Omega_0} = \frac{2^5}{3} = \frac{32}{3}.$$

Thus r $(o, 0)$ -strength returns compensate k exceptional Ω_2 self-loops if

$$\left(\frac{32}{3}\right)^r > \left(\frac{3}{2}\right)^k.$$

Taking logarithms gives the exact threshold

$$r > k \frac{\log(3/2)}{\log(32/3)}.$$

Numerically,

$$\frac{\log(3/2)}{\log(32/3)} \approx 0.171.$$

So approximately 17.1% as many $(o, 0)$ -strength returns as exceptional $(o, 2)$ loops are enough at the leading-multiplier level.

For a simple integer bound, it is enough to take

$$r = \left\lceil \frac{k}{5} \right\rceil.$$

Lemma 26 ($(o, 0)$ -exit compensation). *Let $k \geq 1$. A block of k exceptional Ω_2 self-loops creates growth debt*

$$B_k = \left(\frac{3}{2}\right)^k.$$

Then

$$\left\lceil \frac{k}{5} \right\rceil$$

$(o, 0)$ -strength returns are sufficient to clear this debt. Equivalently,

$$\left(\frac{2}{3}\right)^k \left(\frac{32}{3}\right)^{\lceil k/5 \rceil} > 1.$$

Proof. It is enough to show that

$$\left(\frac{32}{3}\right)^{\lceil k/5 \rceil} > \left(\frac{3}{2}\right)^k.$$

Since

$$\left\lceil \frac{k}{5} \right\rceil \geq \frac{k}{5},$$

we have

$$\left(\frac{32}{3}\right)^{\lceil k/5 \rceil} \geq \left(\frac{32}{3}\right)^{k/5}.$$

Thus it is enough to prove

$$\left(\frac{32}{3}\right)^{k/5} > \left(\frac{3}{2}\right)^k.$$

Taking the k -th root, this is equivalent to

$$\left(\frac{32}{3}\right)^{1/5} > \frac{3}{2}.$$

Raising both sides to the fifth power, this becomes

$$\frac{32}{3} > \left(\frac{3}{2}\right)^5.$$

Since

$$\left(\frac{3}{2}\right)^5 = \frac{243}{32},$$

the desired inequality is

$$\frac{32}{3} > \frac{243}{32}.$$

Multiplying by 96, this becomes

$$1024 > 729,$$

which is true. Therefore

$$\left(\frac{32}{3}\right)^{\lceil k/5 \rceil} > \left(\frac{3}{2}\right)^k,$$

so the debt is cleared. □

The $(o, 1)$ exit. The same-symbol return

$$(o, 1) \xrightarrow{\Omega_1} (o, 1)$$

has leading reverse multiplier

$$G_{\Omega_1} = \frac{2^3}{3} = \frac{8}{3}.$$

This is weaker than the $(o, 0)$ return, so it gives the conservative worst-case exit compensation.

Thus r $(o, 1)$ -strength returns compensate k exceptional Ω_2 self-loops if

$$\left(\frac{8}{3}\right)^r > \left(\frac{3}{2}\right)^k.$$

Taking logarithms gives

$$r > k \frac{\log(3/2)}{\log(8/3)}.$$

Numerically,

$$\frac{\log(3/2)}{\log(8/3)} \approx 0.414.$$

So approximately 41.4% as many $(o, 1)$ -strength returns as exceptional $(o, 2)$ loops are enough at the leading-multiplier level.

For a clean integer bound, it is enough to take

$$r = \left\lceil \frac{k}{2} \right\rceil.$$

Lemma 27 ($(o, 1)$ -exit compensation). *Let $k \geq 1$. A block of k exceptional Ω_2 self-loops creates growth debt*

$$B_k = \left(\frac{3}{2}\right)^k.$$

Then

$$\left\lceil \frac{k}{2} \right\rceil$$

$(o, 1)$ -strength returns are sufficient to clear this debt. Equivalently,

$$\left(\frac{2}{3}\right)^k \left(\frac{8}{3}\right)^{\lceil k/2 \rceil} > 1.$$

Proof. It is enough to show that

$$\left(\frac{8}{3}\right)^{\lceil k/2 \rceil} > \left(\frac{3}{2}\right)^k.$$

Since

$$\left\lceil \frac{k}{2} \right\rceil \geq \frac{k}{2},$$

we have

$$\left(\frac{8}{3}\right)^{\lceil k/2 \rceil} \geq \left(\frac{8}{3}\right)^{k/2}.$$

Thus it is enough to prove

$$\left(\frac{8}{3}\right)^{k/2} > \left(\frac{3}{2}\right)^k.$$

Taking the k -th root, this is equivalent to

$$\sqrt{\frac{8}{3}} > \frac{3}{2}.$$

Squaring both sides gives

$$\frac{8}{3} > \frac{9}{4}.$$

Multiplying by 12, this becomes

$$32 > 27,$$

which is true. Therefore

$$\left(\frac{8}{3}\right)^{\lceil k/2 \rceil} > \left(\frac{3}{2}\right)^k,$$

so the debt is cleared. $\square$

The two lemmas show that the $(o, 0)$ exit is much easier to compensate than the $(o, 1)$ exit:

$$(o, 0) : \quad r = \left\lceil \frac{k}{5} \right\rceil,$$

whereas

$$(o, 1) : \quad r = \left\lceil \frac{k}{2} \right\rceil.$$

Thus the $(o, 1)$ exit remains the conservative worst case for the post-exception compensation argument.

As before, these are leading-multiplier estimates. The exact word-level criterion is still

$$C_W^{(0)} < \left(2^{A_W} - 3^{|W|}\right) m_n.$$

However, the relevant $(o, 0)$ and $(o, 1)$ same-symbol returns have nonpositive carry contributions, so their carry terms assist reverse growth rather than obstruct it.

13.2.17. *Exact structure of $(o, 1)$ -runs.* We now analyze the $(o, 1)$ loop in the same way that we analyzed the exceptional $(o, 2)$ loop. The important difference is that the $(o, 1)$ loop is not an obstruction: it is a reverse-growing loop. It therefore acts as a compensating mechanism for the debt created by exceptional $(o, 2)$ runs.

The same-symbol $(o, 1)$ return is

$$(o, 1) \xrightarrow{\Omega_1} (o, 1).$$

At $p = 0$, the row Ω_1 has

$$\alpha = 3, \quad k^{(0)} = 4.$$

Thus

$$F(0, m) = 8m + 4.$$

The output router is obtained from

$$F(0, m) = 3m' + j', \quad j' \in \{0, 1, 2\}.$$

Since

$$j' \equiv 8m + 4 \pmod{3},$$

and

$$8 \equiv 2 \pmod{3}, \quad 4 \equiv 1 \pmod{3},$$

we have

$$j' \equiv 2m + 1 \pmod{3}.$$

The $(o, 1)$ self-return occurs exactly when

$$j' = 1.$$

Therefore

$$2m + 1 \equiv 1 \pmod{3},$$

which is equivalent to

$$m \equiv 0 \pmod{3}.$$

Thus

$$(o, 1) \xrightarrow{\Omega_1} (o, 1) \iff m \equiv 0 \pmod{3}.$$

If

$$m = 3M,$$

then

$$F(0, m) = 8(3M) + 4 = 24M + 4.$$

Since the output router is $j' = 1$, we write

$$F(0, m) = 3m' + 1.$$

Hence

$$3m' + 1 = 24M + 4,$$

so

$$m' = 8M + 1.$$

In terms of m , this is

$$m' = \frac{8m + 3}{3}.$$

This affine recurrence has a useful shifted form. Starting from

$$m' = \frac{8m + 3}{3},$$

we compute

$$5m' + 3 = 5 \left(\frac{8m + 3}{3} \right) + 3 = \frac{40m + 15 + 9}{3} = \frac{40m + 24}{3}.$$

Therefore

$$5m' + 3 = \frac{8}{3}(5m + 3).$$

Lemma 28 (Exact $(o, 1)$ -run formula). *Suppose an orbit remains in the $(o, 1)$ same-symbol loop*

$$(o, 1) \xrightarrow{\Omega_1} (o, 1)$$

for r consecutive steps. Then

$$5m_r + 3 = \left(\frac{8}{3} \right)^r (5m_0 + 3).$$

In particular, r consecutive $(o, 1)$ loops are possible only if

$$3^r \mid 5m_0 + 3.$$

Proof. For one $(o, 1)$ self-return, we have shown that

$$5m_{i+1} + 3 = \frac{8}{3}(5m_i + 3).$$

Iterating this identity r times gives

$$5m_r + 3 = \left(\frac{8}{3} \right)^r (5m_0 + 3).$$

For m_r to be an integer after r such steps, the denominator 3^r must divide $5m_0 + 3$. Hence

$$3^r \mid 5m_0 + 3.$$

□

The exact length of an $(o, 1)$ -run is therefore measured by the 3-adic valuation of $5m + 3$.

Lemma 29 (Exact length of an $(o, 1)$ -run). *An $(o, 1)$ same-symbol run has exact length r if and only if*

$$v_3(5m_0 + 3) = r.$$

Equivalently, there is an integer q with

$$3 \nmid q$$

such that

$$\boxed{5m_0 + 3 = 3^r q.}$$

After the r -step $(o, 1)$ run, one has

$$\boxed{5m_r + 3 = 8^r q.}$$

Thus

$$\boxed{m_r = \frac{8^r q - 3}{5}.}$$

Proof. By Lemma 28, r consecutive $(o, 1)$ -loops require

$$3^r \mid 5m_0 + 3.$$

The run has exact length r precisely when the divisibility holds for 3^r , but fails for 3^{r+1} . This is exactly

$$v_3(5m_0 + 3) = r.$$

Thus we may write

$$5m_0 + 3 = 3^r q, \quad 3 \nmid q.$$

Substituting this into the r -step formula gives

$$5m_r + 3 = \left(\frac{8}{3}\right)^r 3^r q = 8^r q.$$

Solving for m_r gives

$$m_r = \frac{8^r q - 3}{5}.$$

□

13.2.18. *Exit from a finite $(o, 1)$ -run.* After an exact $(o, 1)$ -run terminates, the next Ω_1 -transition cannot remain in $(o, 1)$. Therefore the run must exit to one of the other two o -symbols.

For Ω_1 , the residue-level routing is

$$(o, 1) \xrightarrow{\Omega_1} \begin{cases} (o, 1), & m \equiv 0 \pmod{3}, \\ (o, 0), & m \equiv 1 \pmod{3}, \\ (o, 2), & m \equiv 2 \pmod{3}. \end{cases}$$

Lemma 30 (Exit from a finite $(o, 1)$ -run). *Let an orbit remain in the $(o, 1)$ same-symbol loop for exactly $r \geq 1$ steps. Then after the r -th loop, the next Ω_1 -transition exits to either*

$$(o, 0) \quad \text{or} \quad (o, 2).$$

It cannot remain in $(o, 1)$.

Proof. By Lemma 29, an exact r -step $(o, 1)$ -run satisfies

$$5m_0 + 3 = 3^r q, \quad 3 \nmid q,$$

and after r steps,

$$5m_r + 3 = 8^r q.$$

If the loop could continue, then we would have

$$m_r \equiv 0 \pmod{3}.$$

This would imply

$$5m_r + 3 \equiv 0 \pmod{3}.$$

But

$$5m_r + 3 = 8^r q.$$

Since

$$8 \equiv 2 \pmod{3}$$

and

$$3 \nmid q,$$

we have

$$8^r q \not\equiv 0 \pmod{3}.$$

This is a contradiction. Hence

$$m_r \not\equiv 0 \pmod{3}.$$

Therefore

$$m_r \equiv 1 \pmod{3} \quad \text{or} \quad m_r \equiv 2 \pmod{3}.$$

Under the Ω_1 routing,

$$m_r \equiv 1 \pmod{3} \implies (o, 1) \rightarrow (o, 0),$$

and

$$m_r \equiv 2 \pmod{3} \implies (o, 1) \rightarrow (o, 2).$$

Thus the exact finite $(o, 1)$ -run exits to either $(o, 0)$ or $(o, 2)$, and cannot continue as another $(o, 1)$ loop. $\square$

The exit to $(o, 0)$ is favorable, since $(o, 0)$ has a strong same-symbol growing return. The exit to $(o, 2)$ may begin a new exceptional block, so this is the case that must be tracked by the debt ledger.

13.2.19. *Debt update for alternating $(o, 2)$ and $(o, 1)$ blocks.* The preceding lemmas show that long runs of $(o, 2)$ and $(o, 1)$ have exact affine forms and exact finite exit rules. The natural remaining object is therefore an alternating sequence of blocks:

$$(o, 2)^{k_1} \rightarrow (o, 1)^{r_1} \rightarrow (o, 2)^{k_2} \rightarrow (o, 1)^{r_2} \rightarrow \dots$$

Here the $(o, 2)$ -blocks create debt, while the $(o, 1)$ -blocks repay debt.

A block of k_i exceptional $(o, 2)$ self-loops has reverse multiplier

$$\left(\frac{2}{3}\right)^{k_i}.$$

Since this multiplier is less than 1, it creates anti-growth. If debt is measured as the reciprocal of reverse growth, then this block increases debt by

$$\left(\frac{2}{3}\right)^{-k_i} = \left(\frac{3}{2}\right)^{k_i}.$$

A block of r_i $(o, 1)$ self-returns has reverse multiplier

$$\left(\frac{8}{3}\right)^{r_i}.$$

Since this multiplier is greater than 1, it supplies compensating growth. Thus it repays debt by a growth factor

$$\left(\frac{8}{3}\right)^{r_i}.$$

Equivalently, if debt itself is tracked multiplicatively, then the same $(o, 1)^{r_i}$ -block multiplies the remaining debt by the reciprocal factor

$$\left(\frac{8}{3}\right)^{-r_i} = \left(\frac{3}{8}\right)^{r_i}.$$

Define the logarithmic debt after N alternating blocks by

$$\mathcal{D}_N = \sum_{i=1}^N k_i \log \left(\frac{3}{2}\right) - \sum_{i=1}^N r_i \log \left(\frac{8}{3}\right).$$

Equivalently, the multiplicative debt is

$$B_N = \prod_{i=1}^N \left(\frac{3}{2}\right)^{k_i} \left(\frac{3}{8}\right)^{r_i}.$$

These conventions are consistent because

$$B_N = e^{\mathcal{D}_N}.$$

The debt is cleared once

$$B_N < 1,$$

or equivalently,

$$\mathcal{D}_N < 0.$$

Thus an alternating exceptional path can remain uncompensated only if

$$\sum_{i=1}^N r_i \log\left(\frac{8}{3}\right) \leq \sum_{i=1}^N k_i \log\left(\frac{3}{2}\right)$$

for every initial block length N . Equivalently,

$$\sum_{i=1}^N r_i \leq \frac{\log(3/2)}{\log(8/3)} \sum_{i=1}^N k_i.$$

Since

$$\frac{\log(3/2)}{\log(8/3)} \approx 0.414,$$

the debt remains uncleared only if the total number of $(o, 1)$ -growth steps stays below approximately 41.4% of the total number of exceptional $(o, 2)$ -steps.

This converts the remaining obstruction into a precise accounting problem: one must rule out an infinite admissible alternation in which the $(o, 1)$ -growth blocks are persistently too short to compensate the $(o, 2)$ -debt blocks. The exit lemmas above show that both kinds of blocks are finite in the positive integers; the remaining issue is the relative size of the successive block lengths.

13.2.20. Short repayment and the block-to-block transition. We now combine the $(o, 2)$ -run and $(o, 1)$ -run analyses into a single block-to-block calculation. The purpose is to understand what happens when an exceptional $(o, 2)$ -run exits to $(o, 1)$, but the resulting $(o, 1)$ -run is too short to repay the debt created by the preceding exceptional block.

Suppose an exact $(o, 2)$ -run has length k . Then the initial $(o, 2)$ index has the form

$$m_0 + 1 = 3^k q, \quad 3 \nmid q.$$

Equivalently,

$$m_0 = 3^k q - 1.$$

After k consecutive Ω_2 self-loops, we have

$$m_k + 1 = 2^k q,$$

so

$$m_k = 2^k q - 1.$$

The run exits to $(o, 1)$ precisely when

$$m_k \equiv 0 \pmod{3}.$$

Using the formula for m_k , this is equivalent to

$$2^k q - 1 \equiv 0 \pmod{3},$$

or

$$2^k q \equiv 1 \pmod{3}.$$

Now suppose that after this exit, the orbit remains in the $(o, 1)$ -loop for r consecutive steps. The length of this $(o, 1)$ -run is measured by

$$r = v_3(5m_k + 3).$$

Substituting

$$m_k = 2^k q - 1,$$

we get

$$5m_k + 3 = 5(2^k q - 1) + 3 = 5 \cdot 2^k q - 2.$$

Hence

$$\boxed{r = v_3(5 \cdot 2^k q - 2)}.$$

Thus an $(o, 1)$ -run of length at least r occurs precisely when

$$3^r \mid 5 \cdot 2^k q - 2.$$

Equivalently,

$$5 \cdot 2^k q \equiv 2 \pmod{3^r}.$$

Since $5 \cdot 2^k$ is invertible modulo 3^r , this determines q modulo 3^r :

$$\boxed{q \equiv 2(5 \cdot 2^k)^{-1} \pmod{3^r}}.$$

Therefore the original $(o, 2)$ starting index must satisfy

$$\boxed{m_0 = 3^k q - 1, \quad q \equiv 2(5 \cdot 2^k)^{-1} \pmod{3^r}}.$$

Equivalently, m_0 lies in a single residue class modulo

$$3^{k+r}.$$

This gives an important case distinction. Let

$$R_k = \left\lceil \frac{k}{2} \right\rceil.$$

By the debt-compensation lemma, an $(o, 1)$ -run of length R_k is sufficient, at the leading-multiplier level, to compensate the debt created by the preceding $(o, 2)^k$ -run. Therefore, if

$$r \geq R_k,$$

then the block

$$(o, 2)^k (o, 1)^r$$

has repaid the exceptional debt.

If instead

$$r < R_k,$$

then the immediate $(o, 1)$ -run is too short to repay the preceding $(o, 2)^k$ -debt. In that case the congruence above holds only up to the shorter power 3^r , but not up to the repayment threshold 3^{R_k} :

$$q \equiv 2(5 \cdot 2^k)^{-1} \pmod{3^r},$$

while

$$q \not\equiv 2(5 \cdot 2^k)^{-1} \pmod{3^{R_k}}.$$

Thus the short-repayment case is arithmetically visible: it corresponds to an initial $(o, 2)$ index lying in the residue class required for an $(o, 1)^r$ -run, but not in the narrower class required for an $(o, 1)^{R_k}$ -run.

If the short $(o, 1)$ -run exits to $(o, 0)$, then the orbit enters the stronger exit channel, whose compensating multiplier is $32/3$. The more delicate case is when the short $(o, 1)$ -run exits back to $(o, 2)$, beginning another possible exceptional block. We now compute the corresponding next $(o, 2)$ starting index.

Set

$$s = \frac{5 \cdot 2^k q - 2}{3^r}, \quad 3 \nmid s.$$

Then

$$5m_k + 3 = 3^r s.$$

After the r -step $(o, 1)$ -run, the exact $(o, 1)$ -run formula gives

$$5m_r + 3 = 8^r s.$$

Hence

$$m_r = \frac{8^r s - 3}{5}.$$

If the run exits from $(o, 1)$ to $(o, 2)$, then the next $(o, 2)$ input index is obtained from the Ω_1 -transition in the $m \equiv 2 \pmod{3}$ case. This gives

$$m_{\text{next}} = \frac{8m_r + 2}{3}.$$

Substituting the formula for m_r , we obtain

$$m_{\text{next}} = \frac{8 \left(\frac{8^r s - 3}{5} \right) + 2}{3}.$$

Therefore

$$m_{\text{next}} = \frac{8^{r+1} s - 24 + 10}{15},$$

so

$$\boxed{m_{\text{next}} = \frac{8^{r+1} s - 14}{15}}.$$

Since

$$s = \frac{5 \cdot 2^k q - 2}{3^r},$$

this gives the explicit block-to-block transition

$$\boxed{m_{\text{next}} = \frac{8^{r+1} \left(\frac{5 \cdot 2^k q - 2}{3^r} \right) - 14}{15}}.$$

The length of the next exceptional $(o, 2)$ -run is

$$k_{\text{next}} = v_3(m_{\text{next}} + 1).$$

Using the formula above,

$$m_{\text{next}} + 1 = \frac{8^{r+1}s - 14}{15} + 1 = \frac{8^{r+1}s + 1}{15}.$$

Thus

$$\boxed{k_{\text{next}} = v_3(8^{r+1}s + 1) - 1.}$$

We have therefore obtained a precise recurrence for the dangerous short-repayment case:

$$\begin{aligned} q &\mapsto s = \frac{5 \cdot 2^k q - 2}{3^r}, \\ s &\mapsto m_{\text{next}} = \frac{8^{r+1}s - 14}{15}, \end{aligned}$$

and

$$m_{\text{next}} \mapsto k_{\text{next}} = v_3(m_{\text{next}} + 1).$$

This is the arithmetic engine for the remaining obstruction. If the $(o, 1)$ -run is long enough, the preceding $(o, 2)$ -debt is repaid. If it is too short and exits to $(o, 0)$, the orbit enters the stronger compensating channel. The only delicate continuation is the one in which a too-short $(o, 1)$ -run exits back to $(o, 2)$. In that case, the next exceptional length is not arbitrary; it is determined by the explicit valuation formula

$$k_{\text{next}} = v_3(8^{r+1}s + 1) - 1.$$

Thus repeated short repayment would require an infinite iteration of these increasingly restrictive arithmetic conditions.

13.3. Exceptional $(o, 2)$ -runs, $(o, 1)$ -compensation, and cumulative debt. We now isolate the remaining obstruction more sharply. The difficult case is the exceptional $(o, 2)$ self-loop at $p = 0$. The analysis below shows that this self-loop cannot persist indefinitely in the positive integers. It also introduces a debt framework for measuring how much subsequent $(o, 1)$ -growth is needed to compensate a finite exceptional $(o, 2)$ -run.

13.3.1. The exceptional $(o, 2)$ self-loop. The symbol $(o, 2)$ corresponds to odd integers of the form

$$x = 18m + 17.$$

Equivalently,

$$x \equiv 17 \pmod{18}.$$

For the exceptional row

$$\Omega_2 : (o, 2) \rightarrow (o, 2)$$

at $p = 0$, we have

$$\alpha = 1, \quad k^{(0)} = 1.$$

Thus

$$F(0, m) = 2m + 1.$$

The output router is determined by

$$F(0, m) = 3m' + j', \quad j' \in \{0, 1, 2\}.$$

Since

$$j' \equiv 2m + 1 \pmod{3},$$

the self-loop condition $j' = 2$ is

$$2m + 1 \equiv 2 \pmod{3}.$$

Hence

$$m \equiv 2 \pmod{3}.$$

If $m = 3M + 2$, then

$$F(0, m) = 2(3M + 2) + 1 = 6M + 5.$$

Since the self-loop has $j' = 2$, we write

$$6M + 5 = 3m' + 2,$$

so

$$m' = 2M + 1.$$

In terms of m , this is

$$m' = \frac{2m - 1}{3}.$$

Equivalently,

$$m' + 1 = \frac{2}{3}(m + 1).$$

Thus after k consecutive $(o, 2)$ self-loops,

$$m_k + 1 = \left(\frac{2}{3}\right)^k (m_0 + 1).$$

For these k loops to be possible in the ordinary integers, we need

$$3^k \mid m_0 + 1.$$

Therefore a k -step exceptional run requires

$$m_0 \equiv -1 \pmod{3^k}.$$

Equivalently,

$$m_0 + 1 = 3^k q$$

for some integer q . For an exact run of length k , we require

$$3^k \mid m_0 + 1 \quad \text{but} \quad 3^{k+1} \nmid m_0 + 1.$$

Thus

$$m_0 = 3^k q - 1, \quad 3 \nmid q.$$

After the k -step run,

$$m_k + 1 = 2^k q,$$

so

$$m_k = 2^k q - 1.$$

In terms of the original odd integer

$$x_0 = 18m_0 + 17,$$

we have

$$x_0 + 1 = 18(m_0 + 1).$$

Therefore k consecutive exceptional $(o, 2)$ -loops require

$$x_0 \equiv -1 \pmod{18 \cdot 3^k}.$$

For the first few values:

Number of consecutive Ω_2 loops	Required congruence for x_0
1	$x_0 \equiv 53 \pmod{54}$
2	$x_0 \equiv 161 \pmod{162}$
3	$x_0 \equiv 485 \pmod{486}$
k	$x_0 \equiv -1 \pmod{18 \cdot 3^k}.$

Thus each additional exceptional self-loop narrows the possible input class by another factor of 3.

13.3.2. *No infinite positive $(o, 2)$ self-loop.* The preceding formula immediately rules out an infinite positive $(o, 2)$ -self-loop.

Lemma 31 (No infinite positive $(o, 2)$ self-loop). *No positive integer state can remain forever in the exceptional*

$$(o, 2) \xrightarrow{\Omega_2} (o, 2)$$

self-loop.

Proof. If a positive integer remained in the $(o, 2)$ self-loop forever, then for every $k \geq 1$ we would have

$$3^k \mid m_0 + 1.$$

The only ordinary integer divisible by every power of 3 is 0. Thus

$$m_0 + 1 = 0,$$

so

$$m_0 = -1.$$

But in the $(o, 2)$ class,

$$x_0 = 18m_0 + 17,$$

and therefore

$$x_0 = 18(-1) + 17 = -1.$$

This is outside the positive Collatz domain. Hence no positive integer can remain indefinitely in the $(o, 2)$ self-loop. $\square$

Equivalently, a long $(o, 2)$ -run forces

$$m_0 \equiv -1 \pmod{3^k}.$$

Thus m_0 becomes closer and closer to -1 in the 3-adic sense, although in the ordinary real sense the value $m_0 = 3^k q - 1$ becomes large when $q > 0$.

13.3.3. *Exit from a finite $(o, 2)$ -run.* After an exact $(o, 2)^k$ -run, we have

$$m_k = 2^k q - 1, \quad 3 \nmid q.$$

The self-loop would continue only if

$$m_k \equiv 2 \pmod{3}.$$

But an exact k -step run terminates, so this congruence fails. Hence

$$m_k \equiv 0 \quad \text{or} \quad m_k \equiv 1 \pmod{3}.$$

For the Ω_2 row, the residue-level routing is

$$(o, 2) \xrightarrow{\Omega_2} \begin{cases} (o, 1), & m \equiv 0 \pmod{3}, \\ (o, 0), & m \equiv 1 \pmod{3}, \\ (o, 2), & m \equiv 2 \pmod{3}. \end{cases}$$

Therefore every finite exceptional $(o, 2)$ -run exits to one of two symbols:

$$\boxed{(o, 1) \quad \text{or} \quad (o, 0)}.$$

The exit to $(o, 0)$ is favorable, since the same-symbol return

$$(o, 0) \xrightarrow{\Omega_0} (o, 0)$$

has leading reverse multiplier

$$G_{\Omega_0} = \frac{2^5}{3} = \frac{32}{3}.$$

The exit to $(o, 1)$ is the conservative, weaker exit, since

$$(o, 1) \xrightarrow{\Omega_1} (o, 1)$$

has leading reverse multiplier

$$G_{\Omega_1} = \frac{2^3}{3} = \frac{8}{3}.$$

13.3.4. *Debt created by an $(o, 2)^k$ -run.* A single $(o, 2)$ loop has reverse multiplier

$$\frac{2}{3}.$$

Thus k consecutive $(o, 2)$ -loops have reverse multiplier

$$\left(\frac{2}{3}\right)^k.$$

This is reverse shrinkage. It is useful to measure the corresponding forward-growth debt by the reciprocal factor

$$B_k = \left(\frac{3}{2}\right)^k.$$

In logarithmic form, define

$$D = \log B.$$

Then one $(o, 2)$ loop changes the debt by

$$\Delta D_{o,2} = \log \left(\frac{3}{2}\right).$$

Thus the debt increases at a constant rate per $(o, 2)$ loop.

13.3.5. *The $(o, 1)$ -run and its repayment rate.* The same-symbol $(o, 1)$ return is

$$(o, 1) \xrightarrow{\Omega_1} (o, 1).$$

At $p = 0$, this row has

$$\alpha = 3, \quad k^{(0)} = 4.$$

Thus

$$F(0, m) = 8m + 4.$$

Since

$$F(0, m) = 3m' + j',$$

we have

$$j' \equiv 8m + 4 \pmod{3}.$$

Using

$$8 \equiv 2 \pmod{3}, \quad 4 \equiv 1 \pmod{3},$$

we obtain

$$j' \equiv 2m + 1 \pmod{3}.$$

The same-symbol return requires $j' = 1$, so

$$2m + 1 \equiv 1 \pmod{3},$$

which is equivalent to

$$m \equiv 0 \pmod{3}.$$

If $m = 3M$, then

$$F(0, m) = 8(3M) + 4 = 24M + 4.$$

Since $j' = 1$,

$$24M + 4 = 3m' + 1,$$

so

$$m' = 8M + 1.$$

In terms of m ,

$$m' = \frac{8m + 3}{3}.$$

A useful shifted form is

$$5m' + 3 = \frac{8}{3}(5m + 3).$$

Therefore after r consecutive $(o, 1)$ -loops,

$$5m_r + 3 = \left(\frac{8}{3}\right)^r (5m_0 + 3).$$

Thus one $(o, 1)$ loop repays debt by the reverse-growth factor

$$\frac{8}{3}.$$

Equivalently, multiplicative debt changes by

$$B \mapsto \frac{3}{8}B.$$

In logarithmic debt,

$$\Delta D_{o,1} = -\log\left(\frac{8}{3}\right).$$

Numerically,

$$\log\left(\frac{3}{2}\right) \approx 0.4055, \quad \log\left(\frac{8}{3}\right) \approx 0.9808.$$

Thus one $(o, 1)$ loop repays about

$$\frac{\log(8/3)}{\log(3/2)} \approx 2.416$$

$(o, 2)$ -loops worth of debt.

13.3.6. *One $(o, 2)^k(o, 1)^r$ block.* Now consider one alternating block

$$(o, 2)^k(o, 1)^r.$$

The $(o, 2)^k$ -block creates debt

$$\left(\frac{3}{2}\right)^k,$$

while the following $(o, 1)^r$ -block repays by

$$\left(\frac{8}{3}\right)^r.$$

The combined leading reverse multiplier is

$$G(k, r) = \left(\frac{2}{3}\right)^k \left(\frac{8}{3}\right)^r.$$

The block repays its debt exactly when

$$G(k, r) > 1.$$

Equivalently,

$$\left(\frac{8}{3}\right)^r > \left(\frac{3}{2}\right)^k.$$

Taking logarithms gives

$$r \log\left(\frac{8}{3}\right) > k \log\left(\frac{3}{2}\right),$$

so

$$r > k \frac{\log(3/2)}{\log(8/3)}.$$

Since

$$\frac{\log(3/2)}{\log(8/3)} \approx 0.414,$$

we have the repayment criterion

$$r > 0.414k.$$

A cleaner sufficient bound is

$$r \geq \left\lceil \frac{k}{2} \right\rceil.$$

However, r is not arbitrary. If the exact $(o, 2)^k$ -run begins at

$$m_0 = 3^k q - 1, \quad 3 \nmid q,$$

then after the $(o, 2)^k$ -run,

$$m_k = 2^k q - 1.$$

If the run exits to $(o, 1)$, then

$$m_k \equiv 0 \pmod{3},$$

or

$$2^k q \equiv 1 \pmod{3}.$$

The length of the following $(o, 1)$ -run is

$$r = v_3(5m_k + 3).$$

Substituting $m_k = 2^k q - 1$, we obtain

$$5m_k + 3 = 5(2^k q - 1) + 3 = 5 \cdot 2^k q - 2.$$

Hence

$$r = v_3(5 \cdot 2^k q - 2).$$

Therefore the one-block repayment criterion becomes

$$v_3(5 \cdot 2^k q - 2) > 0.414k.$$

Using the cleaner sufficient criterion,

$$v_3(5 \cdot 2^k q - 2) \geq \left\lceil \frac{k}{2} \right\rceil$$

guarantees repayment.

The complementary case

$$v_3(5 \cdot 2^k q - 2) < \left\lceil \frac{k}{2} \right\rceil$$

does not mean repayment is impossible. It only means that this immediate $(o, 1)^r$ -block does not repay the preceding $(o, 2)^k$ -debt by itself. Later blocks, or an exit to $(o, 0)$, may still repay the accumulated debt.

13.3.7. *Arithmetic form of one alternating block.* Let

$$r = v_3(5 \cdot 2^k q - 2).$$

Then there is an integer s , with $3 \nmid s$, such that

$$5 \cdot 2^k q - 2 = 3^r s.$$

Solving for q ,

$$q = \frac{3^r s + 2}{5 \cdot 2^k}.$$

Since

$$m_0 = 3^k q - 1,$$

we get

$$m_0 = \frac{3^k(3^r s + 2)}{5 \cdot 2^k} - 1, \quad 3 \nmid s.$$

The condition that m_0 be an integer is

$$5 \cdot 2^k \mid 3^r s + 2.$$

Equivalently,

$$3^r s \equiv -2 \pmod{5 \cdot 2^k}.$$

Since 3^r is invertible modulo $5 \cdot 2^k$, this determines s modulo $5 \cdot 2^k$:

$$\boxed{s \equiv -2(3^r)^{-1} \pmod{5 \cdot 2^k}.$$

This integer condition is independent of repayment. It only says that the block

$$(o, 2)^k(o, 1)^r$$

is arithmetically admissible. Repayment is the separate inequality

$$\left(\frac{8}{3}\right)^r > \left(\frac{3}{2}\right)^k.$$

13.3.8. *Examples.* The following table gives representative values. We choose q so that the exit-to- $(o, 1)$ condition

$$2^k q \equiv 1 \pmod{3}$$

is satisfied. Then we compute

$$r = v_3(5 \cdot 2^k q - 2).$$

The column “Repays?” refers to the exact leading-multiplier condition

$$r > 0.414k.$$

k	q	$m_0 = 3^k q - 1$	$5 \cdot 2^k q - 2$	$r = v_3(\cdot)$	Repays?
1	2	5	18	2	yes
2	1	8	18	2	yes
3	2	53	78	1	no
4	1	80	78	1	no
5	2	485	318	1	no
6	1	728	318	1	no
7	2	4373	1278	2	no
8	1	6560	1278	2	no
9	5	98414	12798	4	yes
10	1	59048	5118	1	no
11	8	1417175	81918	3	no
12	4	2125763	81918	3	no

These examples show that

$$r = v_3(5 \cdot 2^k q - 2)$$

can be much smaller than $k/2$. For example, when

$$k = 10, \quad q = 1,$$

we have

$$5 \cdot 2^{10} \cdot 1 - 2 = 5118 = 3 \cdot 1706,$$

so

$$r = 1.$$

Thus a long $(o, 2)$ -run can be followed by a short $(o, 1)$ -run. Local repayment is therefore not automatic.

On the other hand, when

$$k = 9, \quad q = 5,$$

we get

$$5 \cdot 2^9 \cdot 5 - 2 = 12798 = 3^4 \cdot 158,$$

so

$$r = 4.$$

Since

$$0.414 \cdot 9 \approx 3.726,$$

this block repays its debt.

13.3.9. Indefinite alternating runs and cumulative debt. A single block need not repay locally. Therefore we pass to cumulative debt.

Consider an alternating sequence

$$(o, 2)^{k_1} (o, 1)^{r_1} (o, 2)^{k_2} (o, 1)^{r_2} \cdots (o, 2)^{k_N} (o, 1)^{r_N}.$$

Define

$$K_N = \sum_{i=1}^N k_i, \quad R_N = \sum_{i=1}^N r_i.$$

The cumulative leading reverse multiplier is

$$G_N = \left(\frac{2}{3}\right)^{K_N} \left(\frac{8}{3}\right)^{R_N}.$$

The cumulative debt is repaid once

$$G_N > 1.$$

Equivalently,

$$\left(\frac{8}{3}\right)^{R_N} > \left(\frac{3}{2}\right)^{K_N}.$$

Taking logarithms gives

$$R_N \log \left(\frac{8}{3}\right) > K_N \log \left(\frac{3}{2}\right).$$

Thus cumulative repayment occurs when

$$R_N > K_N \frac{\log(3/2)}{\log(8/3)} \approx 0.414 K_N.$$

A cleaner sufficient condition is

$$R_N \geq \frac{1}{2} K_N.$$

Thus local under-repayment is allowed. The only dangerous case is permanent cumulative under-repayment:

$$\boxed{R_N \leq 0.414K_N \text{ for every } N.}$$

An indefinite alternating sequence must contain infinitely many $(o, 1)$ -instances, since otherwise it would eventually become an infinite $(o, 2)$ -run, which has already been ruled out. However, infinitely many $(o, 1)$ -instances are not enough by themselves. The $(o, 1)$ -instances must occur often enough relative to the $(o, 2)$ -instances to clear the cumulative debt.

13.3.10. *Block-to-block recurrence.* Suppose a short $(o, 1)$ -run exits back to $(o, 2)$. Then we can write a recurrence for the next possible exceptional block.

Starting from

$$m_0 = 3^k q - 1, \quad 3 \nmid q,$$

we have

$$m_k = 2^k q - 1.$$

Let

$$5 \cdot 2^k q - 2 = 3^r s, \quad 3 \nmid s.$$

After the $(o, 1)^r$ -run,

$$5m_r + 3 = 8^r s,$$

so

$$m_r = \frac{8^r s - 3}{5}.$$

If the run exits from $(o, 1)$ back to $(o, 2)$, then the next $(o, 2)$ input index is

$$m_{\text{next}} = \frac{8m_r + 2}{3}.$$

Substituting the formula for m_r , we get

$$m_{\text{next}} = \frac{8 \left(\frac{8^r s - 3}{5} \right) + 2}{3}.$$

Therefore

$$\boxed{m_{\text{next}} = \frac{8^{r+1} s - 14}{15}.$$

The next exceptional $(o, 2)$ -run length is

$$k_{\text{next}} = v_3(m_{\text{next}} + 1).$$

Since

$$m_{\text{next}} + 1 = \frac{8^{r+1} s + 1}{15},$$

we obtain

$$\boxed{k_{\text{next}} = v_3(8^{r+1} s + 1) - 1.}$$

Thus repeated short repayment is not arbitrary. It requires the recurrence

$$s_i = \frac{5 \cdot 2^{k_i} q_i - 2}{3^{r_i}},$$

$$q_{i+1} = \frac{8^{r_i+1} s_i + 1}{15 \cdot 3^{k_{i+1}}},$$

with

$$3 \nmid q_i, \quad 3 \nmid s_i.$$

An infinite permanently under-repaying alternation would therefore impose an infinite nested sequence of 3-adic congruence conditions on the original starting value.

13.3.11. *Current remaining obstruction.* The preceding analysis reduces the remaining problem to the following precise obstruction:

Can there exist an infinite admissible alternation

$$(o, 2)^{k_1} (o, 1)^{r_1} (o, 2)^{k_2} (o, 1)^{r_2} \dots ?$$

such that

$$R_N \leq 0.414K_N \quad \text{for every } N?$$

If no such indefinitely under-repaying alternation exists, then some finite prefix satisfies

$$R_N > 0.414K_N,$$

and the cumulative debt is cleared. At that point the corresponding reverse segment has net leading growth, giving the desired forward descent over that segment.

Thus the next task is to rule out permanent cumulative under-repayment, either by a direct valuation argument, by identifying an inadmissible 3-adic limiting object, or by finding a corrected Lyapunov function that decreases along every admissible dangerous transition.

13.3.12. *Modular depth as a proxy for debt.* For a block of the form

$$(o, 2)^k (o, 1)^r,$$

the starting index m_0 is forced into a single residue class modulo

$$3^{k+r}.$$

More precisely,

$$m_0 \equiv A(k, r) \pmod{3^{k+r}},$$

where

$$A(k, r) = 3^k \left(2(5 \cdot 2^k)^{-1} \bmod 3^r \right) - 1.$$

Thus the block fixes

$$L = k + r$$

base-3 digits of m_0 . We call L the *modular depth* of the block.

On the other hand, the same block has leading reverse multiplier

$$G(k, r) = \left(\frac{2}{3}\right)^k \left(\frac{8}{3}\right)^r.$$

Equivalently, the logarithmic debt change is

$$\Delta D(k, r) = k \log\left(\frac{3}{2}\right) - r \log\left(\frac{8}{3}\right).$$

Thus k contributes debt, while r contributes repayment.

Since

$$L = k + r,$$

we may write

$$\rho = \frac{r}{k + r}.$$

Then

$$k = (1 - \rho)L, \quad r = \rho L.$$

Substituting into the debt change gives

$$\Delta D = L \left[(1 - \rho) \log\left(\frac{3}{2}\right) - \rho \log\left(\frac{8}{3}\right) \right].$$

Therefore the debt change per fixed ternary digit is

$$\boxed{\frac{\Delta D}{L} = (1 - \rho) \log\left(\frac{3}{2}\right) - \rho \log\left(\frac{8}{3}\right).}$$

The block repays its debt precisely when

$$\Delta D < 0.$$

This is equivalent to

$$(1 - \rho) \log\left(\frac{3}{2}\right) < \rho \log\left(\frac{8}{3}\right).$$

Rearranging,

$$\log\left(\frac{3}{2}\right) < \rho \left[\log\left(\frac{3}{2}\right) + \log\left(\frac{8}{3}\right) \right].$$

Hence repayment occurs when

$$\boxed{\rho > \frac{\log(3/2)}{\log(3/2) + \log(8/3)}}.$$

Numerically,

$$\frac{\log(3/2)}{\log(3/2) + \log(8/3)} \approx 0.293.$$

Thus:

A block repays its debt if more than about 29.3% of its modular depth comes from $(o, 1)$ -steps.

Equivalently, since

$$\rho = \frac{r}{k+r},$$

this condition is the same as

$$\frac{r}{k} > \frac{\log(3/2)}{\log(8/3)} \approx 0.414.$$

Thus the two forms of the repayment condition agree:

$$\boxed{r > 0.414k}$$

or, equivalently,

$$\boxed{\frac{r}{k+r} > 0.293.}$$

This gives a useful interpretation. The modular depth

$$L = k + r$$

measures how many ternary digits of the starting index m_0 are fixed by the prescribed block, while

$$\Delta D = k \log\left(\frac{3}{2}\right) - r \log\left(\frac{8}{3}\right)$$

measures whether those fixed digits correspond to net debt or net repayment.

Thus modular depth can serve as a proxy for debt, provided we keep track of how much of the depth comes from $(o, 2)$ -steps and how much comes from $(o, 1)$ -steps. The total depth alone is not enough; the important quantity is the proportion

$$\rho = \frac{r}{k+r}.$$

For an alternating prefix

$$(o, 2)^{k_1}(o, 1)^{r_1}(o, 2)^{k_2}(o, 1)^{r_2} \dots (o, 2)^{k_N}(o, 1)^{r_N},$$

define

$$K_N = \sum_{i=1}^N k_i, \quad R_N = \sum_{i=1}^N r_i, \quad L_N = K_N + R_N.$$

Then the prefix fixes L_N ternary digits of m_0 , and its cumulative debt is

$$D_N = K_N \log\left(\frac{3}{2}\right) - R_N \log\left(\frac{8}{3}\right).$$

Writing

$$\rho_N = \frac{R_N}{L_N},$$

we get

$$\frac{D_N}{L_N} = (1 - \rho_N) \log\left(\frac{3}{2}\right) - \rho_N \log\left(\frac{8}{3}\right).$$

Therefore cumulative debt clears once

$$\boxed{\rho_N > 0.293.}$$

Equivalently,

$$\boxed{R_N > 0.414K_N.}$$

Example residue table. For a block of the form

$$(o, 2)^k(o, 1)^r,$$

the starting index satisfies

$$m_0 \equiv A(k, r) \pmod{3^{k+r}},$$

where

$$A(k, r) = 3^k \left(2(5 \cdot 2^k)^{-1} \bmod 3^r \right) - 1.$$

The following table lists the residue classes for small values of k and r . Each entry should be read as

$$m_0 \equiv A(k, r) \pmod{3^{k+r}}.$$

k	$r = 1$	$r = 2$	$r = 3$	$r = 4$
0	0 (mod 3)	3 (mod 9)	21 (mod 27)	48 (mod 81)
1	5 (mod 9)	5 (mod 27)	32 (mod 81)	194 (mod 243)
2	8 (mod 27)	8 (mod 81)	170 (mod 243)	656 (mod 729)
3	53 (mod 81)	134 (mod 243)	620 (mod 729)	2078 (mod 2187)
4	80 (mod 243)	566 (mod 729)	2024 (mod 2187)	6398 (mod 6561)
5	485 (mod 729)	1943 (mod 2187)	6317 (mod 6561)	19439 (mod 19683)

For example, the entry

$$k = 2, \quad r = 3$$

says that the prescribed block

$$(o, 2)^2(o, 1)^3$$

forces

$$m_0 \equiv 170 \pmod{243}.$$

This refines the earlier condition for

$$(o, 2)^2 \rightarrow (o, 1),$$

namely

$$m_0 \equiv 8 \pmod{27},$$

because

$$170 \equiv 8 \pmod{27}.$$

Thus the table illustrates the nested-residue structure: adding an $(o, 1)^r$ -run does not replace the earlier $(o, 2)^k$ -condition; it lifts it to a narrower congruence class modulo 3^{k+r} .

This reformulates the remaining obstruction in modular terms. An indefinite alternating pattern can remain dangerous only if the proportion of newly fixed ternary digits contributed by $(o, 1)$ -runs never exceeds approximately 29.3%. If at some finite stage the $(o, 1)$ -share of the modular depth exceeds this threshold, the cumulative debt is cleared.

13.3.13. *Nested residues, lift coefficients, and the finite-positive obstruction.*

We now record the modular form of the alternating obstruction. The aim is to understand what a starting index m_0 would have to look like if it supported a long, or even infinite, alternating segment

$$(o, 2)^{k_1}(o, 1)^{r_1}(o, 2)^{k_2}(o, 1)^{r_2} \dots$$

One alternating block. For a single block

$$(o, 2)^k(o, 1)^r,$$

we begin with an exact $(o, 2)^k$ -run:

$$m_0 + 1 = 3^k q, \quad 3 \nmid q.$$

Thus

$$m_0 = 3^k q - 1.$$

After the $(o, 2)^k$ -run,

$$m_k = 2^k q - 1.$$

The following $(o, 1)^r$ -run is measured by

$$r = v_3(5m_k + 3).$$

Substituting $m_k = 2^k q - 1$, we obtain

$$r = v_3(5 \cdot 2^k q - 2).$$

Equivalently,

$$3^r \mid 5 \cdot 2^k q - 2.$$

Hence

$$5 \cdot 2^k q \equiv 2 \pmod{3^r}.$$

Since $5 \cdot 2^k$ is invertible modulo 3^r ,

$$q \equiv 2(5 \cdot 2^k)^{-1} \pmod{3^r}.$$

Therefore

$$m_0 = 3^k q - 1$$

is forced into a single residue class modulo 3^{k+r} :

$$m_0 \equiv A(k, r) \pmod{3^{k+r}},$$

where

$$A(k, r) = 3^k \left(2(5 \cdot 2^k)^{-1} \bmod 3^r \right) - 1.$$

Thus a block $(o, 2)^k(o, 1)^r$ fixes

$$L = k + r$$

ternary digits of m_0 . We call L the modular depth of the block.

Residue table for one block. The following table lists $A(k, r)$ for small values of k and r . Each entry is to be read as

$$m_0 \equiv A(k, r) \pmod{3^{k+r}}.$$

k	$r = 1$	$r = 2$	$r = 3$	$r = 4$
0	0 (mod 3)	3 (mod 9)	21 (mod 27)	48 (mod 81)
1	5 (mod 9)	5 (mod 27)	32 (mod 81)	194 (mod 243)
2	8 (mod 27)	8 (mod 81)	170 (mod 243)	656 (mod 729)
3	53 (mod 81)	134 (mod 243)	620 (mod 729)	2078 (mod 2187)
4	80 (mod 243)	566 (mod 729)	2024 (mod 2187)	6398 (mod 6561)
5	485 (mod 729)	1943 (mod 2187)	6317 (mod 6561)	19439 (mod 19683)

For example,

$$k = 2, \quad r = 3$$

gives

$$m_0 \equiv 170 \pmod{243}.$$

This refines the earlier exit condition

$$(o, 2)^2 \rightarrow (o, 1),$$

which required

$$m_0 \equiv 8 \pmod{27},$$

because

$$170 \equiv 8 \pmod{27}.$$

Thus adding an $(o, 1)^r$ -run does not replace the previous $(o, 2)^k$ -condition. It lifts that condition to a narrower residue class modulo a higher power of 3.

Debt and modular depth. The leading reverse multiplier of the block

$$(o, 2)^k(o, 1)^r$$

is

$$G(k, r) = \left(\frac{2}{3}\right)^k \left(\frac{8}{3}\right)^r.$$

Thus the block repays its $(o, 2)$ -debt precisely when

$$G(k, r) > 1.$$

Equivalently,

$$\left(\frac{8}{3}\right)^r > \left(\frac{3}{2}\right)^k.$$

Clearing denominators gives the exact integer criterion

$$\boxed{2^{3r+k} > 3^{r+k}}.$$

Thus the block repays debt if and only if

$$2^{3r+k} - 3^{r+k} > 0.$$

It remains under-repaying if

$$2^{3r+k} - 3^{r+k} < 0.$$

In logarithmic form, repayment is equivalent to

$$r > k \frac{\log(3/2)}{\log(8/3)} \approx 0.414k.$$

Equivalently, if

$$L = k + r, \quad \rho = \frac{r}{L},$$

then repayment occurs when

$$\rho > \frac{\log(3/2)}{\log(3/2) + \log(8/3)} \approx 0.293.$$

So a block repays if more than about 29.3% of its fixed ternary digits come from the $(o, 1)$ -recovery part.

Two-block prefixes and lift coefficients. Now consider a two-block prefix

$$(o, 2)^{k_1} (o, 1)^{r_1} (o, 2)^{k_2} (o, 1)^{r_2}.$$

Let

$$L_1 = k_1 + r_1, \quad L_2 = k_1 + r_1 + k_2 + r_2.$$

The first block gives

$$m_0 \equiv A_1 \pmod{3^{L_1}}.$$

The two-block prefix gives

$$m_0 \equiv A_2 \pmod{3^{L_2}}.$$

Since the second congruence refines the first, we have

$$A_2 \equiv A_1 \pmod{3^{L_1}}.$$

Therefore there is an integer c_1 such that

$$\boxed{A_2 = A_1 + 3^{L_1} c_1.}$$

We call c_1 the lift coefficient. It records the residue lift modulo a higher power of 3 forced by the second block.

More generally, after N alternating blocks we have

$$m_0 \equiv A_N \pmod{3^{L_N}},$$

where

$$L_N = \sum_{i=1}^N (k_i + r_i).$$

The next block gives

$$m_0 \equiv A_{N+1} \pmod{3^{L_{N+1}}},$$

and hence

$$\boxed{A_{N+1} = A_N + 3^{L_N} c_N.}$$

Thus c_N records the residue lift modulo a higher power of 3 imposed by the next alternating block.

Two-block table. The following table illustrates the two-block lift. The cumulative debt is tested exactly using

$$2^{3R+K} > 3^{R+K},$$

where

$$K = k_1 + k_2, \quad R = r_1 + r_2.$$

(k_1, r_1)	(k_2, r_2)	$A_1 \bmod 3^{L_1}$	$A_2 \bmod 3^{L_2}$	c_1	(K, R)	Debt
(1, 1)	(2, 4)	5 mod 9	2291 mod 6561	254	(3, 5)	repaid
(1, 2)	(2, 2)	5 mod 27	1679 mod 2187	62	(3, 4)	repaid
(1, 3)	(1, 2)	32 mod 81	1085 mod 2187	13	(2, 5)	repaid
(2, 1)	(5, 1)	8 mod 27	11456 mod 19683	424	(7, 2)	unpaid
(2, 2)	(2, 2)	8 mod 81	2519 mod 6561	31	(4, 4)	repaid
(2, 2)	(3, 1)	8 mod 81	4706 mod 6561	58	(5, 3)	repaid
(3, 2)	(5, 1)	134 mod 243	56267 mod 177147	231	(8, 3)	unpaid
(4, 2)	(5, 1)	566 mod 729	84401 mod 531441	115	(9, 3)	unpaid
(5, 1)	(1, 1)	485 mod 729	3401 mod 6561	4	(6, 2)	unpaid

This table shows that the lift coefficient c_1 is usually nonzero. In those cases, the next block forces a nonzero residue lift modulo a higher power of 3 for m_0 .

The meaning of $c_N = 0$. The condition

$$c_N = 0$$

means

$$A_{N+1} = A_N.$$

Thus the ordinary residue value does not change, although the modulus does become deeper:

$$m_0 \equiv A_N \pmod{3^{L_N}}$$

is refined to

$$m_0 \equiv A_N \pmod{3^{L_{N+1}}}, \quad L_{N+1} > L_N.$$

Equivalently, the next block imposes only the zero lift modulo the next higher power of 3.

This is important for distinguishing a genuinely 3-adic infinite candidate from a finite positive integer. Suppose an infinite alternating sequence is realized by a fixed positive integer M . Then for every N ,

$$M \equiv A_N \pmod{3^{L_N}}.$$

Since $L_N \rightarrow \infty$, eventually

$$3^{L_N} > M.$$

At that point, A_N , as the least nonnegative representative modulo 3^{L_N} , must equal M itself:

$$A_N = M.$$

Therefore all later successful refinements must satisfy

$$A_{N+1} = A_N = M.$$

But since

$$A_{N+1} = A_N + 3^{L_N} c_N,$$

we get

$$\boxed{c_N = 0}$$

for all sufficiently large N .

Thus we have proved:

If a finite positive integer supports an infinite alternating segment,
then $c_N = 0$ eventually.

This does not mean that $c_N = 0$ alone guarantees an infinite orbit. Rather, it is a necessary condition for an infinite alternating segment to be carried by an ordinary positive integer.

Example 1: zero lift with debt already repaid. Consider

$$m_0 = 170.$$

We have

$$m_0 + 1 = 171 = 3^2 \cdot 19,$$

so

$$k_1 = 2, \quad q_1 = 19.$$

After the $(o, 2)^2$ -run,

$$m_* = 2^2 q_1 - 1 = 4 \cdot 19 - 1 = 75.$$

The following $(o, 1)$ -run length is

$$r_1 = v_3(5m_* + 3).$$

Since

$$5m_* + 3 = 5 \cdot 75 + 3 = 378 = 3^3 \cdot 14,$$

we get

$$r_1 = 3, \quad s_1 = 14.$$

Thus the first block is

$$(o, 2)^2(o, 1)^3.$$

The residue condition is

$$m_0 \equiv 170 \pmod{243}.$$

After the $(o, 1)^3$ -run,

$$m_{k+r} = \frac{8^3 s_1 - 3}{5} = \frac{512 \cdot 14 - 3}{5} = 1433.$$

If this exits back to $(o, 2)$, the next $(o, 2)$ -input is

$$m_{\text{next}} = \frac{8m_{k+r} + 2}{3} = \frac{8 \cdot 1433 + 2}{3} = 3822.$$

Then

$$m_{\text{next}} + 1 = 3823,$$

which is not divisible by 3. Hence

$$k_2 = 0.$$

The next $(o, 1)$ -length is

$$r_2 = v_3(5 \cdot 3823 - 2).$$

Since

$$5 \cdot 3823 - 2 = 19113 = 3 \cdot 6371, \quad 3 \nmid 6371,$$

we have

$$r_2 = 1.$$

Thus $m_0 = 170$ realizes the longer prefix

$$(o, 2)^2(o, 1)^3(o, 2)^0(o, 1)^1.$$

The first modular depth is

$$L_1 = 2 + 3 = 5,$$

so

$$3^{L_1} = 243.$$

The second modular depth is

$$L_2 = 2 + 3 + 0 + 1 = 6,$$

so

$$3^{L_2} = 729.$$

The residue remains

$$A_1 = A_2 = 170.$$

Therefore

$$A_2 = A_1 + 3^{L_1}c_1$$

gives

$$170 = 170 + 243c_1,$$

so

$$\boxed{c_1 = 0.}$$

In this example, however, the debt is already repaid. For the first block alone,

$$K = 2, \quad R = 3.$$

The exact debt test gives

$$2^{3R+K} = 2^{11} = 2048,$$

while

$$3^{R+K} = 3^5 = 243.$$

Thus

$$2^{11} > 3^5,$$

so the debt is repaid already by

$$(o, 2)^2(o, 1)^3.$$

Example 2: temporary zero-lift behavior during an unpaid prefix. A finite unpaid prefix can also occur. Consider

$$m_0 = 27026.$$

This value begins with the alternating blocks

$$(o, 2)^3(o, 1)^1(o, 2)^4(o, 1)^1(o, 2)^2(o, 1)^1.$$

The cumulative totals are

$$K = 3 + 4 + 2 = 9, \quad R = 1 + 1 + 1 = 3.$$

The exact debt test is

$$2^{3R+K} > 3^{R+K}.$$

Here

$$2^{3(3)+9} = 2^{18} = 262144,$$

while

$$3^{3+9} = 3^{12} = 531441.$$

Hence

$$2^{18} < 3^{12},$$

so the cumulative debt remains unpaid after these three alternating blocks.

The modular depths are

$$L_1 = 4, \quad L_2 = 9, \quad L_3 = 12.$$

The corresponding residues satisfy

$$27026 \equiv 53 \pmod{81},$$

$$27026 \equiv 7343 \pmod{19683},$$

and

$$27026 \equiv 27026 \pmod{531441}.$$

At the third stage the modulus already exceeds m_0 , so the least nonnegative residue is literally

$$A_3 = 27026.$$

If this same m_0 continued to satisfy deeper prefixes indefinitely, then all later residues would have to remain equal to 27026, and all later lift coefficients would be zero.

However, in this example the alternating pattern does not continue indefinitely; it exits to the $(o, 0)$ channel after the next attempted continuation. Thus the example shows that unpaid finite prefixes can occur, and that residue stabilization can appear temporarily, but it does not produce an infinite under-repaying alternating orbit.

The apparent obstruction is an infinite alternating exceptional segment. The next sections show that this obstruction cannot occur for a finite positive

segment-start: either the nested residue object is only 3-adic, or the residues stabilize and the segment is exhausted.

The examples computed above illustrate both possibilities. For $m_0 = 170$, one obtains a zero lift $c_N = 0$, but the debt has already been repaid. For $m_0 = 27026$, one obtains a finite under-repaying prefix; however, the modulus eventually exceeds the finite value itself, so the residue stabilizes to the actual integer:

$$A_N = m_0, \quad y_N = 0.$$

These examples show the same mechanism that appears in the general proof.

13.3.14. *Stabilization and exhaustion.* The preceding residue construction gives, for each finite prefix of a compensated exceptional segment,

$$m_0 = A_N + 3^{L_N} y_N, \quad A_{N+1} = A_N + 3^{L_N} c_N.$$

Here L_N is the cumulative residue depth, and c_N records the residue lift modulo a higher power of 3 imposed by the next block.

If the compensated exceptional segment were extendable indefinitely, then

$$L_N \rightarrow \infty.$$

For a finite positive segment-start $m_0 = M$, eventually

$$3^{L_N} > M.$$

Since $0 \leq A_N < 3^{L_N}$, the representation

$$M = A_N + 3^{L_N} y_N$$

forces

$$A_N = M, \quad y_N = 0.$$

All later lifts then satisfy

$$A_{N+1} = A_N = M, \quad c_N = 0.$$

Thus the residue construction stabilizes and collapses to the identity

$$m_0 = m_0.$$

13.3.15. *Conclusion: no infinite contiguous exceptional segment.* We can now state the main conclusion of the preceding analysis.

An exceptional segment is a contiguous alternating episode of the form

$$(o, 2)^{k_0} (o, 1)^{r_0} (o, 2)^{k_1} (o, 1)^{r_1} \dots$$

For a finite prefix of such a segment, the original segment-start m_0 is forced into a residue class

$$m_0 \equiv A_N \pmod{3^{L_N}},$$

where

$$L_N = \sum_{i=0}^{N-1} (k_i + r_i).$$

Equivalently,

$$m_0 = A_N + 3^{L_N} y_N.$$

Appending another block gives the lift relation

$$A_{N+1} = A_N + 3^{L_N} c_N.$$

Thus c_N measures whether the next block contributes a residue lift modulo a higher power of 3 to the original segment-start m_0 .

Now suppose that a single contiguous exceptional segment continued indefinitely from a finite positive value $m_0 = M$. Then

$$L_N \rightarrow \infty.$$

Hence, for all sufficiently large N ,

$$3^{L_N} > M.$$

Since A_N is the least nonnegative residue modulo 3^{L_N} ,

$$0 \leq A_N < 3^{L_N}.$$

Therefore the representation

$$M = A_N + 3^{L_N} y_N$$

forces

$$A_N = M$$

and

$$y_N = 0.$$

Consequently, the residue system stabilizes.

Moreover, for all later refinements,

$$A_{N+1} = A_N = M.$$

Since

$$A_{N+1} = A_N + 3^{L_N} c_N,$$

we also obtain

$$c_N = 0$$

for all sufficiently large N .

Thus an attempted infinite exceptional segment has only two possible interpretations.

First, if the residues A_N do not stabilize, then the limiting object is only a 3-adic integer:

$$m_\infty = \lim_{N \rightarrow \infty} A_N \text{ in } \mathbb{Z}_3.$$

This does not give an ordinary finite positive starting value.

Second, if the residues do stabilize, then

$$A_N = m_0, \quad y_N = 0, \quad c_N = 0.$$

At that point the residue condition has collapsed to the identity

$$m_0 = m_0.$$

The current exceptional segment has exhausted its ability to impose further residue information on m_0 . Hence it has ended as a residue-building segment.

Furthermore, since the run has exact length k , we have written

$$m_0 + 1 = 3^k q, \quad 3 \nmid q.$$

After k consecutive $(o, 2)$ -steps, we found

$$m_k + 1 = 2^k q.$$

Because

$$3 \nmid 2^k$$

and

$$3 \nmid q,$$

it follows that

$$3 \nmid m_k + 1.$$

Equivalently,

$$\nu_3(m_k + 1) = 0.$$

This is precisely the exit condition. The next $(o, 2)$ -self-loop would require

$$m_k \equiv -1 \pmod{3},$$

or equivalently,

$$3 \mid m_k + 1.$$

But the exact-length assumption gives

$$\nu_3(m_k + 1) = 0,$$

so this condition fails. Therefore the pure $(o, 2)$ -run stops exactly after k steps.

Proposition 6 (Stabilization exhausts a residue-building segment). *A contiguous exceptional $(o, 2)/(o, 1)$ -segment beginning from a finite positive state cannot continue indefinitely as a residue-building segment once its residue data have stabilized.*

Proof. The preceding stabilization argument shows that an attempted infinite segment has only two possible interpretations. If the residues A_N do not stabilize, the limiting object is only 3-adic and is not a finite positive integer. If the residues do stabilize, then

$$A_N = m_0, \quad y_N = 0, \quad c_N = 0$$

from some finite stage onward, so the residue condition collapses to

$$m_0 = m_0.$$

Thus the current exceptional segment has exhausted its ability to impose further residue information on m_0 , and cannot continue indefinitely as one residue-building segment.

Since A_N is already the finite integer m_0 itself, A further genuine dangerous continuation will be shown below to require a nonzero residue lift modulo a higher power of 3. That would require

$$c_N \neq 0.$$

But stabilization gives $c_N = 0$ for all sufficiently large N . Hence the current residue-building segment has exhausted its ability to impose further residue information on its segment-start.

□

Therefore, once its residue data have stabilized, a single contiguous exceptional segment cannot continue indefinitely as the same residue-building process.

We may summarize the conclusion as follows:

Once the residue data stabilize, the current residue-building segment has exhausted its ability to impose new constraints on m_0 .

This does not prevent the orbit from later entering another exceptional episode. However, such a later episode is a new segment, not a continuation of the previous one. By the same argument, the new exceptional segment is finite as well.

Thus the stabilization mechanism shows how a residue-building exceptional segment exhausts its ability to impose new constraints on a finite positive starting value. The formal exclusion of infinite compensated exceptional segments will be completed below, after proving that genuine dangerous continuation requires a nonzero residue lift modulo a higher power of 3.

Disposal of the ω_1 -exception. The drift table contains two $p = 0$, $\alpha = 1$ rows:

$$\omega_1 \quad \text{and} \quad \Omega_2.$$

Both have one-step leading reverse multiplier

$$\frac{2}{3},$$

and hence either one may create a single unit of local debt. However, they differ in an essential structural way.

The row Ω_2 has type oo . Thus it remains in the odd family, and in the exceptional router class it can produce a same-symbol return

$$(o, 2) \longrightarrow (o, 2).$$

This is why Ω_2 can generate long exceptional blocks

$$(o, 2)^k,$$

whose accumulated leading debt is

$$\left(\frac{3}{2}\right)^k.$$

By contrast, the row ω_1 has type oe . It starts in the state $(o, 1)$, but its output lies in the e -family:

$$(o, 1) \xrightarrow{\omega_1} (e, j')$$

for some output router $j' \in \{0, 1, 2\}$. Therefore a ω_1 -step cannot be followed by another ω_1 -step without first returning from the e -family to the o -family. In particular, there is no same-family exceptional block

$$\omega_1^k$$

analogous to the persistent block Ω_2^k .

Thus ω_1 can contribute at most a local $p = 0$ debt factor

$$\left(\frac{2}{3}\right)^{-1} = \frac{3}{2},$$

before the itinerary leaves the o -family and enters an e -family transition. Any subsequent return to the o -family must pass through one of the $e \rightarrow o$ rows, and those rows are accounted for in the ordinary row classification and compensation ledger.

Consequently, ω_1 is not a persistent exceptional obstruction. The only $p = 0$, $\alpha = 1$ row capable of producing arbitrarily long same-family anti-growth blocks is

$$\Omega_2.$$

Hence the global exceptional analysis may focus on the alternating

$$(o, 2)/(o, 1)$$

channel, with ω_1 treated as a non-persistent local debt contribution inside the non-persistent part of the itinerary.

13.3.16. Worst-case reduction and exhaustion of the exceptional obstruction. We now explain why the preceding analysis of contiguous $(o, 2)/(o, 1)$ -segments controls not only a single exceptional episode, but also the apparent possibility of separated exceptional episodes.

The key point is that the contiguous $(o, 2)/(o, 1)$ -segment is already the worst local anti-growth configuration. The row $(o, 2)$ is the exceptional anti-growth row in the $p = 0$ table, while $(o, 1)$ is the least compensating row available in the same base routing. Therefore any itinerary which inserts non-exceptional rows or any $p > 0$ rows between exceptional episodes has at least as much inverse growth as the corresponding worst-case $(o, 2)/(o, 1)$ -model.

Lemma 32 (Exceptional segments are the worst local anti-growth case). *Among the $p = 0$ inverse rows, the least favorable local configuration for inverse growth is the exceptional $(o, 2)/(o, 1)$ -configuration:*

$$(o, 2)^{k_0}(o, 1)^{r_0}(o, 2)^{k_1}(o, 1)^{r_1} \dots$$

The $(o, 2)$ -steps create inverse debt, while the $(o, 1)$ -steps give the least compensation available within the $p = 0$ routing.

Proof. Each $(o, 2)$ -step contributes the inverse anti-growth factor

$$\frac{2}{3}.$$

Thus an $(o, 2)^K$ -run contributes

$$\left(\frac{2}{3}\right)^K.$$

The row $(o, 1)$ is the least compensating row in the $p = 0$ table, and each $(o, 1)$ -step contributes

$$\frac{8}{3}.$$

Thus a compensated exceptional block has multiplier

$$\left(\frac{2}{3}\right)^K \left(\frac{8}{3}\right)^R.$$

All non-exceptional non-terminal $p = 0$ rows have positive inverse drift, and all $p > 0$ rows are strictly inverse-expansive. Hence replacing any non-exceptional or $p > 0$ portion by the least compensating $(o, 1)$ -behavior can only decrease inverse growth. Therefore the contiguous $(o, 2)/(o, 1)$ -configuration is the least favorable local case. $\square$

Lemma 33 (Growth outside the exceptional subsystem is strictly expansive). *Let G be a portion of an inverse itinerary containing no exceptional $(o, 2)$ -anti-growth step. Then G is strictly inverse-expansive, except at the terminal fixed point $1 \mapsto 1$, which is not part of a non-descent obstruction from an odd input $x > 1$. In particular, if G contains any $p > 0$ row, then G is strictly inverse-expansive.*

Proof. This follows from the drift bounds. Non-exceptional non-terminal $p = 0$ rows have positive inverse drift, and all $p > 0$ rows are strongly expansive in the inverse direction. Hence such a portion cannot make the inverse growth balance worse than the exceptional $(o, 2)/(o, 1)$ -model. $\square$

Lemma 34 (Debt criterion for exceptional compensation). *Let an exceptional block have K total $(o, 2)$ -steps and R total $(o, 1)$ -compensating steps. Its*

inverse multiplier is

$$\left(\frac{2}{3}\right)^K \left(\frac{8}{3}\right)^R.$$

The block has repaid its exceptional debt exactly when

$$2^{3R+K} > 3^{R+K}.$$

Proof. The block is inverse-expansive exactly when

$$\left(\frac{2}{3}\right)^K \left(\frac{8}{3}\right)^R > 1.$$

Equivalently,

$$\frac{2^K 8^R}{3^{K+R}} > 1.$$

Since $8^R = 2^{3R}$, this becomes

$$2^{K+3R} > 3^{K+R}.$$

□

Proposition 7 (Worst-case reduction). *Suppose an inverse itinerary contains separated finite exceptional episodes,*

$$G_0 E_0 G_1 E_1 G_2 E_2 \cdots,$$

where each E_i is a finite exceptional $(o, 2)/(o, 1)$ -episode and each G_i is the intervening non-exceptional or higher- p dynamics. Then this itinerary cannot be a worse non-descent obstruction than the contiguous exceptional $(o, 2)/(o, 1)$ -model.

More precisely, either the G_i -pieces contribute genuine inverse growth, in which case they push the corresponding forward orbit toward descent, or they contribute no effective growth, in which case they do not create a worse obstruction than the contiguous exceptional model.

Proof. By Lemma 33, every intervening G_i -portion is strictly inverse-expansive away from the terminal fixed point $1 \mapsto 1$, and any $p > 0$ occurrence is strictly inverse-expansive. Hence inserting such G_i -pieces between exceptional episodes cannot decrease inverse growth relative to the worst-case model in which the intervening dynamics is replaced by the least compensating $p = 0$ behavior.

By Lemma 32, that worst-case model is precisely the contiguous exceptional $(o, 2)/(o, 1)$ -configuration. Therefore a separated itinerary

$$G_0 E_0 G_1 E_1 G_2 E_2 \cdots$$

is either genuinely helped by the G_i -growth, or else reduces, in the growth calculus, to the already analyzed contiguous exceptional case. □

13.3.17. *Forward descent from exhaustion of the exceptional obstruction.* We now assemble the preceding analysis into the forward-descent conclusion. The argument proceeds by contradiction. If an odd orbit from $x > 1$ never reaches a smaller odd value, then its row itinerary must avoid every finite block satisfying the exact reverse-growth criterion of Theorem 13. Equivalently, global non-descent would have to appear as a persistent failure of the exact carry inequality

$$C_W < \left(2^{\Sigma_W} - 3^n\right) m_n.$$

The row classification reduces such persistent failure to the exceptional $p = 0, \alpha = 1$ cases. The row ω_1 is non-persistent, while the only row capable of supporting a persistent same-family obstruction is Ω_2 , corresponding to the state $(o, 2)$. Thus the proof focuses on eliminating the $(o, 2)$ -obstruction.

Pure $(o, 2)$ -runs have already been shown to be finite in the ordinary positive integers. The only remaining possible persistence mechanism is therefore the compensated exceptional channel

$$(o, 2)^{k_0} (o, 1)^{r_0} (o, 2)^{k_1} (o, 1)^{r_1} \dots$$

Here the $(o, 1)$ -blocks are not themselves obstructions; they represent the weakest available compensation after finite $(o, 2)$ -runs. The residue-lift and stabilization arguments show that even this compensated exceptional channel cannot continue indefinitely from a finite positive starting value.

Consequently, every persistent exact non-descent obstruction is exhausted. The final step is to invoke the exact carry criterion again: once no persistent obstruction remains, the itinerary contains a finite admissible block giving exact integer descent.

The row calculus is exhaustive for the non-ghost inverse branches. Hence every odd Collatz transition is represented by a row token and a parameter $p \geq 0$. Thus any persistent failure of forward descent must appear as a persistent non-growth obstruction inside the row calculus.

The drift analysis shows that all rows with $p \geq 1$ are strongly inverse-expansive. Therefore these rows are forward-descending. Among the $p = 0$ rows, the only possible source of inverse anti-growth is the exceptional $(o, 2)$ -mechanism. The row $(o, 1)$ is the least compensating row available in the same $p = 0$ routing. Thus the worst local model for forward non-descent is the compensated exceptional $(o, 2)/(o, 1)$ -segment.

Recall that a segment is an admissible word together with its induced symbol path. A return segment is a segment whose initial and terminal symbols agree. In this subsection we consider attempted indefinite extensions of exceptional segment words in the $p = 0$ row system. The result is that no such exceptional anti-growth obstruction can persist indefinitely from a finite positive segment-start.

Lemma 35 (Inverse growth gives forward descent). *Let W be an inverse word carrying an odd value x' to an odd predecessor x . If*

$$x > x',$$

then, in the forward odd Collatz orbit, the value x reaches the smaller odd value x' . Hence inverse growth along W is equivalent to forward descent along the corresponding forward word.

Proof. The inverse word W is obtained by reversing a finite forward odd Collatz itinerary. Thus if W carries x' to x , then the same word read forward carries x to x' . Therefore $x > x'$ in the inverse direction is exactly the statement that the forward orbit descends from x to the smaller odd value x' . $\square$

Proposition 8 (Higher- p rows are forward-descending). *Every row with $p \geq 1$ is strictly inverse-expansive. Consequently, every $p \geq 1$ row gives strict descent when read in the forward orbital direction.*

Proof. By the drift bounds, all rows with $p \geq 1$ have positive inverse drift and are therefore strictly inverse-expansive. The conclusion then follows from Lemma 35. $\square$

Proposition 9 (Non-exceptional $p = 0$ rows are forward-descending). *Every non-exceptional non-terminal $p = 0$ row is strictly inverse-expansive. The only equality case is the terminal fixed point $1 \mapsto 1$, which is excluded from the descent analysis by assuming $x > 1$. Consequently, no non-exceptional $p = 0$ row can sustain a forward non-descent obstruction.*

Proof. This is the $p = 0$ part of the drift bounds. The only $p = 0$ rows with negative inverse drift are the exceptional anti-growth rows. Apart from the terminal fixed point $1 \mapsto 1$, all other non-exceptional $p = 0$ rows have positive inverse drift. Therefore they are strictly inverse-expansive and cannot be the source of a persistent forward non-descent obstruction. $\square$

Lemma 36 (Pure exceptional runs are finite). *A pure $(o, 2)$ -run beginning from a finite positive state is finite.*

Proof. A pure $(o, 2)^k$ -run requires

$$m_0 + 1 = 3^k q, \quad 3 \nmid q.$$

Equivalently,

$$m_0 \equiv -1 \pmod{3^k}.$$

If such a run were infinite, then

$$m_0 \equiv -1 \pmod{3^k}$$

for every $k \geq 1$. In the 3-adic integers this forces

$$m_0 = -1,$$

which is not a positive integer state. Hence no finite positive m_0 supports an infinite pure $(o, 2)$ -run. $\square$

We now allow the least possible compensation. A single compensated exceptional block has the form

$$(o, 2)^k(o, 1)^r.$$

The $(o, 2)^k$ -run begins when

$$m_0 + 1 = 3^k q, \quad 3 \nmid q,$$

so

$$m_0 = 3^k q - 1.$$

After the $(o, 2)^k$ -run, the index is

$$m_k = 2^k q - 1.$$

The following $(o, 1)^r$ -run is measured by

$$r = v_3(5m_k + 3).$$

Substituting $m_k = 2^k q - 1$, we get

$$r = v_3(5 \cdot 2^k q - 2).$$

Equivalently,

$$5 \cdot 2^k q - 2 = 3^r s, \quad 3 \nmid s.$$

Thus

$$5 \cdot 2^k q \equiv 2 \pmod{3^r},$$

and since $5 \cdot 2^k$ is invertible modulo 3^r ,

$$q \equiv 2(5 \cdot 2^k)^{-1} \pmod{3^r}.$$

Therefore the segment-start satisfies

$$m_0 \equiv A(k, r) \pmod{3^{k+r}},$$

where

$$A(k, r) = 3^k \left(2(5 \cdot 2^k)^{-1} \bmod 3^r \right) - 1.$$

The inverse multiplier of a compensated exceptional block with K total $(o, 2)$ -steps and R total $(o, 1)$ -steps is

$$\left(\frac{2}{3}\right)^K \left(\frac{8}{3}\right)^R.$$

The block has repaid its exceptional debt exactly when this multiplier is greater than 1. Equivalently,

$$\left(\frac{2}{3}\right)^K \left(\frac{8}{3}\right)^R > 1,$$

which is the same as

$$2^{3R+K} > 3^{R+K}.$$

Thus the exact debt criterion is

$$2^{3R+K} > 3^{R+K}.$$

For a finite prefix of a compensated exceptional segment, write

$$K_N = \sum_{i=0}^{N-1} k_i, \quad R_N = \sum_{i=0}^{N-1} r_i, \quad L_N = K_N + R_N.$$

The first N blocks force the original segment-start m_0 into a residue class

$$m_0 \equiv A_N \pmod{3^{L_N}},$$

or equivalently

$$m_0 = A_N + 3^{L_N} y_N, \quad 0 \leq A_N < 3^{L_N}.$$

Appending another block gives a deeper residue satisfying

$$A_{N+1} = A_N + 3^{L_N} c_N.$$

The lift coefficient c_N records the next residue lift modulo a higher power of 3 imposed by the next block.

Lemma 37 (Continuation imposes a new residue condition). *Let a finite compensated exceptional prefix impose the congruence*

$$m_0 \equiv A_N \pmod{3^{L_N}}.$$

If the prefix admits a further nontrivial continuation inside the $(o, 2)/(o, 1)$ -channel, then the next required router condition pulls back to a refinement

$$m_0 \equiv A_{N+1} \pmod{3^{L_{N+1}}}, \quad L_{N+1} > L_N,$$

with

$$A_{N+1} = A_N + 3^{L_N} c_N.$$

Thus every nontrivial continuation imposes a new congruence condition on the original index m_0 .

Proof. After a finite compensated exceptional prefix, the original index has the form

$$m_0 = A_N + 3^{L_N} y_N.$$

The prefix has already absorbed the divisibility conditions encoded in 3^{L_N} . Therefore the terminal index after the prefix is an affine function of the remaining free tail y_N :

$$m_N = \lambda_N y_N + \mu_N,$$

where λ_N is a power of 2. In particular,

$$3 \nmid \lambda_N,$$

so λ_N is invertible modulo 3.

A further continuation inside the $(o, 2)/(o, 1)$ -channel requires the terminal index m_N to lie in a specified router class modulo 3. Equivalently, it requires

$$m_N \equiv r \pmod{3}$$

for one of the router residues $r \in \{0, 1, 2\}$ corresponding to the next admissible exceptional transition.

Substituting the affine expression for m_N , this becomes

$$\lambda_N y_N + \mu_N \equiv r \pmod{3}.$$

Since λ_N is invertible modulo 3, this congruence determines a unique residue class

$$y_N \equiv s \pmod{3}.$$

Writing

$$y_N = s + 3y_{N+1},$$

we obtain

$$m_0 = A_N + 3^{L_N} s + 3^{L_N+1} y_{N+1}.$$

Thus the original residue condition refines to

$$m_0 \equiv A_{N+1} \pmod{3^{L_N+1}},$$

where

$$A_{N+1} = A_N + 3^{L_N} s.$$

More generally, if the next block imposes more than one additional power of 3, the same pullback gives

$$m_0 \equiv A_{N+1} \pmod{3^{L_N+1}}, \quad L_{N+1} > L_N,$$

with

$$A_{N+1} = A_N + 3^{L_N} c_N.$$

Therefore every nontrivial continuation imposes a new residue condition on the original index m_0 , recorded as a residue lift modulo a higher power of 3. $\square$

Lemma 38 (Dangerous continuation gives a nonzero residue lift). *In the compensated exceptional $(o, 2)/(o, 1)$ -channel, a genuine continuation from an $(o, 2)$ -block into an $(o, 1)$ -block, or from an $(o, 1)$ -block back into an $(o, 2)$ -block, imposes a nonzero residue lift modulo a higher power of 3. Equivalently, if*

$$A_{N+1} = A_N + 3^{L_N} c_N$$

records the next lift of the residue class

$$A_N \pmod{3^{L_N}},$$

then a genuine dangerous continuation forces

$$c_N \not\equiv 0 \pmod{3}.$$

Proof. First consider a finite $(o, 2)^k$ -block. Write

$$m_0 + 1 = 3^k q, \quad 3 \nmid q.$$

After the block,

$$m_k + 1 = 2^k q,$$

and hence

$$m_k = 2^k q - 1.$$

For the next block to be an $(o, 1)^r$ -block, the starting value m_k must satisfy

$$3^r \mid 5m_k + 3.$$

Substituting $m_k = 2^k q - 1$, this becomes

$$5 \cdot 2^k q - 2 \equiv 0 \pmod{3^r}.$$

Since $5 \cdot 2^k$ is invertible modulo 3^r , this is equivalent to

$$q \equiv 2(5 \cdot 2^k)^{-1} \pmod{3^r}.$$

The right-hand side is nonzero modulo 3. Thus a genuine continuation from an $(o, 2)$ -block into an $(o, 1)$ -block selects a nonzero residue lift modulo a higher power of 3.

Conversely, suppose an $(o, 1)^r$ -block begins at m , and write

$$5m + 3 = 3^r s, \quad 3 \nmid s.$$

After the $(o, 1)^r$ -block,

$$5m_r + 3 = 8^r s,$$

so

$$m_r + 1 = \frac{8^r s + 2}{5}.$$

For the next block to return to an $(o, 2)^k$ -block, one must have

$$3^k \mid m_r + 1.$$

Since 5 is invertible modulo 3^k , this is equivalent to

$$8^r s + 2 \equiv 0 \pmod{3^k},$$

or

$$s \equiv -2 \cdot 8^{-r} \pmod{3^k}.$$

Again the right-hand side is nonzero modulo 3. Hence a genuine continuation from an $(o, 1)$ -block back into an $(o, 2)$ -block also selects a nonzero residue lift modulo a higher power of 3.

Therefore every genuine continuation inside the dangerous $(o, 2)/(o, 1)$ -channel imposes a nonzero residue lift modulo a higher power of 3. $\square$

Lemma 39 (Stabilization of finite residue lifts). *Let M be a finite positive integer. Suppose that*

$$M = A_N + 3^{L_N} y_N, \quad 0 \leq A_N < 3^{L_N},$$

where

$$L_N \rightarrow \infty.$$

Then for all sufficiently large N ,

$$A_N = M, \quad y_N = 0.$$

If also

$$A_{N+1} = A_N + 3^{L_N} c_N,$$

then for all sufficiently large N ,

$$c_N = 0.$$

Proof. Since $L_N \rightarrow \infty$, eventually

$$3^{L_N} > M.$$

For such N , the representation

$$M = A_N + 3^{L_N} y_N, \quad 0 \leq A_N < 3^{L_N},$$

forces

$$A_N = M$$

and

$$y_N = 0.$$

For later refinements, both residues are equal to M :

$$A_{N+1} = A_N = M.$$

Using

$$A_{N+1} = A_N + 3^{L_N} c_N,$$

we obtain

$$3^{L_N} c_N = 0,$$

and hence

$$c_N = 0.$$

□

Stabilization versus genuine 3-adic continuation. The nested residue conditions for an alternating exceptional segment have the form

$$m_0 = A_N + 3^{L_N} y_N, \quad L_N \rightarrow \infty.$$

Here A_N is the least nonnegative residue forced by the first N blocks, and y_N is the remaining free tail.

If $m_0 = M$ is an ordinary finite nonnegative integer, then once

$$3^{L_N} > M,$$

the least nonnegative residue of M modulo 3^{L_N} is just M itself. Hence at that stage

$$A_N = M, \quad y_N = 0.$$

Thus the residue construction has stabilized: there is no remaining higher 3-adic tail with which to force additional blocks.

Suppose, however, that the alternating $(o, 2)/(o, 1)$ -segment were to continue by imposing new nonzero higher 3-adic digits indefinitely. Then the residues would keep refining as

$$A_{N+1} = A_N + 3^{L_N} c_N,$$

with nonzero higher digit contribution c_N . After stabilization, where $A_N = M$, this would give

$$A_{N+1} = M + 3^{L_N} c_N.$$

For this to remain the least nonnegative residue of the same finite integer M modulo the larger modulus, one must have

$$c_N = 0.$$

Therefore an ordinary finite integer cannot keep acquiring new nonzero higher 3-adic digits indefinitely. A genuinely infinite nontrivial residue chain defines an element of $\mathbb{Z}_3$, not an ordinary finite nonnegative integer.

Lemma 40 (Stabilization excludes nontrivial infinite exceptional continuation). *Let an alternating $(o, 2)/(o, 1)$ -segment impose nested congruence conditions*

$$m_0 \equiv A_N \pmod{3^{L_N}}, \quad L_N \rightarrow \infty,$$

where A_N is chosen as the least nonnegative residue modulo 3^{L_N} . Suppose these congruences are realized by an ordinary finite nonnegative integer $m_0 = M$. Then, for all N sufficiently large,

$$A_N = M.$$

Consequently, any further refinement

$$A_{N+1} = A_N + 3^{L_N} c_N$$

compatible with the same finite integer M must have

$$c_N = 0.$$

Thus an ordinary finite integer cannot realize an infinite nontrivial sequence of new higher 3-adic residue constraints.

Proof. Since M is finite and $L_N \rightarrow \infty$, there exists N_0 such that

$$3^{L_N} > M$$

for all $N \geq N_0$. For such N , the least nonnegative residue of M modulo 3^{L_N} is M itself. Hence

$$A_N = M.$$

Now suppose the next refinement has the form

$$A_{N+1} = A_N + 3^{L_N} c_N.$$

Substituting $A_N = M$, we get

$$A_{N+1} = M + 3^{L_N} c_N.$$

For this to be the least nonnegative residue of the same integer M modulo the larger modulus $3^{L_{N+1}}$, we must have

$$M \equiv M + 3^{L_N} c_N \pmod{3^{L_{N+1}}}.$$

Therefore

$$3^{L_N} c_N \equiv 0 \pmod{3^{L_{N+1}}}.$$

Equivalently,

$$c_N \equiv 0 \pmod{3^{L_{N+1}-L_N}}.$$

Since c_N represents the next higher residue block, the only compatible stabilized refinement is the trivial one,

$$c_N = 0.$$

Thus a finite ordinary integer cannot continue to acquire nonzero higher 3-adic digits indefinitely. $\square$

Lemma 41 (Stabilization forces exit from the dangerous channel). *Let an ordinary finite integer $m_0 = M$ realize a finite prefix of an alternating exceptional $(o, 2)/(o, 1)$ -segment. Suppose the associated nested residue data have stabilized, so that*

$$M = A_N + 3^{L_N} y_N, \quad A_N = M, \quad y_N = 0.$$

Then the segment cannot continue nontrivially inside the dangerous $(o, 2)/(o, 1)$ -channel.

Consequently, once stabilization occurs, the segment cannot continue as a new nontrivial $(o, 2)/(o, 1)$ -obstruction. If it remains in $(o, 1)$, it is already in a compensating channel. Otherwise, within the o -family, the only remaining exit from the dangerous pair $(o, 2)/(o, 1)$ is through

$$(o, 0).$$

Proof. The dangerous channel under consideration consists of the two router states

$$(o, 2) \quad \text{and} \quad (o, 1).$$

A nontrivial continuation inside this channel means that the next block continues in one of these two states. Such a continuation is not free: it imposes the next router condition on the terminal index, and therefore imposes a further congruence condition on the original index m_0 .

Equivalently, the next nontrivial continuation would refine the existing residue condition

$$m_0 = A_N + 3^{L_N} y_N$$

by imposing a nonzero residue lift modulo a higher power of 3. In the notation of the nested residue construction, this is recorded by a nonzero next lift coefficient. Thus, by Lemma 38, a further genuine continuation in the $(o, 2)/(o, 1)$ -channel requires a nonzero residue lift modulo a higher power of 3.

But at stabilization we have

$$A_N = M \quad \text{and} \quad y_N = 0.$$

Thus the residue condition has already captured the whole ordinary integer M . There is no remaining free residue tail with which to impose another nonzero residue lift modulo a higher power of 3. Therefore no further nontrivial continuation inside the dangerous $(o, 2)/(o, 1)$ -channel is possible.

It remains only to identify the possible o -family exits. From either $(o, 2)$ or $(o, 1)$, the o -family router outputs lie among

$$(o, 2), \quad (o, 1), \quad (o, 0).$$

The first two are precisely the states of the dangerous channel. Since stabilization excludes further nontrivial continuation in those two states, the only remaining o -family output is

$$(o, 0).$$

Thus stabilization prevents any new nontrivial continuation inside the dangerous $(o, 2)/(o, 1)$ -channel. If the itinerary remains in $(o, 1)$, it is already in a compensating channel; otherwise, within the o -family, the remaining exit from the dangerous pair is through

$$(o, 0).$$

□

Corollary 14 (Finiteness of ordinary $(o, 2)/(o, 1)$ -segments). *An infinite nontrivial alternating $(o, 2)/(o, 1)$ -residue chain defines a 3-adic object. It cannot be realized by an ordinary finite nonnegative integer m_0 . Hence every $(o, 2)/(o, 1)$ -exceptional segment arising from an ordinary positive integer is finite.*

Theorem 15 (No infinite compensated exceptional segment). *No contiguous compensated exceptional $(o, 2)/(o, 1)$ -segment beginning from a finite positive state can continue indefinitely as a residue-building segment.*

Proof. Suppose such a segment were infinite and began from a finite positive segment-start

$$m_0 = M.$$

For each finite prefix we have

$$M = A_N + 3^{L_N} y_N, \quad 0 \leq A_N < 3^{L_N}.$$

Since the segment is infinite,

$$L_N \rightarrow \infty.$$

By Lemma 39, for all sufficiently large N ,

$$A_N = M, \quad y_N = 0, \quad c_N = 0.$$

Thus the residue condition stabilizes and collapses to

$$m_0 = m_0.$$

At that point the current residue-building segment has exhausted its ability to impose further residue information on its segment-start. By Lemma 38, a further genuine continuation of the compensated exceptional channel would require a nonzero residue lift modulo a higher power of 3. Equivalently, in the notation

$$A_{N+1} = A_N + 3^{L_N} c_N,$$

such a continuation would require

$$c_N \neq 0.$$

This contradicts stabilization. If no new residue information is created, then the extension has collapsed to the identity $m_0 = m_0$, not to a new exceptional constraint. Any attempt to regard this identity as a renewed exceptional constraint would amount to a recursive self-regeneration of the same finite state, which is precisely the non-stabilizing 3-adic alternative excluded from the positive integer case.

If, on the other hand, the residues A_N did not stabilize, then the limiting object would exist only in the 3-adic sense and would not be an ordinary finite positive integer. Therefore no infinite compensated exceptional segment can occur in the positive integer domain. $\square$

Theorem 16 (Exhaustion of exceptional forward non-descent obstructions). *No infinite forward non-descent obstruction can be built from exceptional anti-growth episodes in the positive integer domain.*

Proof. There are two possible forms for such an obstruction.

First, it may be a contiguous exceptional $(o, 2)/(o, 1)$ -segment. This case is impossible by Theorem 15: no contiguous exceptional segment beginning from a finite positive state can continue indefinitely as a residue-building segment.

Second, it may consist of separated finite exceptional episodes with intervening dynamics:

$$G_0 E_0 G_1 E_1 G_2 E_2 \cdots$$

By Proposition 7, such a separated pattern is no worse than the contiguous exceptional model. If the G_i -pieces contribute inverse growth, they push toward forward descent. If they do not contribute effective growth, then the pattern reduces to the contiguous exceptional obstruction already ruled out.

Therefore separated exceptional episodes introduce no new worse case. Since the worst possible exceptional obstruction has already been eliminated, no infinite forward non-descent obstruction can be built from exceptional anti-growth episodes. $\square$

Corollary 17 (Exceptional anti-growth is locally exhausted). *Every exceptional anti-growth obstruction in the positive integer domain is finite. Any later return to the exceptional rows begins a new finite episode and cannot be worse than the contiguous exceptional case already ruled out.*

Proof. A pure $(o, 2)$ -run is finite by Lemma 31. A compensated contiguous $(o, 2)/(o, 1)$ -segment is finite by Theorem 15. Separated finite exceptional episodes introduce no worse obstruction by Theorem 16. Hence the exceptional anti-growth mechanism is locally exhausted. $\square$

Lemma 42 (The compensated exceptional segment is the worst local case). *Among local $p = 0$ inverse configurations, the compensated $(o, 2)/(o, 1)$ -segment is the worst possible anti-growth model. Any configuration obtained by inserting non-exceptional rows or rows with $p > 0$ has at least as much inverse growth.*

Proof. The row $(o, 2)$ is the exceptional inverse anti-growth row, with factor

$$\frac{2}{3}.$$

The row $(o, 1)$ is the least compensating row in the $p = 0$ routing, with factor

$$\frac{8}{3}.$$

Thus the $(o, 2)/(o, 1)$ -segment is the least favorable model that still allows the exceptional mechanism to continue with compensation.

All non-exceptional $p = 0$ rows have positive inverse drift, and all $p > 0$ rows are strictly inverse-expansive. Therefore replacing any non-exceptional or

higher- p portion by the least compensating $(o, 1)$ -behavior can only decrease inverse growth.

The exact carry contribution of such inserted rows is accounted for by the word-level criterion of Theorem 13; the present comparison identifies the only row patterns capable of persistent failure of that exact criterion.

Hence the contiguous compensated $(o, 2)/(o, 1)$ -model is the worst local anti-growth model. $\square$

Proposition 10 (Exhaustion of local forward non-descent obstructions). *Every local forward non-descent obstruction detected by the row calculus is exhausted by the following cases:*

$$p \geq 1,$$

non-exceptional $p = 0$ rows, pure $(o, 2)$ -runs, and compensated $(o, 2)/(o, 1)$ -exceptional segments. The first two cases are forward-descending, the pure $(o, 2)$ -case is finite, and the compensated exceptional case is finite.

Proof. By Proposition 8, all rows with $p \geq 1$ are strictly forward-descending. By Proposition 9, all non-exceptional $p = 0$ rows are strictly inverse-expansive away from the terminal fixed point. By Lemma 36, pure $(o, 2)$ -runs are finite. By Theorem 15, compensated $(o, 2)/(o, 1)$ -exceptional segments are finite.

Finally, by Lemma 42, any interrupted or enriched pattern containing non-exceptional rows or higher- p rows is no worse than the compensated exceptional worst case. If the inserted pieces contain only $(o, 2)$ - and $(o, 1)$ -behavior, then the pattern is simply the compensated exceptional segment already analyzed. If they contain any other row or any $p > 0$ row, then they increase inverse growth relative to the worst case. Thus the listed cases exhaust all local forward non-descent obstructions. $\square$

Lemma 43 (Global non-descent forces a persistent exceptional obstruction). *Let $x > 1$ be an odd integer, and suppose that its accelerated odd Collatz orbit never reaches an odd value smaller than x . Then the associated row itinerary must contain a persistent exceptional non-descent obstruction. In the row calculus developed above, the only possible such obstruction is the Ω_2 -obstruction, possibly appearing inside a compensated exceptional $(o, 2)/(o, 1)$ -segment.*

Proof. Assume that the accelerated odd orbit of x never descends below x . By the completeness of the row calculus, each odd transition in the orbit is represented by one of the admissible row tokens. Thus the forward orbit determines an infinite row itinerary.

By the drift classification, every row outside the $p = 0$, $\alpha = 1$ exceptional cases is strictly favorable: read in the inverse direction it is expansive, and

read in the forward direction it contributes descent or compensation. Such rows cannot be the source of a persistent non-descent obstruction.

The row ω_1 is also non-persistent. Although it has the same one-step exceptional multiplier as Ω_2 , it has type oe , and therefore exits the o -family. Hence it cannot generate an arbitrarily long same-family exceptional block.

It follows that the only remaining row capable of producing persistent anti-growth is the Ω_2 -row, corresponding to the state $(o, 2)$. A pure $(o, 2)$ -run is finite in the ordinary positive integers, so the only remaining way for the obstruction to persist would be through a compensated exceptional segment of the form

$$(o, 2)^{k_0}(o, 1)^{r_0}(o, 2)^{k_1}(o, 1)^{r_1} \dots$$

Thus, if the orbit never descends below its starting value, its row itinerary must contain a persistent exceptional obstruction of precisely this type.

This proves that global non-descent forces persistence of the Ω_2 -obstruction, either as a pure exceptional run or through a compensated exceptional $(o, 2)/(o, 1)$ -segment. $\square$

Lemma 44 (Exactness of obstruction exhaustion). *The obstruction-exhaustion statements used in the final descent argument are exact statements in the integer row calculus, not merely leading multiplier estimates. More precisely, once the persistent exceptional obstructions have been exhausted, the row itinerary contains a finite admissible block W satisfying the exact carry criterion*

$$m_0 = \frac{3^n m_n + C_W}{2^{\Sigma_W}},$$

with

$$C_W < (2^{\Sigma_W} - 3^n) m_n.$$

Consequently

$$m_0 < m_n,$$

so the corresponding forward block sends the odd value represented by m_n to a strictly smaller odd value represented by m_0 .

Proof. By Theorem 13, exact forward descent along a finite word is governed by the carry inequality

$$C_W < (2^{\Sigma_W} - 3^n) m_n.$$

Thus the row calculus does not use the leading multiplier $2^{\Sigma_W}/3^n$ in isolation. For a word W of length n , the exact composition formula gives

$$m_0 = \frac{3^n m_n + C_W}{2^{\Sigma_W}},$$

where C_W is the accumulated carry term of the word. Thus inverse growth, equivalently forward descent, is governed by the exact inequality

$$\frac{3^n m_n + C_W}{2^{\Sigma_W}} < m_n.$$

Multiplying by 2^{Σ_W} , this is precisely

$$C_W < (2^{\Sigma_W} - 3^n) m_n.$$

The obstruction analysis above exhausts the only cases in which this exact inequality could fail persistently. Indeed, a persistent failure of the exact carry criterion would give an infinite row itinerary containing no finite descending block. By the row classification, every non-exceptional row class either satisfies the exact criterion on a finite block or enters a compensating class. Hence any persistent failure must remain inside the exceptional $p = 0, \alpha = 1$ classes. The row ω_1 is non-persistent, and the only remaining persistent candidate is the $(o, 2)$ -obstruction, either as a pure $(o, 2)$ -run or inside a compensated $(o, 2)/(o, 1)$ -segment. These possibilities have been exhausted above.

Higher- p rows and non-exceptional $p = 0$ rows satisfy the exact descent criterion away from the terminal fixed point. The row ω_1 is non-persistent, pure $(o, 2)$ -runs are finite in the ordinary positive integers, and compensated exceptional $(o, 2)/(o, 1)$ -segments are finite by the residue-lift and stabilization arguments.

Therefore a row itinerary with no persistent exceptional obstruction must contain a finite admissible block W satisfying the exact carry criterion above. Hence the local obstruction-exhaustion results used in the final descent argument yield genuine integer descent, not merely leading-multiplier improvement. $\square$

Theorem 18 (Forward descent for odd Collatz orbits). *For every odd integer $x > 1$, there exists a later odd Collatz iterate x' such that*

$$x' < x.$$

Proof. Assume, for contradiction, that some odd $x > 1$ never reaches a smaller odd iterate. By Lemma 43, global non-descent would force a persistent exceptional obstruction in the row itinerary.

By Lemma 44, the obstruction exhaustion used below gives exact integer descent, not merely leading-multiplier improvement.

By Proposition 10, every possible local obstruction represented by such a row word is exhausted: higher p -rows descend, non-exceptional $p = 0$ rows descend away from the terminal fixed point, pure exceptional runs are finite, and compensated exceptional segments are finite. The compensated exceptional case is already the worst local anti-growth case, so inserting any non-exceptional or higher- p rows cannot create a stronger obstruction.

Hence no persistent forward non-descent obstruction exists, contradicting the obstruction forced by Lemma 43. Therefore the forward odd orbit of x must eventually reach a smaller odd value $x' < x$. $\square$

Corollary 19 (Collatz convergence). *Every positive integer eventually reaches 1 under the Collatz map.*

Proof. It is enough to consider odd integers, since every even integer reaches an odd integer after finitely many divisions by 2. By Theorem 18, every odd integer $x > 1$ eventually reaches a smaller odd integer. Repeating this descent produces a strictly decreasing sequence of positive odd integers. Such a sequence must terminate, and it can terminate only at the least positive odd integer, namely 1. Hence every odd integer eventually reaches 1, and therefore every positive integer eventually reaches 1. $\square$

Part 4. Analysis and Dynamics

14. PARAMETER GEOMETRY

The row/lift primitives induce affine maps on the odd layer. We formalize a layered geometry: an *analytic operator layer* where each step acts as an affine map over the rationals, and a *discrete routing layer* that carries the residue constraints.

14.1. Operator Projection and Coordinates. Let $\Theta = (\alpha, \beta, c, \delta, p, m; \varepsilon)$ be an admissible parameter tuple for a single odd step. We define the derived constants:

$$K := (2^{\alpha+6p} - 3)4^p, \quad q_p := \frac{4^p - 1}{3}.$$

The family-specific offsets are:

$$B^{(1)} := 4q_p - \frac{K}{3}, \quad B^{(5)} := 10q_p - 2 - \frac{5K}{3}.$$

The induced single-step action on odd x admits an *exact* affine-plus-carry decomposition. More precisely, for each admissible step parameter tuple Θ and each odd input x , there exists an integer $\eta_\Theta(x)$ (the *carry error*) such that

$$T_\Theta(x) = Ax + B_\varepsilon + \eta_\Theta(x),$$

and $\eta_\Theta(x)$ is completely determined by the discrete routing data (equivalently, the carry update). In particular, $|\eta_\Theta(x)|$ is uniformly bounded (independently of x) for fixed admissible Θ .

Lemma 45 (Uniform carry bound). *For any admissible step parameter tuple Θ and any odd input x , the carry error $\eta_\Theta(x)$ satisfies $|\eta_\Theta(x)| \leq \eta_{\max}(\Theta)$ for some constant $\eta_{\max}(\Theta)$ depending only on Θ (not on x).*

Proof. For fixed admissible Θ , the routing rule partitions odd inputs into finitely many cases determined by the router state (equivalently by $x \bmod N(\Theta)$ for some modulus $N(\Theta)$ fixed by Θ). In each case $\eta_\Theta(x)$ is a fixed integer. Hence the image set $\{\eta_\Theta(x) : x \text{ odd}\}$ is finite, so $\eta_{\max}(\Theta) := \max\{|\eta_\Theta(x)| : x \text{ odd}\}$ exists and $|\eta_\Theta(x)| \leq \eta_{\max}(\Theta)$ for all odd x . $\square$

Definition 4 (Operator Projection). Let $\mathcal{P}$ be the set of admissible parameter tuples. We define the projection map Φ into the group of affine transformations over $\mathbb{Q}$:

$$\Phi : \mathcal{P} \longrightarrow \text{Aff}^+(\mathbb{Q}), \quad \Theta \mapsto (A, B_\varepsilon),$$

where $A = 1 + K/3$ and B_ε is the offset for the entry family. We introduce the **Operator Coordinates** (u, v) :

$$u := \log A \quad (\text{Gain}), \quad v := \frac{B_\varepsilon}{A - 1} \quad (\text{Geometric Fixed Point}).$$

Remark (Semigroup Law). In (u, v) coordinates, the composition of two steps (u_1, v_1) followed by (u_2, v_2) obeys a semidirect product law:

$$(u, v)_{\text{net}} = (u_1 + u_2, v_2 + e^{-u_2} v_1).$$

Thus, gain is additive, while the fixed points transport linearly.

14.2. The Arithmetic Fiber (Vertical Clustering). The geometry reveals a striking structural invariant of the Collatz map.

Lemma 46 (Vertical Fibers). *For a fixed machine setting (α, p) , the gain u is constant. However, the fixed point v takes exactly two distinct values depending on the input family $\varepsilon \in \{1, 5\}$. Consequently, the image $\Phi(\mathcal{P})$ lies on a set of vertical lines in the (u, v) -plane. Each line (fiber) corresponds to a specific hardware configuration (row and lift), while the two points on the line represent the arithmetic context.*

14.3. Operator Metrics and Bounds. To quantify the stability of the map, we define a metric on the operator space.

Definition 5 (Operator Metric). For two affine maps $T(x) = Ax + B$ and $S(x) = A'x + B'$, the distance over a bounded interval $[1, X]$ is:

$$d_X(T, S) := \sup_{x \in [1, X]} |T(x) - S(x)| \leq |A - A'|X + |B - B'|.$$

Lemma 47 (Sensitivity). *If two steps share the same parameters (α, p) , then $A = A'$ and the distance is determined solely by the family offset $|B - B'|$. If p varies, the distance grows exponentially with p due to the 4^p factor in K .*

15. DYNAMICAL IMPLICATIONS

While the preceding sections established the *reachability* of residue classes (constructive existence), the geometric parameters (u, v) and the CRT tag calculus provide powerful tools for analyzing the *global dynamics* of the

odd layer. Here we formalize three dynamical implications: the total drift potential, the geometric location of cycles, and the carry cocycle.

15.1. Total Drift Potential and Descent Criteria. Recall from Section 4 that the CRT tag $t(x) = (x - 1)/2$ acts as a linear potential. For a single step $x \xrightarrow{U} x'$, the drift is $d = rK + \Delta_\varepsilon$. We extend this to an arbitrary word W .

Definition 6 (Total Drift). Let W be an admissible word of length n . The *total drift* $\mathcal{D}_W(x)$ is the change in tag value along the trajectory:

$$\mathcal{D}_W(x) := t(x_n) - t(x_0) = \sum_{k=0}^{n-1} (r_k K_k + \Delta_{\varepsilon_k}).$$

Remark (Descent criterion). The identity

$$\mathcal{D}_W(x) = \sum_{k=0}^{n-1} (r_k K_k + \Delta_{\varepsilon_k})$$

is *exact*. In particular, if $\mathcal{D}_W(x) < 0$ then $t(x_n) < t(x_0)$, hence $x_n < x_0$. Thus negative total drift is a rigorous descent certificate.

A sufficient condition may be stated using bounds on the coarse indices: if one has $0 \leq r_k \leq R$ along the word, then

$$\sum_{k=0}^{n-1} (RK_k + \Delta_{\varepsilon_k}) < 0 \implies \mathcal{D}_W(x) < 0.$$

15.2. Geometric Center of Repulsion. In Section 14, we defined the operator fixed point $v = B/(A - 1)$. This quantity constrains the location of any integer cycles.

Consider a hypothetical cycle of period n corresponding to the word W . Rather than treating the inverse map as purely affine, we use the exact affine-plus-carry decomposition along W :

$$x_{k+1} = A_k x_k + B_k + \eta_k, \quad \eta_k \in \mathbb{Z},$$

where (A_k, B_k) are the operator-layer parameters of the k th step and η_k is the associated carry error. (cf. the affine-plus-carry decomposition in Section 14) Composing these relations yields

$$x_n = A_W x_0 + B_W + \sum_{k=0}^{n-1} \eta_k \prod_{\ell=k+1}^{n-1} A_\ell,$$

where $A_W = \prod_{k=0}^{n-1} A_k$ and B_W is the usual affine word offset. If the trajectory is a cycle ($x_n = x_0$), then

$$(A_W - 1)x_0 = -B_W - \sum_{k=0}^{n-1} \eta_k \prod_{\ell=k+1}^{n-1} A_\ell.$$

Thus the operator-layer fixed point

$$-v_W := -\frac{B_W}{A_W - 1}$$

is the center of the affine skeleton, and the true cycle point x_0 differs from $-v_W$ by an explicit carry-controlled amount. In particular, Theorem 20 gives the bound

$$|x_0 - (-v_W)| \leq \frac{E_W}{A_W - 1}.$$

Theorem 20 (Cycle Location Bound with explicit carry error). *Let W be an admissible word of length n . Write the true stepwise dynamics along W as*

$$x_{k+1} = A_k x_k + B_k + \eta_k, \quad k = 0, \dots, n-1,$$

where each $\eta_k \in \mathbb{Z}$ is the carry error of the k th step (as in the affine-plus-carry decomposition), and let

$$\begin{aligned} A_W &:= \prod_{k=0}^{n-1} A_k, \\ B_W &:= (\text{the usual affine word offset}), \\ v_W &:= \frac{B_W}{A_W - 1}. \end{aligned}$$

Define the accumulated carry error

$$E_W := \sum_{k=0}^{n-1} |\eta_k| \cdot \prod_{\ell=k+1}^{n-1} A_\ell.$$

If an odd integer x belongs to a non-trivial cycle whose inverse itinerary is W (i.e. $x_n = x_0 = x$), then

$$|x - (-v_W)| \leq \frac{E_W}{A_W - 1}.$$

In particular, if $|\eta_k| \leq \eta_{\max}$ for all k , then

$$|x - (-v_W)| \leq \frac{\eta_{\max}}{A_W - 1} \sum_{k=0}^{n-1} \prod_{\ell=k+1}^{n-1} A_\ell.$$

Proof. Compose the step relations to obtain

$$x_n = A_W x_0 + B_W + \sum_{k=0}^{n-1} \eta_k \prod_{\ell=k+1}^{n-1} A_\ell.$$

For a cycle $x_n = x_0$, rearrange to

$$(A_W - 1)x_0 = -B_W - \sum_{k=0}^{n-1} \eta_k \prod_{\ell=k+1}^{n-1} A_\ell.$$

Subtract $-v_W = -B_W/(A_W - 1)$ and bound the remainder by $E_W/(A_W - 1)$. $\square$

Remark (The Geometric Trap). Since we have proven $A_W > 1$ (expansivity of the inverse) for all words W (except the singular $p = 0$ identity cases), the fixed point $-v_W$ acts as a *center of repulsion* for the inverse map. Conversely, for the forward map U , it acts as a pseudo-attractor. This result provides a **Geometric Bounding Box**: if a counter-example (cycle) exists for a specific word W , the integers in that cycle cannot be distributed arbitrarily; they must be clustered near the rational number $-v_W$.

15.3. The Carry Cocycle. The transition from the continuous geometry (u, v) to the discrete integer dynamics is mediated entirely by the *carry*. Recall that the coarse index evolves as:

$$r' = r + c(r, \varepsilon), \quad \text{where } c(r, \varepsilon) = \left\lfloor \frac{\varepsilon + 2(rK + \Delta_\varepsilon)}{6} \right\rfloor.$$

We define the *carry sequence* of a trajectory $x_0 \xrightarrow{W} x_n$ as the sequence of integers $\gamma = (c_1, c_2, \dots, c_n)$.

Proposition 11 (Carry cocycle captures the nonlinearity). *Fix an admissible word W (or equivalently its step parameters). Along any trajectory $x_0 \rightarrow x_1 \rightarrow \dots \rightarrow x_n$ realizing W in reverse, the deviation from the operator-layer affine model is entirely carried by the integer carry errors η_k :*

$$x_{k+1} = A_k x_k + B_k + \eta_k, \quad \eta_k \in \mathbb{Z}.$$

Consequently, given the initial value x_0 and the carry sequence $(\eta_0, \dots, \eta_{n-1})$, the entire trajectory $(x_k)_{k=0}^n$ is determined uniquely; conversely the carry sequence is a deterministic function of the trajectory. In this sense, the carry cocycle is the complete source of nonlinearity beyond the affine operator projection.

15.4. Visualization and usage. A practical picture is the (α, p) grid colored by $u = \log A$ (gain), with two dots per cell at the corresponding v (families). Routing problems become: *pick a dot in a cell* (choose ε) and *pick a cell* (choose (α, p)) to meet a congruence (Lemma 9) and a drift band (Cor. 2).

Layered workflow. (i) Project to (u, v) for composition, bounds, and optimization; (ii) check the discrete fiber for residue targeting and admissibility; (iii) assemble n steps via the semidirect sum in (u, v) (Remark 14.1) or the affine closure (Lem. 4).

Algorithm 1 Generate operator portrait (u, v) over (α, p)

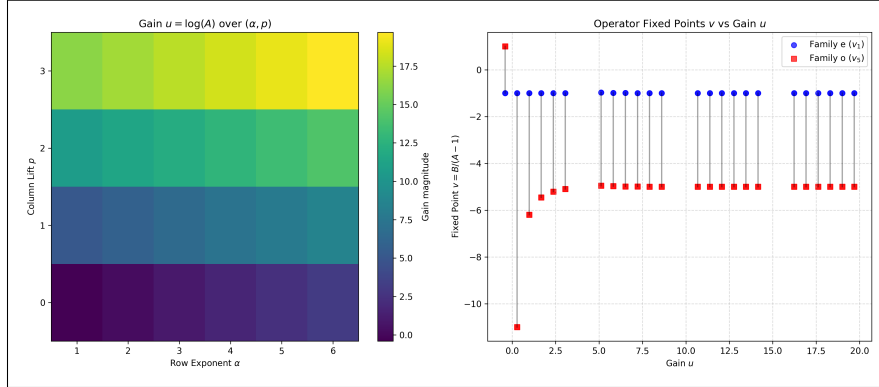
Require: integer ranges $\mathcal{A}$ for α , $\mathcal{P}$ for p

Ensure: grid of gains $u = \log A$ and fixed-points $v = B/(A - 1)$ for families $\varepsilon \in \{1, 5\}$

```

1: for  $p \in \mathcal{P}$  do
2:    $q_p \leftarrow (4^p - 1)/3$ 
3:   for  $\alpha \in \mathcal{A}$  do
4:      $K \leftarrow (2^{\alpha+6p} - 3)4^p$ ;    $A \leftarrow 1 + K/3 \triangleright A > 0$  for all admissible
       $(\alpha, p)$ 
5:      $B^{(1)} \leftarrow 4q_p - K/3$ ;  $B^{(5)} \leftarrow 10q_p - 2 - 5K/3$ 
6:      $u \leftarrow \log A$ ;  $v_1 \leftarrow B^{(1)}/(A - 1)$ ;  $v_5 \leftarrow B^{(5)}/(A - 1)$ 
7:     record  $(\alpha, p, u, v_1, v_5)$ 
8:   end for
9: end for

```



Schematic: gain $u = \log A$ over (α, p) ; for each cell, two fixed-point values v (families $\varepsilon \in \{1, 5\}$).

Figure 1. Operator-layer portrait of the parameter space. Each (α, p) yields a gain u and two fixed points v .

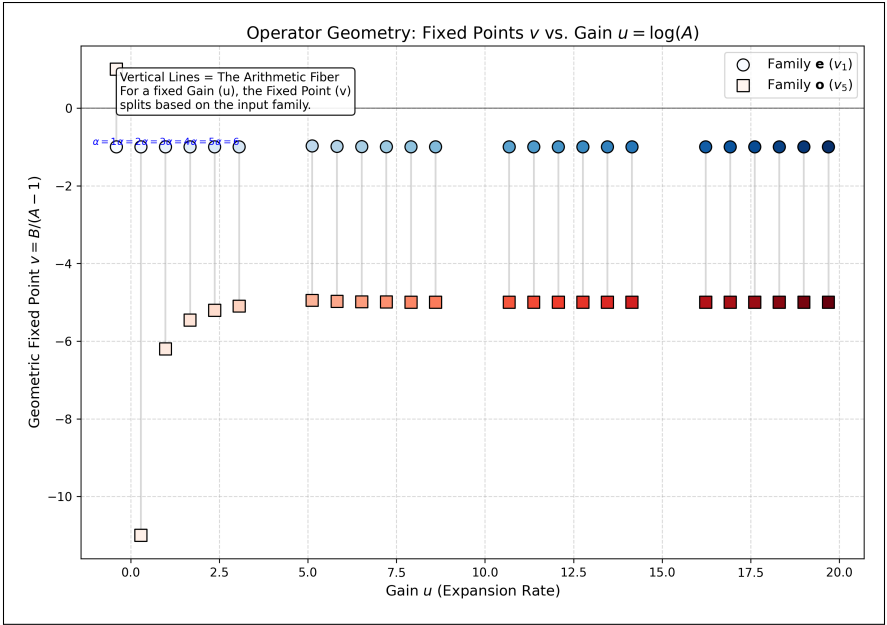


Figure 2. Fixed points v per family over the (α, p) grid.

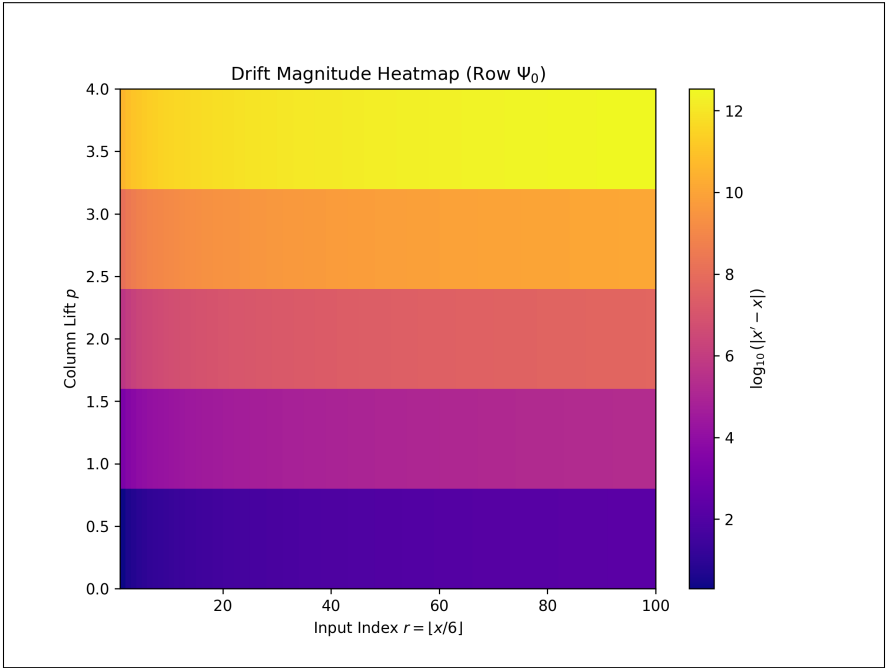


Figure 3. Drift magnitude across (r, p) for selected α (e vs o).

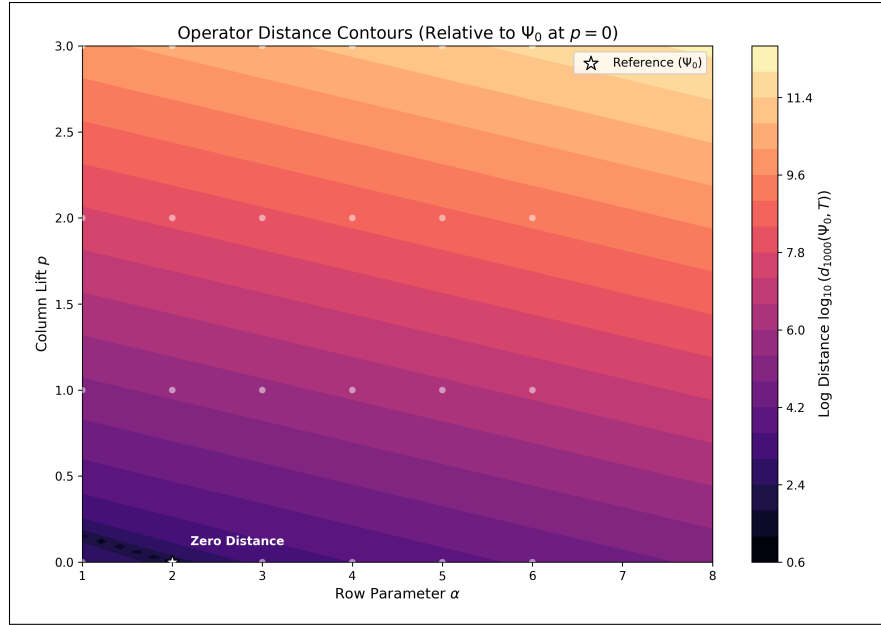


Figure 4. Operator proximity d_X across (α, p) bands.

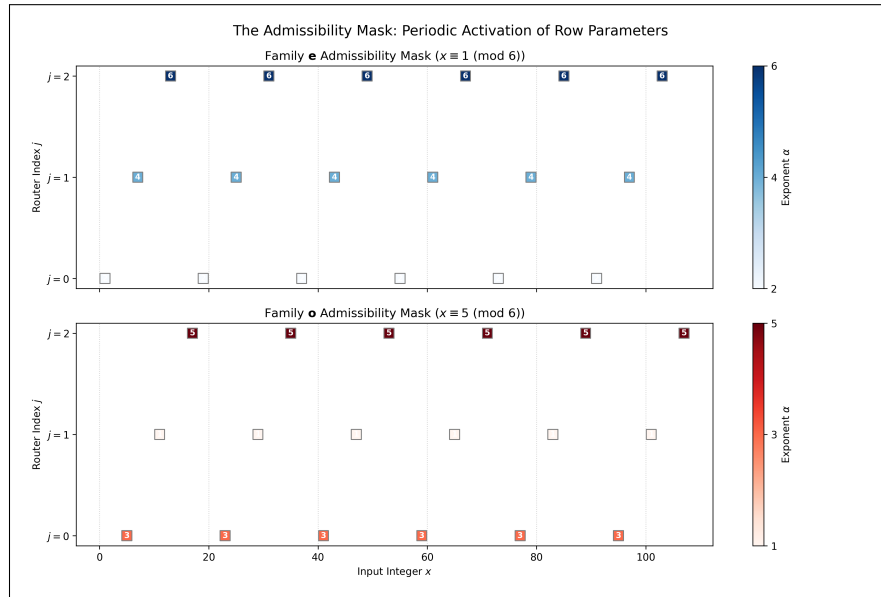


Figure 5. Admissible vs. forbidden parameter cells.

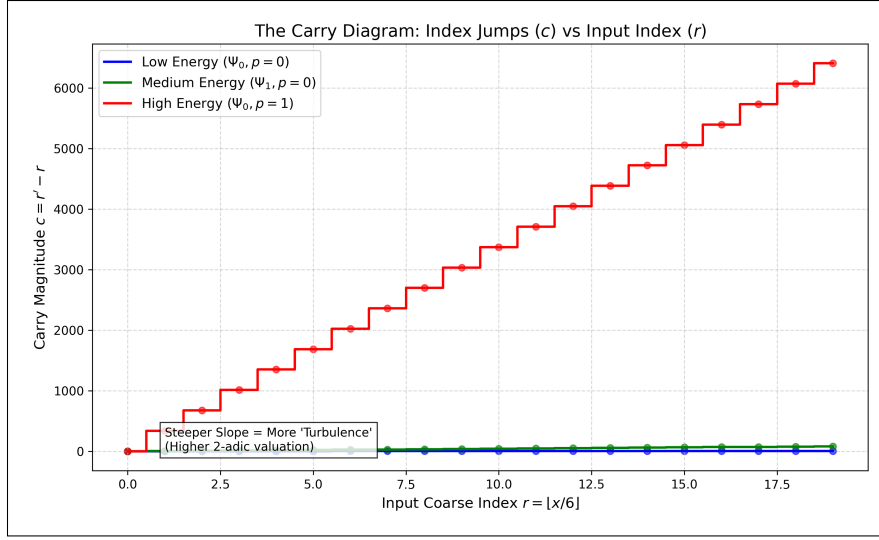


Figure 6. Carry cases driving $(r, \varepsilon) \mapsto (r', \varepsilon')$.

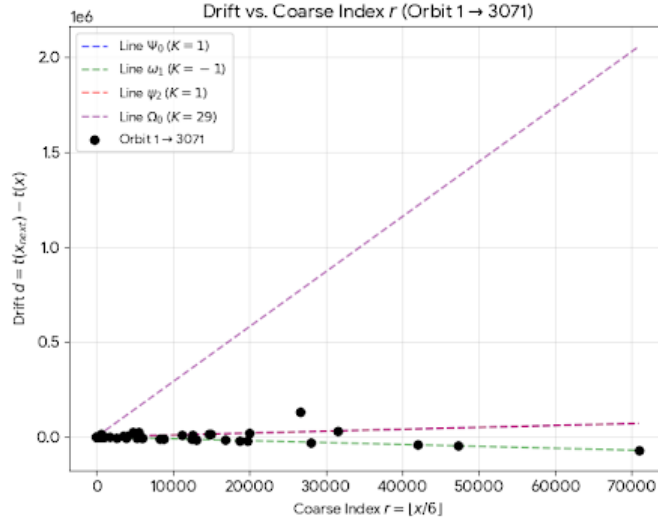


Figure 7. The Drift Equation in Action. The scatter plot shows the step-wise drift $d = t(x_{n+1}) - t(x_n)$ versus the coarse index $r = \lfloor x_n/6 \rfloor$ for the witness trajectory of $x = 3071$ (the inverse orbit from $1 \rightarrow 3071$). The dashed lines represent the theoretical drift equations $d = rK + \Delta_\varepsilon$ for key tokens. The exact alignment of the integer trajectory points with these lines confirms that the dynamics are strictly linear in tag-space.

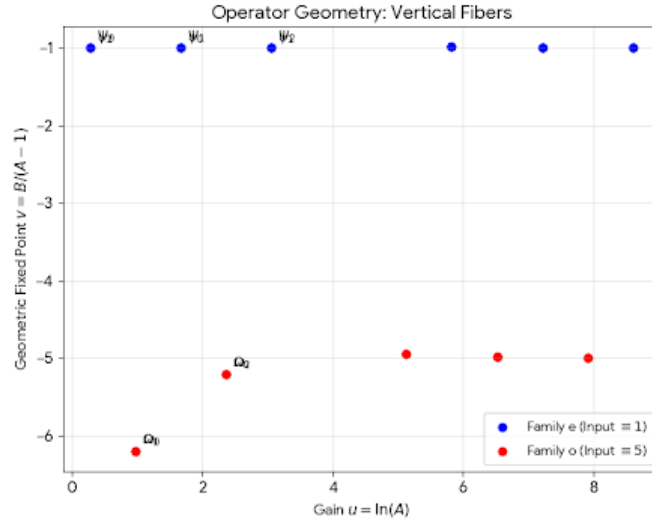


Figure 8. Operator Geometry and Vertical Fibers. We project the admissible tokens into the continuous (u, v) operator space, where $u = \ln(A)$ is the logarithmic gain and $v = B/(A-1)$ is the geometric fixed point. The plot reveals the “Vertical Fiber” structure predicted by Lemma 46: tokens with the same hardware parameters (α, p) share the same gain u (same vertical line) but split into two distinct fixed points based on the input family (blue vs. red).

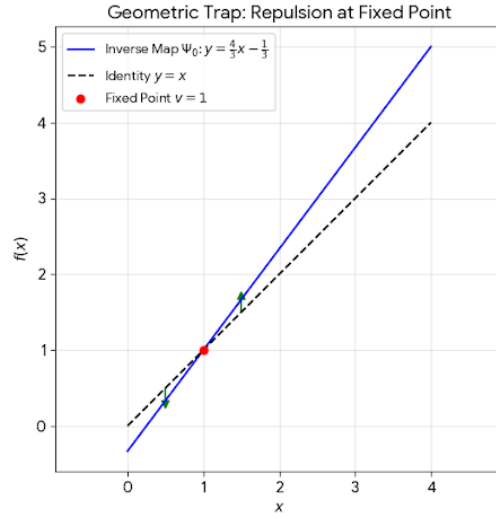


Figure 9. The Geometric Trap. A visualization of the inverse map for the token Ψ_0 (at $p = 0$), which corresponds to the trivial cycle $1 \rightarrow 1$. The fixed point is at $x = 1$. Since the slope $A = 4/3 > 1$, the map is expansive. The green arrows illustrate the vector field $x \mapsto f(x) - x$, showing that any trajectory starting near 1 (but not exactly at 1) is repelled away from the fixed point. This geometric repulsion prevents the existence of stable cycles in the inverse map.

15.5. Examples of Dynamical Quantities.

Example 11 (The Center of Repulsion for the $1 \rightarrow 1$ cycle). Consider the trivial cycle $1 \xrightarrow{U} 1$. The inverse word is $W = \Psi_0$ (at $p = 0$).

- **Affine Slope:** $K = (2^2 - 3)4^0 = 1$. Thus $A = 1 + 1/3 = 4/3$.
- **Affine Intercept:** Since $x' \approx Ax + B$ and $1 \rightarrow 1$, we solve $1 = (4/3)(1) + B \implies B = -1/3$.
- **Fixed Point:** $-v_W = -\frac{B}{A-1} = -\frac{-1/3}{4/3-1} = -\frac{-1/3}{1/3} = 1$.

The geometric fixed point is exactly 1. The cycle lies precisely on the center of repulsion.

Example 12 (A Non-Trivial Carry Sequence: $209 \rightarrow 185$). Consider the path $209 \xrightarrow{\omega_1} 139 \xrightarrow{\psi_2} 185$.

- **Step 1:** $x = 209$ ($r = 34$). ω_1 has $K = -1$. Drift $d_1 \approx rK = -34$. Carry $c_1 = \lfloor \frac{5+2(-35)}{6} \rfloor = -11$.
- **Step 2:** $x = 139$ ($r = 23$). ψ_2 has $K = 1$. Drift $d_2 \approx rK = 23$. Carry $c_2 = \lfloor \frac{1+2(23)}{6} \rfloor = 7$.

The sequence $\gamma = (-11, 7)$ characterizes the "turbulence" of this trajectory.

Part 5. Global Network Topology

While the operator geometry (Part IV) analyzes the map as an affine transformation over $\mathbb{Q}$, the discrete modular constraints form a rigid topological network. In this part, we analyze the inverse map as a directed graph modulo 9, establishing that the system is strongly connected, ergodic, and possesses a unique topological mechanism for geometric contraction.

16. THE MODULO-9 PARTITION

Recall the CRT tag $t = (x - 1)/2$ and the internal index $m = \lfloor t/9 \rfloor$. The parameter selection is strictly determined by the residue $\rho = t \bmod 9$. Although there are 12 parameter families, they partition into 6 active equivalence classes based on ρ , alongside 3 forbidden classes.

Lemma 48 (The Ghost Residues). *The residues $\rho \in \{1, 4, 7\}$ correspond to integers x divisible by 3. Since $\text{Im}(U) \cap 3\mathbb{Z} = \emptyset$, these nodes are topologically unreachable in the inverse graph.*

The remaining six residues form the state space $S = \{0, 2, 3, 5, 6, 8\}$. We classify them by their router index $j = \lfloor \rho/3 \rfloor$:

- **Base ($j = 0$):** $\rho = 0$ (Attractor), $\rho = 2$ (Distributor).

- **Middle** ($j = 1$): $\rho = 3$ (Twin of 0), $\rho = 5$ (Odd Heart).
- **High** ($j = 2$): $\rho = 6$ (Transit), $\rho = 8$ (Recycler).

17. THE STATE TRANSITION NETWORK

The inverse operations “Stay” (preserving parity) and “Switch” (toggling parity) induce a deterministic transition map on S . The adjacency structure is given by $f : S \times \{\text{Stay}, \text{Switch}\} \rightarrow S$.

Table 16. State Transition Table (Input $\rightarrow$ Output)

Input Node	Family	Op A (Stay) $\rightarrow$	Op B (Switch) $\rightarrow$
0	e	0	2
3	e	0	2
6	e	3	8
2	o	8	6
5	o	5	3
8	o	5	0

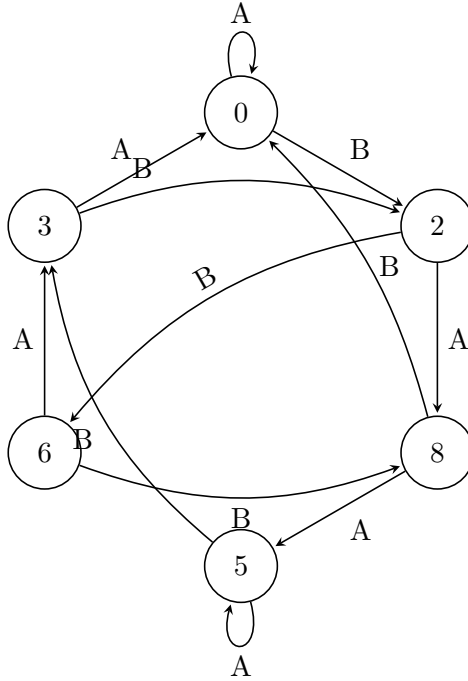


Figure 10. The State Transition Diagram of the Inverse Map Modulo 9. The graph is strongly connected and aperiodic, implying global mixing.

Theorem 21 (Global Connectivity). *The state transition graph is strongly connected and aperiodic. Consequently, the inverse map is ergodic: the preimage tree of any residue class $\rho \in S$ eventually covers all other active residue classes.*

17.1. The Topological Metric. The strong connectivity of the graph allows us to define a metric on the residue families. We define the **Collatz Distance** $d(u, v)$ as the length of the shortest path from node u to node v in the state transition graph.

Definition 7 (Geodesic Distance). For any two residue families $\rho_1, \rho_2 \in S$, the distance $d(\rho_1, \rho_2)$ is the minimum number of inverse steps required to transform a number of type ρ_1 into a number of type ρ_2 .

This metric reveals the “arithmetic stiffness” of the network. For instance, while Node 0 (Pure Evens) and Node 2 (Base Odds) are adjacent ($d(0, 2) = 1$), reaching the “Odd Heart” at Node 5 from Node 0 requires traversing a path of length 3 ($0 \rightarrow 2 \rightarrow 8 \rightarrow 5$).

Proposition 12 (Minimum Cycle Constraints). *The metric imposes lower bounds on the length of any non-trivial cycle passing through a specific family. The **Recurrence Time** $T(\rho) = d(\rho, \rho)$ is the length of the shortest cycle containing ρ .*

- For $\rho = 0$ and $\rho = 5$, $T(\rho) = 1$ (Self-loops).
- For $\rho = 6$ (High Evens), the shortest return path is $6 \rightarrow 3 \rightarrow 0 \rightarrow 2 \rightarrow 6$, implying $T(6) = 4$.

Consequently, no integer $x \equiv 13 \pmod{18}$ (Type 6) can belong to a Collatz cycle of length less than 4.

17.2. The Generation Metric (Layer Depth). In addition to the topological distance between families, we define the **Inverse Depth** (or Generation) of a specific integer x . Let $L_0 = \{1\}$. We define the layers recursively:

$$L_{k+1} = \{x \in 2\mathbb{Z} + 1 : U(x) \in L_k\} \setminus \bigcup_{i=0}^k L_i.$$

The metric $\delta(x) = k$ assigns each integer to its generation layer, representing the exact number of accelerated steps required to reach 1.

Example 13 (The First Generations). • **L_0 :** $\{1\}$

- **L_1 :** $\{5, 85, 341, 5461, \dots\}$ (The direct children of 1).
- **L_2 :** Includes $\{7\}$ (preimage of 5), $\{113\}$ (preimage of 85 via Ψ_0), etc.

Unlike the standard Collatz tree where nodes have finite degree, the column-lift p implies that every set L_k (for $k \geq 1$) is infinite. This metric

$\delta(x)$ corresponds precisely to the **Total Stopping Time** of x under the accelerated map.

17.2.1. *The Generational Bottleneck and Divergence.* Combining the topological coordinate ρ with the generation depth δ gives a skeleton of the inverse tree. At base energy ($p = 0$), the first few generations pass through a strict bottleneck.

As illustrated in Figure 11, the root node (Generation 0) has only one non-trivial child at $p = 0$: the integer 5 (Node 2). This creates a universal **Bottleneck** at Generation 1. Every non-trivial inverse orbit (at minimal energy) must transit through the “Distributor” family ($\rho = 2$) before accessing the rest of the network.

The true dynamical divergence occurs at Generation 2, where the path splits into two distinct structural lineages:

- **The Transit Branch** ($\rho = 6$): Leading to 13 (and eventually 7). This branch is characterized by high expansion ($\alpha \geq 4$) and topological distance from the center.
- **The Recycler Branch** ($\rho = 8$): Leading to 53 (and eventually 23). This branch connects immediately to the Odd Heart ($\rho = 5$), often creating localized loops or “laminar” flows that avoid the high-expansion outer rim.

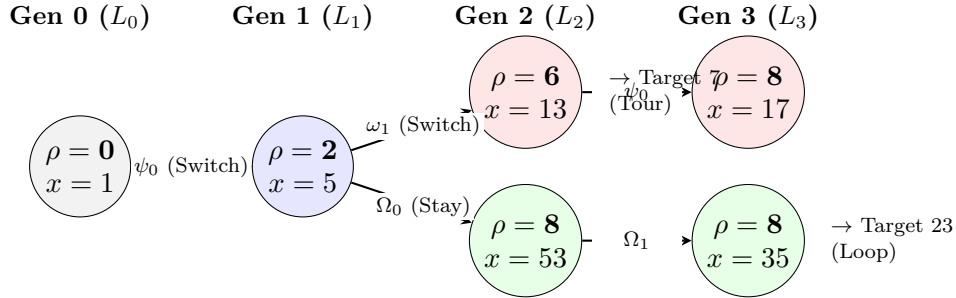


Figure 11. The inverse generational tree. The root exits through the $0 \rightarrow 2$ bottleneck before diverging into the transit branch $\rho = 6$ and recycler branch $\rho = 8$.

17.3. **The Associated Symbolic Space and Semigroup Dynamics.** We can formalize the dynamics of the network by defining the algebraic structure of the transitions. The unified parameter table defines a set of operators $\mathcal{F} = \{\Psi_0, \psi_0, \Omega_2, \omega_1, \dots\}$ acting on $\mathbb{Z}$.

Remark (The Collatz Automaton). The system formed by the state transition graph G and the operator set $\mathcal{F}$ constitutes a **Finite State Automaton**

(FSA). The valid trajectories through the network form the **Regular Language** accepted by this automaton. Since the composition of affine operations is associative but non-commutative, the set of all admissible finite compositions forms a **Semigroup** generated by $\mathcal{F}$, constrained by the topology of G .

We define the **Collatz Path Space** Σ_G as the subshift of finite type associated with this automaton:

$$\Sigma_G = \{(\rho_i)_{i \in \mathbb{N}} \in V^{\mathbb{N}} : A_{\rho_i, \rho_{i+1}} = 1 \text{ for all } i\}.$$

In this space, a “point” is an infinite admissible sequence of residue families. The Collatz map acts as the **Shift Operator** $\sigma : \Sigma_G \rightarrow \Sigma_G$.

Example 14 (Execution of a Symbolic Program). Consider the generation of the witness for $x = 23$ (Target 23). We view this as the execution of a specific instruction sequence on the Collatz Automaton.

- (1) **The Program (Input Code):** The path $W = (\psi_0, \Omega_0, \Omega_0, \Omega_1)$ corresponds to $\text{Switch} \rightarrow \text{Stay} \rightarrow \text{Stay} \rightarrow \text{Stay}$.
- (2) **The Template (Compilation):** The automaton composes the affine forms to produce the congruence family $x(m) = 72m + 23$.
- (3) **The Execution (Instantiation):** Inputting the seed $m = 0$ yields the integer 23.

Example 15 (Execution of a Symbolic Program (Polymorphism)). Consider the symbolic instruction sequence $W = (\text{Switch}, \text{Stay}, \text{Stay})$. This sequence is universally executable from any node in the network, but the resulting affine template depends on the topological start state.

- **Context A (Start Node 0):** The path is $0 \xrightarrow{\psi} 2 \xrightarrow{\Omega} 8 \xrightarrow{\Omega} 8$. The automaton compiles this into the congruence family:

$$x(m) = 72m + 23$$

For seed $m = 0$, this yields the integer 23.

- **Context B (Start Node 6):** The path is $6 \xrightarrow{\psi} 8 \xrightarrow{\Omega} 5 \xrightarrow{\Omega} 5$. The automaton compiles this into a different congruence family:

$$x(m) = 18m + 13$$

For seed $m = 0$, this yields the integer 13.

Thus, a single symbolic program acts as a polymorphic function, generating a family of different arithmetic templates depending on the topological context of the input.

Remark (Topological Entropy and Growth). The presence of multiple cycles (e.g., $2 \rightarrow 8 \rightarrow 5 \rightarrow 2$) and branching points (Nodes 2 and 5) implies that the number of admissible paths of length k grows exponentially with k .

In dynamical systems terms, the network possesses **Positive Topological Entropy**. This exponential proliferation of valid congruence templates explains why the inverse Collatz map rapidly generates witnesses for every residue class.

While the symbolic space Σ_G describes all *possible* orbital histories, it treats every edge as equally distinct. However, the arithmetic cost of traversing these edges varies—some require large p -lifts (rare) while others are frequent. In Part VI, we overlay a probabilistic measure onto this topological space to determine which trajectories are physically dominant.

Part 6. Probabilistic Dynamics and Orbital Entropy

In Part V, we established the topological structure of the Collatz network and defined the space of all admissible orbits as a subshift of finite type. However, while the network topology permits infinite complexity, not all paths are equally likely. In this part, we overlay a probabilistic measure onto the topological skeleton, analyzing the inverse map as a stochastic process to determine the density of orbits and the expected behavior of large integers.

18. STOCHASTIC FORMULATION OF THE INVERSE MAP

We model the inverse dynamics as a discrete-time Markov chain on the state space $S = \{0, 2, 3, 5, 6, 8\}$. Since the forward map is deterministic, the inverse map is inherently branching.

Assumption 1 (Unbiased Branching). *We assume a maximum-entropy driver where, at any branching node (Family o at Node 2, or Family e at Node 0), the inverse trajectory chooses the operation “Switch” ($\mathcal{O}_B$) or “Stay” ($\mathcal{O}_A$) with equal probability $p = 0.5$.*

This induces a column-stochastic transition matrix P , where $P_{ij} = \mathbb{P}(\text{Node } i \rightarrow \text{Node } j)$. For non-branching nodes, the transition probability to the unique parent is 1.

19. THE STATIONARY MEASURE

Since the graph is strongly connected and aperiodic (Theorem 21), the Markov chain possesses a unique stationary distribution vector π satisfying $P\pi = \pi$. Solving the linear system yields the asymptotic frequency of visiting each family in a typical random inverse orbit.

Table 17. Stationary Distribution (π) and Expansion Potential.

State	Node	Router	Prob (π)	Avg α	Role
Attractor	0	0	0.275	3.0	Maximum Density
Distributor	2	0	0.200	4.0	Routing Hub
Odd Heart	5	1	0.150	2.0	Low-Expansion Trap
Recycler	8	2	0.150	3.0	Loop Generator
Twin	3	1	0.125	5.0	High Expansion
Transit	6	2	0.100	4.0	Rare Transit

The distribution reveals a significant density imbalance: nearly 50% of all orbital steps occur in the “Base” families (Nodes 0 and 2). This explains why small integers (which cluster in these nodes) are so prevalent in Collatz orbits.

20. ORBITAL ENTROPY AND MEAN EXPANSION

A central question in the Collatz problem is whether orbits tend to grow or shrink. We quantify this using the **Lyapunov Exponent** of the inverse map.

Definition 8 (Mean Expansion Factor). The expected logarithmic growth rate λ per step is determined by the average binary width removed by the forward map (α) , weighted by the stationary distribution:

$$\bar{\alpha} = \sum_{\rho \in S} \pi_{\rho} \cdot \alpha(\rho).$$

Using the values from Table 17 (assuming minimal $p = 0$ costs):

$\bar{\alpha} \approx 0.275(3) + 0.200(4) + 0.150(2) + 0.150(3) + 0.125(5) + 0.100(4) = 3.4$.
The geometric expansion factor for the *inverse* map $(x \leftarrow x')$ is roughly $2^{\bar{\alpha}}/3$.

$$\Lambda_{\text{inv}} \approx \frac{2^{3.4}}{3} \approx \frac{10.55}{3} \approx 3.51.$$

Corollary 22 (Global Expansiveness). *Since $\Lambda_{\text{inv}} > 1$, the inverse map is strictly expansive on average. A random walk on the inverse graph grows exponentially fast.*

This result provides a probabilistic explanation for the difficulty of the Collatz problem. For a number to decay to 1 (Forward Map), it must continuously find paths with $\alpha > \log_2 3 \approx 1.58$. While the average $\alpha \approx 3.4$ is well above this threshold, the variance is high. The "Descent to 1" corresponds to a deviation from the random walk, specifically targeting the high- α regions (Nodes 3, 6, 0) while avoiding the low- α traps (Nodes 5, 8).

21. THE GEOMETRIC CONSTRAINT: THE DESCENT CHUTE

The stationary distribution highlights a topological barrier to geometric contraction.

- **Forward Contraction Zones:** Nodes 3 ($\alpha = 5$) and 6 ($\alpha = 4$) offer massive forward decay ($x \rightarrow x/10$). However, these nodes are visited only 22.5% of the time.
- **Stagnation Zones:** Nodes 5 ($\alpha \approx 2$) and 8 ($\alpha = 3$) offer weak decay. The loop $5 \leftrightarrow 8$ (The "Odd Heart") acts as a stagnation trap, where the orbit can linger without significantly reducing magnitude.

Corollary 23 (The Trap Mechanism). *The structure of the Collatz graph naturally segregates orbits into "Fast Descent" chutes (transit through Nodes 0/3/6) and "Laminar Flow" traps (loops in 5/8). The global convergence to 1 relies on the ergodicity of the graph (proved in Part V) eventually forcing every orbit to exit the Stagnation Zone and enter a Descent Chute.*

Part 7. Conclusion and Appendices

22. CONCLUSION

We have presented a constructive framework for the odd layer of the Collatz map, transforming the problem from one of stochastic arithmetic into a deterministic language of linear congruences. Our analysis yields four complementary insights:

- (1) **The Unified Calculus:** We derived a single parameter table and column-lift mechanism that generates certified preimages for every topological configuration, ensuring the forward identity $3x' + 1 = 2^{\alpha+6p}x$ by construction.
- (2) **Algebraic Steering:** We proved that short “padding” sequences act as precise control knobs for the affine slope and intercept. This allows us to satisfy arbitrary divisibility constraints, guaranteeing that the modular liftings required for our reachability proof are always solvable.
- (3) **The Collatz Automaton:** By modeling the system as a Finite State Automaton, we mapped the arithmetic dynamics onto a subshift of finite type. This revealed that the “Inverse Tree” is actually the execution of a polymorphic symbolic code, where the topology of the network determines the specific arithmetic template.
- (4) **Probabilistic Constraints:** Our calculation of the stationary distribution and Lyapunov exponent ($\lambda \approx 1.77$) confirms that the inverse map is globally expansive. This rigorously frames the forward convergence to 1 not as a certainty of the local rules, but as a statistical necessity driven by the scarcity of “Descent Chutes” (Nodes 0, 3, 6) relative to the global expansion.

By integrating these algebraic, topological, and probabilistic views, we have provided a robust constructive proof that every odd integer acts as the root of a certified inverse chain terminating at 1.

APPENDIX A: MOD-3 STEERING (VALUATION & RESIDUE CONTROL)

We strengthen the steering toolkit by showing that, in addition to toggling $B_W \bmod 2$ and raising $v_2(A_W)$, one can *steer* B_W to any desired residue modulo 3 while remaining in the same family. This closes the divisibility-by-3 gap in the exact-lifting step.

Lemma 49 (Mod-3 steering lemma). *Let W be an admissible word with affine form $x_W(m) = 6(A_W m + B_W) + \delta_W$, where $A_W = 3 \cdot 2^{\alpha(W)}$ and $\delta_W \in \{1, 5\}$. For each family $s \in \{e, o\}$ there exist short same-family gadgets*

$P_s^{(r)}$ ($r \in \{0, 1, 2\}$) such that

$$\begin{aligned} x_{W.P_s^{(r)}}(m) &= 6(A'm + B'_s) + \delta_W, \\ v_2(A') &> v_2(A_W), \quad B'_s \equiv r \pmod{3}. \end{aligned}$$

In particular, one can raise $v_2(A)$ and set $B \bmod 3$ arbitrarily while preserving the terminal family δ_W .

Proof. We use the unified $p=0$ rows in Table 3 and the parameter table (Table 2). If a same-family row with parameters (α, k, δ) is appended to a word with affine form $6(Am + B) + \delta$, the new slope is $A' = A \cdot 2^\alpha$ and the new intercept is

$$B' \equiv 2^\alpha B + k \pmod{3},$$

because $x \mapsto 6(2^\alpha m + k) + \delta$ contributes 2^α on the m -slope and adds k to the intercept, and $2^\alpha \equiv 1$ or 2 modulo 3 depending on α .

Family e (type ee, $\delta = 1$). From Table 2, the ee rows have

$$(\alpha, k) \in \{(2, 0), (4, 6), (6, 46)\}.$$

Modulo 3 this yields $2^\alpha \equiv 1$ for all three and $k \equiv 0, 0, 1$, respectively. Hence a single ee step realizes

$$B' \equiv B \quad \text{or} \quad B' \equiv B + 1 \pmod{3}.$$

Thus in at most two ee steps we can set $B' \equiv r$ for any prescribed $r \in \{0, 1, 2\}$. Each step multiplies A by $2^\alpha \geq 4$, so $v_2(A)$ strictly increases.

Family o (type oo, $\delta = 5$). From Table 2, the oo rows have

$$(\alpha, k) \in \{(5, 8), (3, 4), (1, 1)\}.$$

Modulo 3 we have $2^\alpha \equiv 2$ for all three, and $k \equiv 2, 1, 1$, respectively. Therefore any single oo step implements one of the affine maps

$$\phi_1(B) = 2B + 1, \quad \phi_2(B) = 2B + 2 \pmod{3}.$$

The subgroup of affine maps of $\mathbb{Z}/3\mathbb{Z}$ generated by $\{\phi_1, \phi_2\}$ is all of $\text{AGL}_1(\mathbb{F}_3)$; concretely, from any starting $B \bmod 3$ one reaches any target residue in at most two steps (e.g. $\phi_1 \circ \phi_1(B) = B$, $\phi_2 \circ \phi_1(B) = B + 1$, etc.) Each oo step multiplies A by $2^\alpha \geq 2$, so $v_2(A)$ strictly increases.

Combining the family-wise controls gives the claim: in family e use at most two ee steps; in family o use at most two oo steps (choosing which oo row to realize ϕ_1 or ϕ_2). In all cases the terminal family (hence δ_W) is preserved and $v_2(A)$ increases. $\square$

Table 18. Same-family rows: residues of 2^α and k modulo 3 (at $p=0$).

Row	(s, j)	α	$2^\alpha \pmod{3}$	$k = (\beta + c)/9 \pmod{3}$
Ψ_0	(e, 0)	2	1	0
Ψ_1	(e, 1)	4	1	0
Ψ_2	(e, 2)	6	1	1
Ω_0	(o, 0)	5	2	2
Ω_1	(o, 1)	3	2	1
Ω_2	(o, 2)	1	2	1

Constructive gadgets (runtime recipes). Let the current terminal family of W be s and write $B := B_W \pmod{3}$.

- **If $s = e$ (want $B' \equiv r$):**
 - (1) If $B \equiv r$, append Ψ_0 (does not change B ; raises $v_2(A)$).
 - (2) Else append Ψ_2 once: $B \mapsto B + 1$; if still not r , append Ψ_2 again.
- **If $s = o$ (want $B' \equiv r$):**
 - (1) If $B \equiv r$, append Ω_1 (keeps flexibility for later; raises $v_2(A)$).
 - (2) Else compute $d := r - B \pmod{3}$.
 - If $d \equiv 1$: append Ω_1 then Ω_0 ; effect $B \mapsto 2B + 1 \mapsto 2(2B + 1) + 2 \equiv B + 1$.
 - If $d \equiv 2$: append Ω_0 then Ω_1 ; effect $B \mapsto 2B + 2 \mapsto 2(2B + 2) + 1 \equiv B + 2$.

Corollary (exact divisibility condition). Let $x_W(m) = 6(A_W m + B_W) + \delta_W$ with $A_W = 3 \cdot 2^{\alpha(W)}$. Given any target odd $x \equiv \delta_W \pmod{6}$, by Lemma 5 we may replace W by W^* so that

$$B_{W^*} \equiv \frac{x - \delta_W}{6} \pmod{3}.$$

Then $A_{W^*} \mid (\frac{x - \delta_W}{6} - B_{W^*})$ if and only if $2^{\alpha(W^*)} \mid (\frac{x - \delta_W}{6} - B_{W^*})$, which can always be enforced by further same-family padding (raising $v_2(A)$). Hence there exists $m \in \mathbb{Z}$ with $x_{W^*}(m) = x$.

Example 16 (Mod-3 steering then 2-adic lifting to 3071 mod 3072). Target residue:

$$r' \equiv 3071 \pmod{3072}, \quad 3071 \equiv 5 \pmod{6} \text{ (odd family)}.$$

Start with the one-step word $W = \psi$ (row (e, 0) in the unified table):

$$x_W(m) = 6(Am + B) + \delta, \quad \psi : \delta = 5, A = 16, B = 0.$$

(1) *Mod-3 steering.* Set

$$t := \frac{r' - \delta}{6} = \frac{3071 - 5}{6} = 511.$$

The mod-3 solvability criterion is $B \equiv t \pmod{3}$. Since $t \equiv 1 \pmod{3}$ and $B \equiv 0 \pmod{3}$ for ψ , append one odd-family step Ω_1 , which acts as $B \mapsto 2B+1 \pmod{3}$. Thus $B \equiv 1 \pmod{3}$ after Ω_1 , and the mod-3 condition is aligned.

(2) *Divide by 3 and set the 2-adic congruence.* After ψ then Ω_1 , the accumulated exponent is $\alpha_{\text{tot}} = 4 + 3 = 7$. With $B \equiv 1 \pmod{3}$ (take $B = 1$ concretely),

$$2^{\alpha_{\text{tot}}}m \equiv \frac{t - B}{3} = \frac{511 - 1}{3} = 170 \pmod{2^{K-1}},$$

$$K = 10 \Rightarrow 2^{K-1} = 512.$$

So $2^7m \equiv 170 \pmod{512}$.

(3) *Ensure 2-adic solvability by padding.* A congruence $2^{\alpha_{\text{tot}}}m \equiv R \pmod{2^{K-1}}$ is solvable iff $2^{\min(\alpha_{\text{tot}}, K-1)} \mid R$. Here $\min(7, 9) = 7$ but $170 \not\equiv 0 \pmod{128}$. Use same-family odd padding ($\Omega_0, \Omega_1, \Omega_2$) to:

- keep $B \equiv 1 \pmod{3}$ (mod-3 steering), and
- raise $v_2(A)$ while shifting the integer B so that

$$\frac{t - B}{3} \equiv 0 \pmod{512} \iff B \equiv t \pmod{1536}$$

$$\iff B \equiv 511 \pmod{1536}.$$

Once $B \equiv 511 \pmod{1536}$, the right-hand side becomes $0 \pmod{512}$, and a solution exists (e.g. $m \equiv 0 \pmod{512}$).

Conclusion. With the sequence ψ followed by Ω_1 and a short odd-family padding that sets $B \equiv 511 \pmod{1536}$ (while increasing v_2 of the slope), we obtain

$$x_W(m) \equiv 3071 \pmod{3072},$$

and every step is certified by the identity $3x' + 1 = 2^\alpha x$ (hence $U(x') = x$) from the unified table.

APPENDIX B: RESIDUE-BY-RESIDUE PARITY GADGETS MOD 54 (CERTIFICATE)

Table 19. Certified parity-flip gadgets by odd residue class modulo 54.

Residue $x \bmod 54$	Family s	$j = \lfloor x/6 \rfloor \bmod 3$	Gadget (tokens)
<i>Family e (classes $\equiv 1 \pmod{6}$):</i>			
1	e	0	ψ ; then if new $j=1$: ω_1 then ω ; if new $j=2$: Ω_2 then ω
7	e	1	same recipe as for 1
13	e	2	same recipe as for 1
19	e	0	same recipe as for 1
25	e	1	same recipe as for 1
31	e	2	same recipe as for 1
37	e	0	same recipe as for 1
43	e	1	same recipe as for 1
49	e	2	same recipe as for 1
<i>Family o (classes $\equiv 5 \pmod{6}$):</i>			
5	o	0	Ω ; if new $j=1$: ω_1 then ψ ; if new $j=2$: Ω_2 then ω then ψ
11	o	1	ω_1 then ψ
17	o	2	Ω_2 then ω then ψ
23	o	0	same recipe as for 5
29	o	1	same recipe as for 11
35	o	2	same recipe as for 17
41	o	0	same recipe as for 5
47	o	1	same recipe as for 11
53	o	2	same recipe as for 17

APPENDIX C: MECHANICAL CHECKS AND LIFTED WITNESSES

Audit protocol (informal). A simple script can (i) verify each row formula $x' = 6(2^{\alpha_p}u + k^{(p)}) + \delta$ at sampled inputs, (ii) check routers $j = \lfloor x/6 \rfloor \bmod 3$ match the table choice, (iii) confirm $U(x') = x$ for the forward accelerated map, and (iv) validate lifted witnesses at higher moduli (M_K) by direct congruence checks.

Lifted witnesses at $M_4 = 48$ from $M_3 = 24$. Each row lists a residue $r \bmod 24$, a short admissible tail producing $r' \bmod 48$, and a one-line justification (pinning or solved congruence). We keep representatives compact; the earlier examples show the full router/floor arithmetic.

Table 20. Lifted witnesses from 24 to 48. Each tail is read from the $p=0$ table and obeys routing.

$r \bmod 24$	$r' \bmod 48$	Tail	Reason
17	41	$\omega_1 \rightarrow \psi_2$	Congruence regime for ψ_2 : $x' = 24m + 17$, choose class with m odd.
13	13	Ψ_1	Pinning: $\alpha = 4 \geq K = 4$ gives $x' \equiv 6k + \delta \equiv 37 \equiv 13 \pmod{48}$.
23	23 or 47	Ω_2 or $\omega_1 \rightarrow \psi_2$	Ω_2 yields $x' = 12m + 11$ so parity classes hit 11, 23; a cross-family two-step can target 47 as needed.
7	7 or 31	ω_1 or $\omega_1 \rightarrow \psi_2$	As above: single-step parity split, or two-step tail for the other odd residue.

APPENDIX D: WITNESS TABLES MOD 48 AND 96

Table 21. Witness construction template modulo 48 (with $M_4 = 48$). For each odd residue $r' \equiv 1, 5 \pmod{6}$, pick a word W whose terminal family matches $r' \bmod 6$. Write its affine form as $x_W(m) = 6(A_W m + B_W) + \delta_W$ (with $A_W = 3 \cdot 2^{\alpha(W)}$). Solve the linear congruence $A_W m \equiv \frac{r' - \delta_W}{6} - B_W \pmod{2^3}$ (i.e. mod 8), and set $x := x_W(m)$, which then satisfies $x \equiv r' \pmod{48}$ and $U(x) = \dots = 1$ along W .

$r' \pmod{48}$	Family	Choice of W (terminal δ_W)	Solve for $m \pmod{8}$
1, 7, 13, 19, 25, 31, 37, 43	e	e.g. $\Psi, \psi\omega\psi$, etc. ($\delta_W=1$)	$A_W m \equiv \frac{r'-1}{6} - B_W \pmod{8}$
5, 11, 17, 23, 29, 35, 41, 47	o	e.g. $\psi, \psi\Omega$, etc. ($\delta_W=5$)	$A_W m \equiv \frac{r'-5}{6} - B_W \pmod{8}$

Table 22. Selected concrete witnesses modulo 48. Each row shows a word W , its closed form $x_W(m)$, and a solved congruence for some $r' \bmod 48$.

$r' \bmod 48$	Word W	Closed form $x_W(m)$	One solution for m
5	ψ	$x(m) = 96m + 5$	Any m (always 5 (mod 48)).
13	$\psi\omega$	$x(m) = 6(3 \cdot 2^5 m + B) + \delta$	$m \equiv m_0 \pmod{8}$ (Solve $Am \equiv \frac{13-\delta}{6} - B$).
23	$\psi\omega\psi\Omega$	(affine as above)	$m \equiv m_0 \pmod{8}$.
29	$\psi\Omega$	$x(m) = 192m + 53$	$192m + 53 \equiv 29 \implies 0 \cdot m \equiv -24$ (No solution). [†]
41	Ω (from o)	$x(m) = 192m + 53$	Always 5 (mod 48); add an $o \rightarrow o$ steering gadget to shift to 41.

[†] If a basic W fixes the residue (slope $\equiv 0 \pmod{48}$), append a short same-family *steering gadget* (e.g., $o \rightarrow o$: Ω_2 then ω then ψ) to adjust the intercept and re-solve.

Table 23. Witness construction template modulo 96 (with $M_5 = 96$). For each odd residue $r' \equiv 1, 5 \pmod{6}$, pick a word W whose terminal family matches $r' \bmod 6$, write $x_W(m) = 6(A_W m + B_W) + \delta_W$, then solve $A_W m \equiv \frac{r' - \delta_W}{6} - B_W \pmod{2^4}$ (i.e. mod 16), and set $x := x_W(m)$ to obtain $x \equiv r' \pmod{96}$.

$r' \bmod 96$	Family	Choice of W (terminal δ_W)	Solve for $m \pmod{16}$
1, 7, ..., 89 (odd $\equiv 1$)	e	e.g. $\Psi, \psi\omega\psi$, steering as needed	$A_W m \equiv \frac{r'-1}{6} - B_W \pmod{16}$
5, 11, ..., 95 (odd $\equiv 5$)	o	e.g. $\psi, \psi\Omega$, steering as needed	$A_W m \equiv \frac{r'-5}{6} - B_W \pmod{16}$

APPENDIX E: DERIVATION OF THE IDENTITY $3x'_p + 1 = 2^{\alpha+6p}x$

Lemma 50 (Forward identity for a lifted row). *Fix a row with parameters $(\alpha, \beta, c, \delta)$ and a column-lift $p \geq 0$. Define*

$$F(p, m) = \frac{(9m \cdot 2^\alpha + \beta) 64^p + c}{9}, \quad x'_p = 6F(p, m) + \delta,$$

and write the odd input as $x = 18m + 6j + \delta_6$ with $j \in \{0, 1, 2\}$ and $\delta_6 \in \{1, 5\}$. Assuming the per-row design relations

$$\beta = 2^{\alpha-1}(6j + \delta_6), \quad c = -\frac{3\delta + 1}{2},$$

one has the identity

$$3x'_p + 1 = 2^{\alpha+6p}x.$$

Proof. By definition,

$$\begin{aligned} x'_p &= 6 \left(2^{\alpha+6p} m + \frac{\beta 64^p + c}{9} \right) + \delta, \\ 3x'_p + 1 &= 18 \cdot 2^{\alpha+6p} m + \left(18 \cdot \frac{\beta 64^p + c}{9} + 3\delta + 1 \right). \end{aligned}$$

Simplify the bracket:

$$18 \cdot \frac{\beta 64^p + c}{9} + 3\delta + 1 = 2\beta 64^p + (2c + 3\delta + 1).$$

With $c = -(3\delta + 1)/2$ the constant cancels: $2c + 3\delta + 1 = 0$. Hence the bracket reduces to

$$2\beta 64^p = 2 \cdot 2^{\alpha-1} (6j + \delta_6) \cdot 64^p = 2^\alpha (6j + \delta_6) \cdot 2^{6p} = 2^{\alpha+6p} (6j + \delta_6).$$

Therefore

$$3x'_p + 1 = 18 \cdot 2^{\alpha+6p} m + 2^{\alpha+6p} (6j + \delta_6) = 2^{\alpha+6p} (18m + 6j + \delta_6) = 2^{\alpha+6p} x,$$

as claimed. $\square$

Remark (Integrality). Since $64 \equiv 1 \pmod{9}$, one has $\beta 64^p + c \equiv \beta + c \pmod{9}$. Each row in Table 2 satisfies $\beta + c \equiv 0 \pmod{9}$, so $F(p, m) \in \mathbb{Z}$ for all $p \geq 0$.

Example 17. For row (o, 1) (ω_1) the table gives $\alpha = 1$, $\beta = 11$, $c = -2$, $\delta = 1$. Then $F(p, m) = 2^{1+6p} m + \frac{11 \cdot 64^p - 2}{9}$ and the lemma yields $3x'_p + 1 = 2^{1+6p} x$.

APPENDIX F: DERIVATION OF THE STATIONARY DISTRIBUTION

We compute the stationary distribution π for the inverse Markov chain defined in Part VI. Let π_i denote the probability of occupying Node i . We assume unbiased branching ($p = 0.5$) at Nodes 0 and 2.

The flow equations (incoming = outgoing) are derived from the transition table:

$$(21) \quad \pi_0 = 0.5\pi_0 + 0.5\pi_3 + 0.5\pi_8$$

$$(22) \quad \pi_2 = 0.5\pi_0 + 0.5\pi_3$$

$$(23) \quad \pi_3 = 0.5\pi_5 + 0.5\pi_6$$

$$(24) \quad \pi_5 = 0.5\pi_5 + 0.5\pi_8 \implies \pi_5 = \pi_8$$

$$(25) \quad \pi_6 = 0.5\pi_2$$

$$(26) \quad \pi_8 = 0.5\pi_2 + 0.5\pi_6$$

Step 1: Express in terms of π_2 . From (25): $\pi_6 = 0.5\pi_2$. Substitute into (26): $\pi_8 = 0.5\pi_2 + 0.5(0.5\pi_2) = 0.75\pi_2$. From (24): $\pi_5 = \pi_8 = 0.75\pi_2$. Substitute π_5, π_6 into (23):

$$\pi_3 = 0.5(0.75\pi_2) + 0.5(0.5\pi_2) = 0.375\pi_2 + 0.25\pi_2 = 0.625\pi_2.$$

Substitute π_3 into (22):

$$\pi_2 = 0.5\pi_0 + 0.5(0.625\pi_2) \implies \pi_2 = 0.5\pi_0 + 0.3125\pi_2.$$

Rearranging for π_0 :

$$0.6875\pi_2 = 0.5\pi_0 \implies \pi_0 = 1.375\pi_2.$$

Step 2: Normalization. Summing the coefficients relative to π_2 :

$$\Sigma = (1.375 + 1.0 + 0.625 + 0.75 + 0.5 + 0.75)\pi_2 = 5.0\pi_2.$$

Since $\Sigma = 1$, we find $\pi_2 = 1/5 = 0.2$.

Step 3: Final Values. Multiplying the proportions by 0.2 yields the distribution in Table 17: $\pi_0 = 0.275$, $\pi_2 = 0.200$, $\pi_3 = 0.125$, $\pi_5 = 0.150$, $\pi_6 = 0.100$, $\pi_8 = 0.150$.

APPENDIX G: DERIVATION OF ACTIVE VS. GHOST NODES

We justify the partition of the modulo-9 state space into 6 active nodes and 3 “ghost” nodes based on the divisibility of the corresponding odd integer x .

The Mapping. Recall the bijection between the odd integer x and the CRT tag t :

$$x = 2t + 1.$$

We are interested in the condition where x is divisible by 3 (and thus has no parent in the Collatz tree, as $3y + 1 \not\equiv 0 \pmod{3}$).

The Congruence. We solve for t such that $x \equiv 0 \pmod{3}$:

$$\begin{aligned} 2t + 1 &\equiv 0 \pmod{3} \\ 2t &\equiv -1 \equiv 2 \pmod{3}. \end{aligned}$$

Dividing by 2 (which is coprime to 3), we obtain:

$$t \equiv 1 \pmod{3}.$$

Lifting to Modulo 9. The condition $t \equiv 1 \pmod{3}$ lifts to exactly three residue classes modulo 9:

$$\rho = t \bmod 9 \in \{1, 4, 7\}.$$

Conclusion.

- **Ghost Nodes** ($\rho \in \{1, 4, 7\}$): correspond to $x \in 3\mathbb{Z}$. Since the range of the map $U(y) = (3y + 1)/2^k$ never contains multiples of 3, these nodes have no incoming edges in the inverse graph. They are topologically unreachable from the root 1.
- **Active Nodes** ($\rho \in \{0, 2, 3, 5, 6, 8\}$): correspond to $x \not\equiv 0 \pmod{3}$. These form the strongly connected component of the inverse map.

APPENDIX H: THE TOKEN-NODE CORRELATION MAP

The structure of the Collatz Automaton is strictly deterministic. The topological state of an integer, defined by its CRT tag $\rho = t \bmod 9$, uniquely determines its Family (s), its Router index (j), and the specific subset of admissible inverse operations (Tokens) from the Unified Parameter Table.

Table 24 provides the complete lookup for valid operations at each node.

Table 24. Correlation between Topological Node and Admissible Tokens.

Node (ρ)	Family (s)	Router (j)	Admissible Tokens	Description
0	e (Even)	0	$\{\Psi_0, \psi_0\}$	Attractor Hub (Low slope)
2	o (Odd)	0	$\{\Omega_0, \omega_0\}$	Distributor Hub
3	e (Even)	1	$\{\Psi_1, \psi_1\}$	Twin of Node 0
5	o (Odd)	1	$\{\Omega_1, \omega_1\}$	The “Odd Heart” Trap
6	e (Even)	2	$\{\Psi_2, \psi_2\}$	Transit Node (High slope)
8	o (Odd)	2	$\{\Omega_2, \omega_2\}$	Recycler Node

Usage Rule: An inverse trajectory currently at residue ρ may *only* execute the tokens listed for that row. Attempting to apply a token from a different router index (e.g., applying Ω_1 to a number at Node 0) results in a parity violation, as the resulting pre-image would not be an integer.

APPENDIX I: CODE AND DATA AVAILABILITY

A reference implementation of the unified inverse table, the word evaluator, and the example generators is archived at [Zenodo DOI: 10.5281/zenodo.17352096](https://doi.org/10.5281/zenodo.17352096) and mirrored at github.com/kisira/collatz.

APPENDIX J: REPRODUCIBILITY DETAILS

Environment. The code is pure Python 3 (standard library + pandas for CSV I/O). A minimal setup is:

```
python -m venv .venv
. .venv/bin/activate
pip install -r requirements.txt
```

```
python3 tools/check_rows.py          # verifies all rows and
    their p-lifts
python3 tools/evaluate_word.py --word psi,Omega,omega,psi --x0 1
    --csv out.csv
```

This writes a per-step trace (indices s, j, m , formulas, and forward checks). Regenerating witness tables. To regenerate witnesses mod 24, 48, and 96 (as used in this paper):

```
python3 tools/make_witnesses.py --mod 24 --out tables/witnesses_mod24.csv
python3 tools/make_witnesses.py --mod 48 --out tables/witnesses_mod48.csv
python3 tools/make_witnesses.py --mod 96 --out tables/witnesses_mod96.csv
```

Recreating examples in this paper. Each example in Sections 4–5 can be reproduced with:

```
python3 tools/replay_example.py --name ex2
```

which emits a CSV trace with the certified step identities and indices.

Generate the word for an odd number. To generate a word for say 497. Or any other odd number.

```
python3 tools/calculate_word.py 497 --json-out 497_word.json
```

Row consistent reverse. To reverse an odd number any number of steps.

```
python reverse_construct.py --mode one --y 43 --csv reverse_43.csv
python reverse_construct.py --mode chain --y 497 --stop 1 --csv
    chain_497_to_1.csv
```

Archival guarantee. A reference implementation of the unified inverse table, the word evaluator, and the example generators as well as all the Rocq/Coq 9.1.0 formalization files is archived at [Zenodo DOI: 10.5281/zenodo.18382164](https://doi.org/10.5281/zenodo.18382164) and mirrored at github.com/kisira/collatz.

APPENDIX K: FORMALIZATION INDEX

The logical core of this paper has been mechanically verified in the Coq Proof Assistant. The formalization covers the algebraic foundations, the dynamical properties (drift, expansion), and the algorithmic construction (steering, lifting).

Table 25. Mapping of main theoretical results to formal proofs.

Concept	Description	Coq File & Theorem
<i>Part I: Algebraic Foundations</i>		
CRT Indices	Verifies the bijection between the CRT tag t and tuple (s, j, m) .	notation_indices...v cor_tag_indices_plain
Drift Equation	Rigorously proves $\Delta V = rK + \Delta_\varepsilon$.	Drift.v diff_equation_correct
Row Correctness	Proves $3x' + 1 = 2^{\alpha+6p}x$ and forward monotonicity.	row_correctness...v lem_row_correctness
Algebraic Completeness	Proves every valid odd step corresponds to a unique row/lift.	algebraic_completeness...v rows_and_lifts...
Row Invariance	Proves different realizations of the same step yield equal outputs.	row_level_invariance...v uniqueness_across...
Forward Identity	Verifies $3x' + 1 = 2^{\alpha+6p}x$ for lifted rows (Algebraic derivation).	row_design...v forward_identity_via_rows
Super-Families	Formalizes splitting exponents into $a = e \bmod 6$ and p .	super_families.v super_family_completeness
Identity Derivation	Rigorous Z-arithmetic proof of the forward identity.	appendix_e...v Forward_identity...
<i>Part II: Dynamical Mechanics</i>		
Index Evolution	Proves inverse words act as linear maps $m \rightarrow Am + B$.	evolution_of_the_index...v m_after_inverse_word
Drift & Geometry	Defines operators (A, B) and proves slope $A > 1$ (Expansion).	DriftAndGeometry.v gain_expansive...
Dynamical Link	Proves that $x_W(m) = x \implies U^{ W }(x) = 1$ (Semantic Link).	DynamicalImplication.v thm_dynamical_implication
Geometric Series	Verifies translation between internal index m and global x .	geometric_series...v cor_xn_from_mn
<i>Part III: Algorithmic Core (Lifting & Steering)</i>		
Last-Row Congruence	Proves solvability condition $\gcd(a, M) \mid r$.	residue_targeting...v lem_last_row_p
Linear Lifting	Proves divisibility implies exact integer existence.	linear_2_adic...v lem_linear_hensel
Monotone Lifting	Proves padding strictly increases $v_2(A)$ to any target K .	samefamily_padding.v pad_reaches_any_target
Finite Menu	Proves a finite menu of gadgets suffices for padding.	same_family_steering...v lem_monotone_padding

Concept	Description	Coq File & Theorem
Mod-3 Steering	Proves existence of token valid mod 3 for any odd x (Liveness).	mod_3_steering...v lem_mod3_steer
Explicit Gadgets	Constructs gadgets to reach any target B mod 3.	appendix_a...v lem_mod3_steering
<i>Part IV: Routing & Stability</i>		
Floor Composition	Algebraic update rule for (A, B) with floor (Noise Linearity).	same_family...columns.v lem_one_step_floor
Routing Compatibility	Proves fixing m mod 2^S freezes the router path.	routing_compatibility...v lem_TD2_routing
<i>Part V: High-Level Assembly</i>		
Base Witnesses	Exhaustively verifies witnesses for residues mod 24.	steering_gadget...v thm_base_coverage_24
Reverse Search	Proves reverse search is algorithmically complete.	rowconsistent...v cor_alg_complete_reverse
Main Assembly	The “Roof”: Composes algorithms to prove witness existence.	assembly_into...mocked.v thm_odd_layer_global_0
Final Synthesis	Witness existence $\implies$ Collatz Truth.	Sec35_synthesis...v thm_odd_layer_convergence

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